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Physics-Informed State Reconstruction and Event-Source Inference From Sparse PMUs Under Limited Observability

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ABSTRACT Sparse phasor measurements leave distinct questions: whether hidden voltages can be reconstructed and whether event sources can be distinguished. We separate them with a hybrid differential-algebraic model. Functional observability governs the requested state target; finite-horizon separation between candidate response manifolds governs event cardinality and source support. On the IEEE 39-bus system with eight voltage/current PMUs, a causal model-based estimator reconstructs hidden voltages with 0.009634% mean relative error under matched operation. A separate Bayesian bank evaluates 137 load-intervention hypotheses. Across 3488 fresh nonlinear trajectories, exact-support Top-1 ranges from 24.4% for weak pairs to 100.0% for finite pairs. Weak-event evidence often resolves cardinality before source identity; profiled response distance explains that ambiguity better than support-only geometry. A separate unknown-onset load/line component obtains 53/54 timely family/source decisions with 4 unmatched alarms in paired continuous-stream development tests. Its nominal ablation retains one more timely event but produces more extras, exposing a model-uncertainty trade-off. These components support bounded state-recovery and source-inference claims. A published physical comparator is implemented, while its common confirmatory comparison remains pending. The complete joint state/event/integrity posterior, changing sensing, unseen-source causal generalization, and field transfer remain unvalidated.

INDEX TERMS dynamic state estimation, event detection, functional observability, limited observability, phasor measurement units, physics-informed inference, source localization.

I. INTRODUCTION

PHASOR measurement units (PMUs) reveal electromechanical transients through synchronized voltage and current phasors, yet transmission networks instrument only a fraction of their buses and branches [1], [2]. An event may start outside the measured region and still alter every available PMU. Missing frames and corrupted channels create a second problem because they can resemble a physical transient. A useful diagnosis must therefore answer separate questions: what is the hidden network state, what changed physically, and which measurements can still be trusted?

Dynamic estimators combine network models with synchrophasors [2], [3]; sparse-PMU classifiers identify events and faulted assets [4], [5]; and recent learning methods reconstruct hidden trajectories before localization [6], [7]. These capabilities do not imply one another. In particular, a low state-reconstruction error says nothing about whether two candidate sources produce distinguishable measurements.

This distinction changes how the problem must be evaluated. A generation change may be visible throughout the network while its source remains ambiguous. A dropped PMU identifies the failed measurement device, yet the lost channel can make a simultaneous physical event harder to localize. Treating both cases as entries in one multiclass label hides the relevant failure mechanism and mixes incompatible denominators.

Our central idea is to separate target recovery from source separation inside the same physical model. A candidate event specifies a family, a sparse support, an onset, and signed severity. Measurement availability and corruption remain separate variables. Each candidate generates a counterfactual DAE response. After admissible state and integrity direc-

tions are profiled, the distance between candidate response manifolds measures the information available for source discrimination [8], [9]. Zero distance means that two candidates intersect; a small positive distance means that they are distinct but difficult to separate with the available horizon and noise.

We test that argument in two stages. The state study asks what eight voltage/current PMUs recover, how the estimate fails when its operating point is stale, and whether a constrained physical recentering repairs that failure. The event study then evaluates all zero-, one-, and two-source load supports without fitting a bus-label classifier. A retrospective geometry audit and a longer-horizon diagnostic explain the weak-source failures. Finally, an earlier supervised ANDES model provides a contrasting test: it performs well when a source appears in training but collapses under complete source holdout. These studies use separate targets and denominators.

The contributions are:

1) a hybrid DAE formulation that keeps physical interventions and measurement integrity compositional, and represents simultaneous sources by a sparse support rather than a new class label;
2) a finite-horizon analysis that separates functional state recovery, sparse event injectivity, and the conditioning supplied by additional PMUs or observation time, and connects profiled separation to Gaussian error and support-recovery limits; and
3) a prospective 137-hypothesis load-source test that connects exact-support recovery to response-manifold geometry, together with independent state-reconstruction and source-holdout evidence.

The evidence yields one consistent picture. Sparse PMUs can recover the requested voltage field under a matched model, but operating-point error must be corrected physically. For simultaneous load events, source severity controls whether the posterior resolves only cardinality or the exact support. Longer observation adds information, yet nearly collinear signatures remain difficult. The implemented batch event estimator therefore supports matched load interventions with at most two sources. It does not yet solve the full multi-family problem shown in Fig. 1.

A separate causal component extends the executed evidence to unknown-onset load and line interventions in continuous streams. It retains conditional recovery states and marginalizes a finite bank of fixed network hypotheses. Its source/family array and increments separate event-evidence evolution from model reweighting. Section VIII-E reports the resulting precision–timeliness trade-off, including failures; it does not identify this component with the complete joint posterior. We implement a published line-outage likelihood comparator and preregister its eligible confirmatory comparison rather than infer general superiority from incompatible task definitions.

II. PRIOR WORK AND RESEARCH GAP

A. STATE, ALGEBRAIC VARIABLE, AND UNKNOWN-INPUT ESTIMATION

Power-system dynamic state estimation combines machine, control, load, and network models with synchronized measurements; model uncertainty and measurement availability remain persistent constraints [2]. Robust filters address non-Gaussian measurement errors [10], [11], whereas joint input/state estimators recover exogenous inputs in linear systems [12], nonlinear DAEs [3], [13], and event-triggered or distributed power networks [14], [15]. These estimators do not define a multi-family physical-source bank with a simultaneous integrity state.

Learning methods reconstruct states from sparse measurements by encoding graph structure or power-flow relations. Examples combine graph learning with physics-based estimates [16], outlier-resistant estimation and outage identification [17], or topology-aware reconstruction under PMU failures and bad data [18]. Robust reconstruction does not, by itself, identify the physical source.

B. SPARSE-PMU EVENT AND SOURCE INFERENCE

PMU event identification uses subspaces, engineered features, transfer learning, and graph models [19]–[22]. Label-light field-stream detection and scarce-label transfer target event presence or family [23], [24]; physical-source recovery is distinct. Prior localization spans joint classification/localization [4], sparse-PMU fault, generation-loss, and outage methods [5], [25]–[27], group-sparse physical dictionaries [28], and topology-constrained graph losses [29].

Model-based outage inference provides the closest classical comparison. Phasor angles support line-outage detection [30]; uncertain pre-event state can be marginalized jointly with outage identity and sensor placement [31]; and sequential likelihoods handle stochastic operating variation [32], [33]. Matched-subspace detectors profile linear nuisance through a generalized likelihood ratio [9]. These are the likelihood and nuisance-elimination precedents used here. Our batch load contract instead uses a sparse voltage/current response dictionary and known onset; it is not a quickest-detection comparison. For the separate causal component, we implement the persistent single-outage recursion of Chen *et al.* with its partial-observation covariance marginalization. Its persistent-line scope determines the eligible comparison; load classification and restoration are not attributed to that algorithm.

Open-set recognition differs from source holdout. Siamese boundaries and generative evidence reject events outside learned classes [34]–[36]. Source generalization instead ranks a physically enumerable but unlabeled asset as a counterfactual source.

The pmuBAGE collection supports standardized analytics but does not certify source holdout [37]. Integrity is a separate variable: robust recovery distinguishes excursions from outliers [38], while utility examples include missing, locked, quantized, and timing-affected synchrophasors [39].

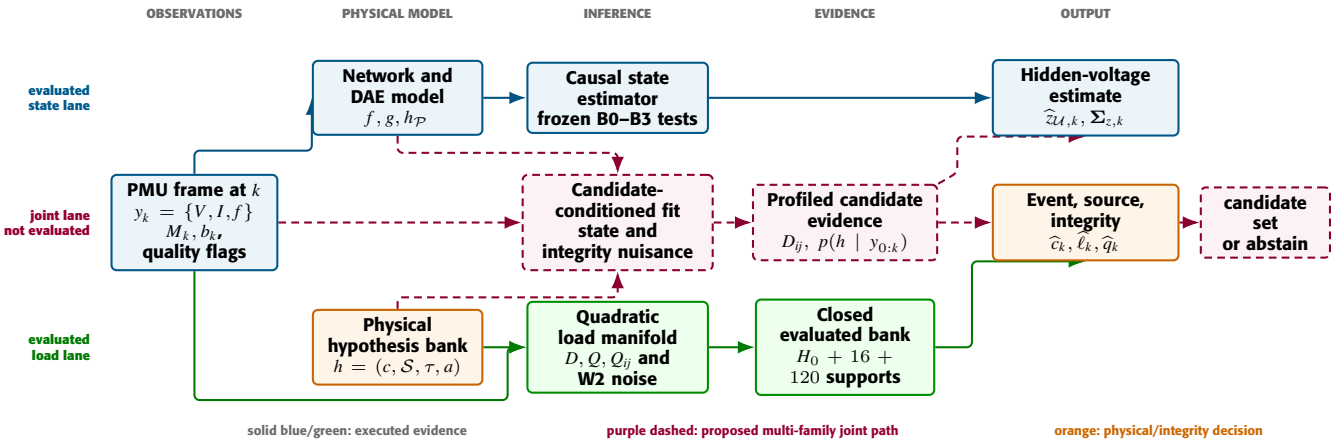


FIGURE 1. Scientific argument and validation boundary. The blue B0–B3 path is an independently evaluated state-reconstruction component. The green path is the implemented batch reference estimator $\mathcal{E}_B$. The dashed path is the proposed joint estimator $\mathcal{E}_J$; its complete posterior remains unevaluated. The exploratory single-event pilot implements only part of that route.

The unresolved issue is the evaluation contract. New replicas from trained assets measure interpolation, not source generalization; nuisance variation can also alter conclusions [40]. Reconstruction followed by localization is prior art [6], [7], but not under the combined held-source, diagnosability, integrity, and mismatch tests used here.

Frequency and rate-of-change-of-frequency (ROCOF) channels are themselves the output of a dynamic estimation problem rather than ideal derivatives of a noiseless phasor angle. A prior benchmark by the authors compared 16 dynamic frequency estimators spanning six practical groups over 34 disturbance scenarios across ten families, including ramps, phase jumps, harmonics, ringdown, and composite inverter-based-resource event sequences, and found substantial disturbance-dependent latency–robustness trade-offs [41]. That result motivates treating reported frequency and ROCOF as measured quantities with their own estimator- and disturbance-dependent error, not as an additional noiseless PMU channel.

C. OBSERVABILITY, DIAGNOSABILITY, AND SENSOR VALUE

Functional observability asks whether a requested function of state can be inferred without full-state observability [42], [43]; nonlinear extensions require the observation fibers to determine the requested functional locally, not merely a favorable Jacobian at one point [44]. Unknown-input observability instead asks whether exogenous inputs can be reconstructed from output histories [45]; fault detectability asks whether a fault can be separated while disturbances are decoupled [46]. Sparse strong observability extends the question to corrupted sensor sets [47], [48], and attack identification characterizes when adversarial input supports are distinguishable from system behavior [49]. Eriksson *et al.* quantify pairwise diagnosability for stochastic linear descriptor models [8]. Likelihood inference with singular first-order information also has established nonregular asymptotics [50]; therefore, a vanishing projected tangent is not by itself a universal impossibility result. PMU placement based on empirical observability Gramians treats sensor value as task dependent [51], [52]. For event sources, Jamei *et al.* derive placement-dependent fault-localization limits [53]; Conte *et al.* partition a radial grid into indistinguishable line clusters [54]; and Yildiz and Abur optimize sparse placement for event detection and identification [55]. These results distinguish hidden-state reconstruction, input reconstruction, and pairwise source separation. They also rule out a general optimal-placement claim from one prescribed eight-PMU set.

D. REMAINING TECHNICAL GAP

Table 1 shows why one accuracy value cannot close the gap. The state path must specify the requested function, and the event path must expose indistinguishable sources before any classifier is trusted. A source-generalization test must also remove the physical asset from every source-labeled fitting step. Under that contract, learning is useful only if it corrects model discrepancy or reduces candidate-search cost without memorizing asset identity.

The experiments cover only part of this intersection. PowerDynamics supports hidden-voltage estimation, physical recentering, and prospective load-source inference. ANDES documents what a supervised representation loses under source holdout. A common multi-family experiment with state and integrity nuisance remains future work.

III. STATE, EVENT, AND SOURCE PROBLEM

A. ONE PHYSICAL MODEL, TWO INFERENCE QUESTIONS

Let $x(t)$ and $z(t)$ denote differential and algebraic states, $q(t)$ the network mode, and $\boldsymbol{\theta}$ the operating parameters. A physical candidate is

$$h = (c, \mathcal{S}, \tau, a, d), \tag{1}$$

where c is the event family, $\mathcal{S}$ its bus or branch support, τ the onset, a the signed severity, and d the temporal profile. The

TABLE 1. Research gap by methodological family. The full paper-by-paper capability matrix is reported in Supplementary Appendix A.

Family	Established capability	Remaining issue for this problem	Role here
Dynamic state and unknown-input estimation [3], [12], [13]	Reconstructs states or unknown inputs from model and PMU histories	Does not usually define a closed physical event-source bank with compositional integrity	Defines the state-reconstruction path and its nuisance variables
Learned event localization [4], [5], [29]	Detects and localizes represented event families under sparse sensing	Source-label training can confound asset memorization with source generalization	Supplies the supervised baseline and held-source stress test
Model-based event inference [30]–[32]	Ranks physical outage hypotheses and can marginalize operating uncertainty	Published contracts are event-specific and do not establish the present simultaneous-load problem	Motivates physical candidate likelihoods and nuisance profiling
Observability and placement [44], [53], [55]	Characterizes state recovery or event separation for a declared sensing task	Rank alone does not quantify finite-horizon support ambiguity	Supplies the target-recovery and diagnosability tests
This work	Reconstructs a requested hidden-voltage field and ranks all load supports with $K \leq 2$	The multi-family state/event/integrity posterior remains untested	Connects separate evaluated paths through one information-based formulation

hybrid plant and measurement boundary are

$$\dot{x} = f_h(x, z, u; \boldsymbol{\theta}) + w_x, \qquad \mathbf{0} = g_h(x, z, u; \boldsymbol{\theta}), \tag{2}$$

$$y_k = M_k[h_{\mathcal{P}_k}(x_k, z_k, q_k; \boldsymbol{\theta}) + b_k] + v_k. \tag{3}$$

The event enters the physical equations through h. In contrast, M_k records channel availability and b_k records measurement corruption. A physical event and a failed PMU can therefore coexist without becoming a new event class.

For a scheduled impulse-free intervention, the physical switch preserves the differential state and re-equilibrates the algebraic state in the post-event mode:

$$x(\tau^+) = x(\tau^-), \qquad g_{h^+}\left(x(\tau^-), z(\tau^+), u; \boldsymbol{\theta}^+\right) = \mathbf{0}. \tag{4}$$

This right limit is not necessarily the array stored inside a simulator callback. The frozen trajectories begin at the first retained 30-Hz frame, so they validate the sampled response rather than the callback microstate. Supplementary Appendix D proves the local consistency expansion and separates the direct algebraic PMU signature from its subsequent dynamic propagation.

The primary estimated electrical target is the rectangular complex-voltage field, not a collection of independently learned PMU channels. Conditional on a physical candidate and its network mode, bus injections, oriented terminal currents, and complex powers are deterministic functionals of the estimated DAE state. The network map uses $I = Y_{\text{bus}}V$ and $S_i = V_i I_i^*$; terminal currents retain the complete branch primitive rather than replacing it by a bus-injection current. Thus a fault changes a shunt, a line outage changes Y_{bus}, and generation or load events change the corresponding injections.

Supplementary Appendix B defines the corresponding electrical output maps and their frequency, ROCOF, and balanced-waveform qualifications. They belong to the proposed joint estimator $\mathcal{E}_{\text{J}}$; the implemented batch event estimator $\mathcal{E}_{\text{B}}$ does not infer a state trajectory or virtual outputs. Only hidden complex voltage is empirically validated by the separate state-reconstruction component.

The first question asks for a function of state. With PMUs at

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}, \tag{5}$$

the requested target is the complex voltage at the remaining 31 buses. The state estimator uses rectangular voltage and controlled terminal-current channels. The second question asks which physical candidate generated the observed response. For the evaluated load family,

$$\mathcal{S} \subseteq \mathcal{L}, \qquad |\mathcal{L}| = 16, \qquad K = |\mathcal{S}| \leq 2, \tag{6}$$

which yields one no-event hypothesis, 16 single-source hypotheses, and 120 two-source supports.

B. PHYSICAL EVENTS AND MEASUREMENT INTEGRITY

The historical event bank has eight abnormal labels, but the label number is not the scientific variable. Table 2 rewrites the bank as four physical interventions, two integrity states, and two prescribed compositions. This representation matters because a missing packet identifies an integrity source directly while reducing the information available for physical-source inference.

The desired decision consists of the hidden-voltage estimate, its uncertainty, the physical family and support, and the integrity state. These components use different correctness tests. Hidden-voltage relative error is averaged over a complete trajectory and the 31 requested buses. Event detection, family classification, exact-support recovery, and integrity localization retain their own populations. A joint decision is correct only when every applicable component is correct.

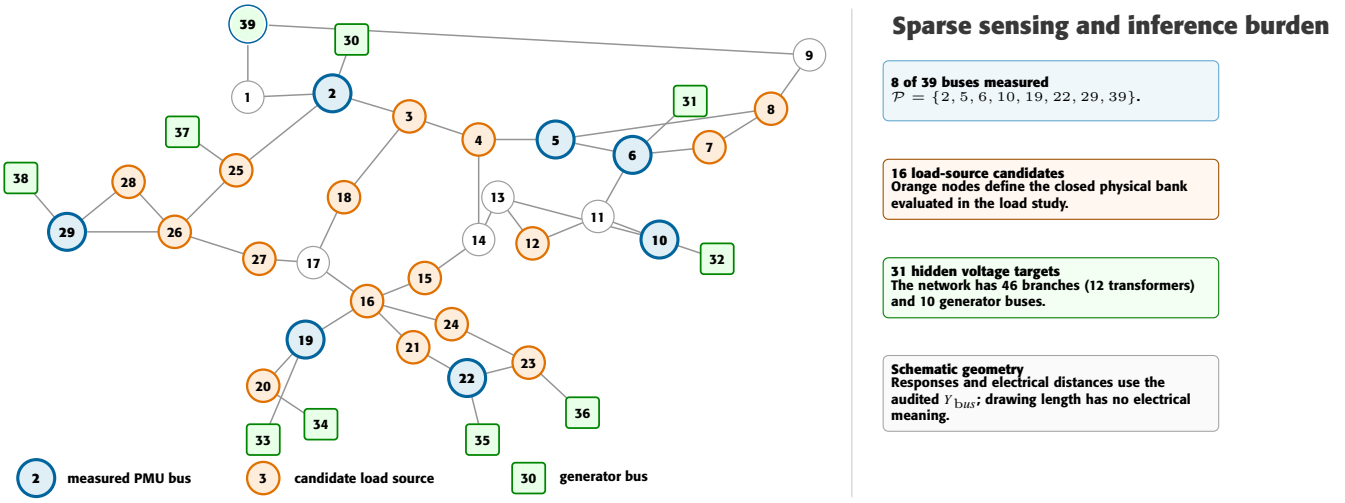


FIGURE 2. **Sensing and candidate geometry on the IEEE 39-bus system. Blue circles mark the eight PMU buses, orange circles the 16 evaluated load-source buses, and green boxes generator buses. The drawing preserves the audited 46-branch connectivity but does not encode geographic or electrical distance.**

TABLE 2. **Event ontology and historical ANDES inventory. Each source configuration has five replicas. Physical and integrity sources remain separate outputs.**

ID	Operational meaning	Physical intervention	Physical source	Integrity state/source	Records
0	Normal operation	none	none	nominal	5
1	Bus fault	fault admittance	bus	nominal	195
2	Line outage	topology removal	branch	nominal	170
3	Generation change	active/reactive injection	generator bus	nominal	50
4	Load change	active/reactive withdrawal	load bus	nominal	90
5	Missing data	none	none	dropout / PMU	40
6	Generation change with missing data	generation change	generator bus	dropout / PMU	50
7	Bad data	none	none	corruption / PMU channel	40
8	Prescribed mixed disturbance	generation plus load changes	declared generator bus	measurement disturbance / PMU	50
	Total				690

Causality and generalization are also explicit. A causal estimate at time k may use only $y_{0:k}$; fixed-lag or complete-window results are retrospective. A held-source test removes the physical asset from every source-labeled fitting and tuning step. Confidence intervals use trajectories, event parents, or rebuilt plants as independent units, never individual 30-Hz samples. Supplementary Appendix C records the exact metrics and event operators.

IV. WHAT SPARSE MEASUREMENTS CAN DETERMINE

The analysis separates two questions before an estimator is chosen. The first asks whether the measured history determines the requested hidden-voltage function. The second asks whether it distinguishes competing event supports. Both depend on the sensing geometry, but they fail for different reasons.

A. RECOVERING A REQUESTED VOLTAGE FUNCTION

Linearizing the index-reduced DAE around a verified equilibrium gives

$$\delta x_{k+1} = A_d \delta x_k + w_k, \qquad \delta y_k = C \delta x_k + v_k, \qquad (7)$$
$$\delta z_{\mathcal{U},k} = L_{\mathcal{U}} \delta x_k.$$

Here $\delta x_k \in \mathbb{R}^{114}$, the 32 measured channels form δy_k, and the 62 rectangular voltage components at the 31 unmeasured buses form the target. For a horizon of N transitions, let

$$\mathcal{O}_N = \mathrm{col}(C, CA_d, \ldots, CA_d^N), \qquad L_r = L_{\mathcal{U}} A_d^r. \qquad (8)$$

Proposition 1 (Finite-horizon functional reconstruction): The noise-free measurement history through N uniquely determines the requested voltage at time $r \le N$ if and only if

$$\ker \mathcal{O}_N \subseteq \ker L_r. \qquad (9)$$

The condition is weaker than full-state observability: unobservable state directions are harmless when they do not change the requested voltage. Under Gaussian noise, the same row-space condition yields the minimum-variance linear unbiased target estimate; its covariance depends on singular values and target alignment, not rank alone [42], [43]. Supplementary Appendix D gives the proof, the covariance expression, and the nonlinear constant-rank qualification [44].

The numerical audit illustrates why that qualification matters. The original implementation formed $\mathcal{O}_N^{\mathsf{T}}\mathcal{O}_N$ and applied a relative eigenvalue threshold. Its pseudo-rank rises and then falls as samples are added, which an exact observability

rank cannot do. Figure 3 therefore separates two conclusions. Longer history improves numerical conditioning, but the tested computation does not certify full-state or exact functional observability. The nonlinear estimator is evaluated independently.

The bus-level residual from this audit was frozen before the nonlinear TEST trajectories were scored. Its association with empirical bus error is used only as an exploratory theory-to-data check because the projector remains threshold sensitive. An SVD-based, noise-tied rank audit is still required for a structural certificate.

B. DISTINGUISHING EVENT SUPPORTS

Source inference introduces a different nuisance space. Under candidate h, stack the available measurements over H samples as

$$Y_H = s_{h,H} + \mathcal{O}_{h,H}\boldsymbol{\xi}_h + \mathcal{E}_{S,H}\boldsymbol{\beta}_h + \boldsymbol{\varepsilon}_H, \tag{10}$$

where $s_{h,H}$ is the physical response, $\boldsymbol{\xi}_h$ an admissible initial-state deviation, and $\mathcal{E}_{S,H}\boldsymbol{\beta}_h$ corruption on integrity support S. Each candidate therefore defines an affine response manifold. For candidates i and j, their profiled separation is

$$D_{ij,S}(H) = \frac{1}{2}\min_{\boldsymbol{\xi}_i,\boldsymbol{\xi}_j,\boldsymbol{\beta}} \big\| s_{i,H} + \mathcal{O}_{i,H}\boldsymbol{\xi}_i - s_{j,H} - \mathcal{O}_{j,H}\boldsymbol{\xi}_j + \mathcal{E}_{S,H}\boldsymbol{\beta} \big\|^2_{R_H^{-1}}. \tag{11}$$

This distance profiles the nuisance directions that either candidate can use to explain the same measurements. It specializes descriptor-model diagnosability and matched-subspace testing to the PMU event bank [8], [9].

Proposition 2 (Local candidate indistinguishability): If $D_{ij,S}(H) = 0$, the two candidate manifolds intersect. At an intersection they induce the same Gaussian measurement law, so no rule can distinguish them uniformly over the declared nuisance set.

A positive distance establishes local separation, not an easy decision. Small distance or near-collinear signatures amplify noise and delay support resolution. Supplementary Appendix D proves the proposition, treats unknown sparse corruption, and shows that adding a conditionally independent PMU cannot reduce the profiled margin when the nuisance contract is fixed.

The infinitesimal version makes the first-order boundary explicit. Let $g_\star$ be the event-response tangent, J_c the tangent of the closest competing nuisance manifold, and $W^\mathsf{T}W = R_H^{-1}$. With

$$\gamma_1^2 = \| (I - P_c)\, W g_\star \|_2^2, \qquad P_c = (WJ_c)(WJ_c)^\dagger, \tag{12}$$

the corresponding local version of (11) satisfies $D_{\star c}(a) = a^2\gamma_1^2/2 + o(a^2)$. Thus $\gamma_1 > 0$ gives first-order separation after nuisance fitting; $\gamma_1 = 0$ only makes the first-order test inconclusive. Relative curvature can then produce an a^4 squared-distance law. The full statement and proof are in Supplementary Appendix D. The present experiments do not yet test the convergence of $2D_{\star c}(a)/a^2$ to γ_1^2.

For multiple events, let G collect the first-order event signatures, N the nuisance tangent, let P_A denote the orthogonal projector onto $\mathrm{range}(A)$, and define $\overline{G} = (I - P_{WN})WG$ the event dictionary after nuisance removal. All first-order event information that survives worst-case nuisance fitting lies in $\mathcal{D} = \mathrm{range}(\overline{G})$. Hence $d_{\mathcal{D}} = \mathrm{rank}(\overline{G})$ is the intrinsic linear diagnostic dimension, not the number of retained PMU channels or reconstructed states. Supplementary Appendix D proves that coordinates on $\mathcal{D}$ preserve every profiled Gaussian likelihood ratio when candidates share the same nuisance range. For true support S, severity direction $\boldsymbol{\alpha}_S$, and competing support C, the support-to-support margin is

$$\gamma_{S\to C}^2(\boldsymbol{\alpha}_S) = \| \left(I - P_{\overline{G}_C}\right) \overline{G}_S \boldsymbol{\alpha}_S \|_2^2. \tag{13}$$

Equation (13) is exact for signed amplitudes or an interior parameter point. At a physical boundary, let $T_{\mathcal{A}_C}(\mathbf{0})$ be the tangent cone of the competitor's admissible amplitude domain. The first-order physical margin is instead

$$\gamma_{\mathrm{phys}}^2 = \min_{u,\, \boldsymbol{\beta}_C \in T_{\mathcal{A}_C}(\mathbf{0})} \| WG_S\boldsymbol{\alpha}_S - WNu - WG_C\boldsymbol{\beta}_C \|_2^2. \tag{14}$$

Because the cone is contained in its span, $\gamma_{\mathrm{phys}} \geq \gamma_{S\to C}$; zero relaxed margin does not prove ambiguity for a one-sided event. Shared sources cancel exactly only when the matched amplitude is feasible and the relevant constraint is inactive. Competitor enrichment still cannot increase the margin, and a nested superset still has zero margin when its added amplitude admits zero. Geometry can therefore rank non-nested explanations, but cardinality among nested supports requires a model-selection penalty or integrated evidence. Supplementary Appendix D proves the subspace and cone statements.

For persistent load events, subtract the common no-event trajectory and write the T-sample response as

$$Y_T = \mathcal{H}_T a + \boldsymbol{\varepsilon}_T. \tag{15}$$

Let $\mathcal{A}_{2K}$ contain vectors with at most $2K$ nonzero entries.

Proposition 3 (Finite-horizon sparse event injectivity): Every event with at most K active sources is uniquely determined in the noise-free model if and only if

$$\ker \mathcal{H}_T \cap \mathcal{A}_{2K} = \{\mathbf{0}\}. \tag{16}$$

Stable amplitude estimation also requires a well-conditioned support Gramian.

The proposition distinguishes uniqueness from finite-information difficulty. With prefix-consistent causal whitening, adding samples cannot reduce the profiled separation between two fixed supports. Separation grows when their steady signatures differ and can saturate when only a finite transient distinguishes them. The experiments therefore ask two questions in order: does the posterior identify how many sources are active, and does the remaining separation identify which sources? Full proofs and the temporal cases appear in Supplementary Appendix D.

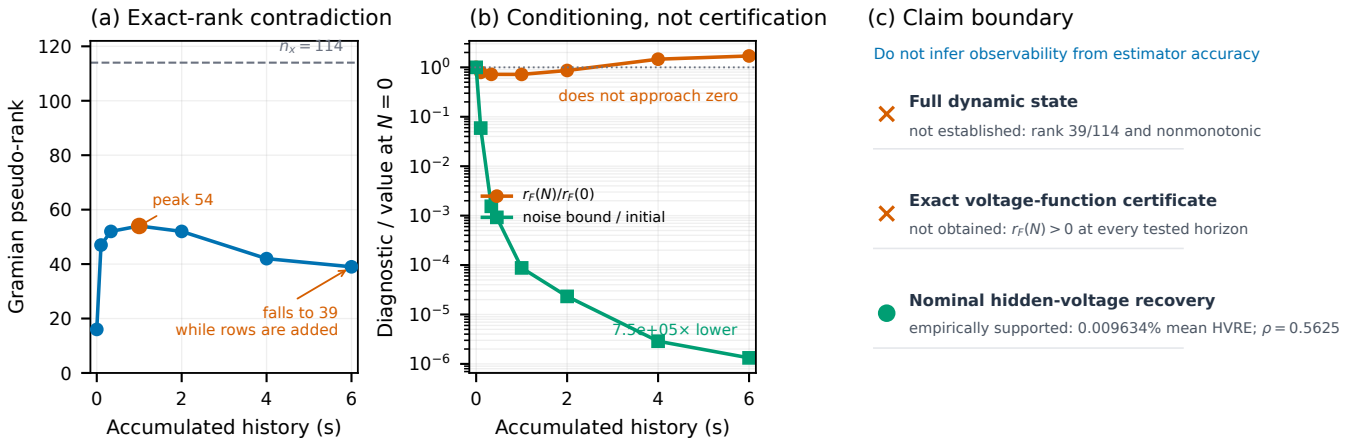


FIGURE 3. Structural audit and claim boundary. Adding history lowers the target-noise bound, but the relative-threshold pseudo-rank is nonmonotone and the target residual remains nonzero. The audit therefore motivates a conditioning analysis; it does not prove full-state or exact hidden-voltage observability.

The separation also has an operational meaning under the corresponding simple equal-covariance Gaussian quotient experiment. If d is the nuisance-profiled Mahalanobis distance, then

$$P_e^\star = \Phi(-d/2), \qquad D_{\mathrm{KL}} = d^2/2. \tag{17}$$

Thus a local order-k pair satisfying $d_T^2(a) = T\mathcal{I}_k|a|^{2k} + o(T|a|^{2k})$, with $\mathcal{I}_k > 0$, has the resolution scale

$$|a|_{\min}(\alpha, T) \sim \left[\frac{2\Phi^{-1}(1-\alpha)}{\sqrt{T\mathcal{I}_k}}\right]^{1/k}. \tag{18}$$

First-order and curvature-only separation therefore improve as $T^{-1/2}$ and $T^{-1/4}$, respectively. Equation (17) is exact for the stated quotient experiment, not for an amplitude-marginalized nonlinear support mixture.

Two limits prevent accuracy from being interpreted without a severity and candidate-set contract. If an added active amplitude may approach zero, its law approaches the nested lower-cardinality law, so uniform exact-support recovery is impossible without a minimum-severity condition or abstention. For M equiprobable supports, Fano's inequality further gives

$$P_e \geq \left[1 - \frac{\overline{D}_{\mathrm{KL}} + \log 2}{\log M}\right]_+, \qquad \overline{D}_{\mathrm{KL}} = \frac{1}{M^2}\sum_{i,j} D_{\mathrm{KL}}(P_i \| P_j). \tag{19}$$

which separates algorithmic failure from insufficient physical information [56], [57]. Supplementary Appendix D states the assumptions, proofs, and the resulting necessary horizon bound.

V. EVALUATED COMPONENTS AND TWO ESTIMATOR CONTRACTS

Figure 1 separates three objects that must not share one name. The blue B0–B3 state reconstructor is an independently evaluated component. The green path is the implemented reference event estimator, $\mathcal{E}_{\mathrm{B}}$, a batch physical-manifold Bayesian bank. The dashed path is the second estimator proposal, $\mathcal{E}_{\mathrm{J}}$, a joint Hybrid-DAE state–event Bayesian model that remains unexecuted. They share the plant and PMU operator, but no fitted posterior is shared in the present results.

A. CAUSAL STATE RECONSTRUCTION

The state experiment changes one information source at a time. It validates a state-reconstruction component, not the joint estimator $\mathcal{E}_{\mathrm{J}}$. B0 returns the equilibrium voltage. B1 conditions that equilibrium on the current PMU frame. B2 propagates the state causally and then applies the PMU correction. B3 is a retrospective fixed-lag smoother used only to test whether later measurements add useful information. This sequence isolates where each improvement comes from.

For the local model in (7), B2 uses the standard Kalman recursion [58]:

$$\begin{aligned} \widehat{x}_{k|k-1} &= A_d \widehat{x}_{k-1|k-1}, \\ P_{k|k-1} &= A_d P_{k-1|k-1} A_d^{\mathsf{T}} + Q, \\ K_k &= P_{k|k-1} C^{\mathsf{T}} (C P_{k|k-1} C^{\mathsf{T}} + R)^{-1}, \\ \widehat{x}_{k|k} &= \widehat{x}_{k|k-1} + K_k \boldsymbol{\nu}_k. \end{aligned} \tag{20}$$

The hidden-voltage estimate is $z_{\mathcal{U},0} + L_{\mathcal{U}} \widehat{x}_{k|k}$. Only measurements through k enter B2. B3 follows the Gaussian backward recursion [59] and is explicitly retrospective.

Two checks separate point accuracy from uncertainty. First, a temperature map fitted on independent calibration trajectories rescales the two rectangular voltage components without changing the state estimate. Second, an observed-innovation statistic tests whether the nominal model has become inadequate. Neither operation repairs the physical model. Their complete equations and negative discrepancy ablations appear in Supplementary Appendix E.

When the operating point is stale, the physical-recentering estimator rebuilds a feasible equilibrium instead of shifting

measured voltages by an unconstrained offset. It solves

$$(\widehat{V},\widehat{d}) = \arg\min_{V,d} \|e_y(V)\|_2^2 + \gamma^2\|g_{\mathrm{AC}}(V,d)\|_2^2 + \lambda q_d^{-2}\|d\|_2^2, \tag{21}$$

where e_y uses only measured voltage and terminal-current phasors, g_{AC} enforces AC balance and bus conventions, and d collects admissible load and generation changes. Development data fix the weights and continuation rule before TEST. Hidden voltages and the true nuisance vector never enter the solve.

B. IMPLEMENTED BATCH PHYSICAL-MANIFOLD BAYES $\mathcal{E}_{\mathrm{B}}$

The reference estimator targets $p_{\mathrm{B}}(S, a_S \mid r, \boldsymbol{\eta}_0, \tau_0, c_{load})$: operating point, onset, and family are fixed, while support and amplitudes are inferred. The physical question is what the measured PMUs would have recorded if a candidate load bus had changed. For each of the 16 buses, centered nonlinear simulations form the unit response

$$D_g = \frac{y_g(+\epsilon) - y_g(-\epsilon)}{2\epsilon}, \qquad \epsilon = 0.005. \tag{22}$$

Symmetric simulations also estimate source self-curvature Q_i and pairwise interaction Q_{ij}. The mean response for support S is

$$\boldsymbol{\mu}_S(a) = \sum_{i\in S}(a_i D_i + a_i^2 Q_i) + \sum_{\substack{i,j\in S \\ i<j}} a_i a_j Q_{ij}. \tag{23}$$

The mixed term matters physically: two simultaneous interventions need not equal the sum of two isolated responses.

After subtracting the normal mean, the residual follows $r = \boldsymbol{\mu}_S(a) + \boldsymbol{\varepsilon}$ under the frozen Gaussian covariance. Each support receives normalized evidence by integrating over its active amplitudes,

$$Z_S = \int \mathcal{N}(r; \boldsymbol{\mu}_S(a), \boldsymbol{\Sigma}) \prod_{i\in S} p(a_i)\, da_S,$$
$$p(S \mid r) \propto Z_S \frac{p(|S|)}{\binom{16}{|S|}}. \tag{24}$$

Gauss–Hermite quadrature evaluates every support; independent Gauss–Kronrod calculations check representative two-dimensional integrals. Priors, covariance estimation, quadrature order, and calibration checks are documented in Supplementary Appendix E. Thus $\mathcal{E}_{\mathrm{B}}$ marginalizes one- and two-dimensional amplitudes rather than substituting a MAP amplitude and local Laplace approximation. The quadrature is accurate for the frozen second-order physical surrogate; it is not an exact posterior for the full nonlinear DAE.

The amplitude integral is structurally necessary, not decorative. A superset can reproduce any smaller support by driving its additional amplitudes to zero, so its pure profiled geometric distance can also be zero. Integrated evidence accounts for the volume of amplitude values that fit the data, while $p(|S|)$ and the within-cardinality prior state the sparsity contract. Cardinality is therefore a Bayesian model-selection question; it cannot be recovered from best-fit distance alone.

The evaluated load bank uses a proper signed Gaussian amplitude prior because load changes may increase or decrease. A one-sided fault, trip, or outage instead requires a proper truncated slab and boundary-aware evidence. In the local single-amplitude Gaussian model, Supplementary Appendix E shows that a half-normal slab multiplies the signed Bayes factor by an exact directional factor $2\Phi(z)$. This extension is part of the proposed $\mathcal{E}_{\mathrm{J}}$ design; it is not an evaluated multi-family result.

The posterior is read in a fixed order. First, $p(K \mid r)$ asks whether no source, one source, or two sources are active. Conditional support ranks then ask which buses are involved. Amplitude intervals answer how large the changes are. This order is essential under limited observability because cardinality can be well supported while several source pairs remain plausible.

The prospective experiment evaluates all 137 supports at a known onset and fixed PMU set. It conditions on the nominal initial state and assumes no corrupted channels. Thus it evaluates the sparse load specialization of Proposition 3, not the full state-and-integrity nuisance model in (10).

C. PROPOSED JOINT HYBRID-DAE BAYES $\mathcal{E}_{\mathrm{J}}$

The second proposal targets a candidate-conditioned joint posterior over state, operating nuisance, event family, onset, support, amplitude, and integrity, with normalized evidence across physical hypotheses. A sequential implementation of $\mathcal{E}_{\mathrm{B}}$ is not a third estimator: under the same prefix and initial AR(1) density, it must reproduce the batch posterior numerically.

More generally, define the restriction $\mathcal{R}_0$ to fix $\boldsymbol{\eta} = \boldsymbol{\eta}_0$, $\tau = \tau_0$, and $c = c_{load}$ and to reuse the same 137-support surrogate, covariance, and priors. The mandatory reduction test is

$$p_{\mathrm{J}}(S, a_S \mid Y, \mathcal{R}_0) = p_{\mathrm{B}}(S, a_S \mid r, \boldsymbol{\eta}_0, \tau_0, c_{load}). \tag{25}$$

Failure of (25) is either an implementation error or an undeclared change of statistical model.

Iterated smoothing, Gauss–Newton steps, moving-horizon optimization, or generalized EM would be numerical realizations of that object, not different scientific estimators. Changing the solver cannot change the declared priors, candidate set, causal data boundary, or evidence normalization. Where the active amplitude remains one- or two-dimensional, the design should retain quadrature marginalization and use optimization or Laplace only for the remaining state and nuisance coordinates. For each candidate, virtual electrical outputs are deterministic functionals of the same state trajectory and network mode. Model averaging must therefore propagate both within-candidate state uncertainty and between-candidate event ambiguity; it does not require a separate regressor for every output channel.

A learned component could then correct candidate residuals or reduce the search, but it must not consume source identity as a shortcut. It must also preserve the directions that distinguish critical physical candidates. After whitening

and nuisance removal, a rank-q discrepancy basis Q_D with variance τ^2 retains the worst-case fraction

$$\rho = 1 - \frac{\tau^2}{1+\tau^2}\sigma^2_{\max}(Q_U^{\mathsf{T}} Q_D) \tag{26}$$

of the information in a protected discriminant subspace Q_U. Supplementary Appendix D derives the corresponding finite-variance rule using the full Gaussian likelihood; maximizing residual accommodation alone would omit its log-determinant penalty. The required common-contract ablation remains learned-only, physics-only, physics plus learned discrepancy, and diagnosability-aware inference with abstention. None of those integrations is presented as an executed result here; Supplementary Appendix E records the full generative objective so the proposed extension is unambiguous.

D. EXPLORATORY NUMERICAL ROUTE TO MULTI-FAMILY INFERENCE

The joint target $\mathcal{E}_{\mathrm{J}}$ in the preceding subsection is a statistical specification, not an executed end-to-end result. A separate single-event pilot implements parts of its physical search and continuous state–severity fit. It uses 32 rectangular voltage/current channels from the same eight PMU buses at 30 Hz; frequency and ROCOF from the historical competition files are not inputs to this likelihood. Its candidate catalog contains H_0, bus faults, load changes, generator-reference changes, and branch outages. The generator and line scenarios in this pilot have recovery semantics that must be matched by each candidate's temporal operator. Supplementary Appendix H records the exact contract and the unverified evidence boundary.

For a fixed family, source, and onset, let X collect the differential trajectory after algebraic projection and let ϑ_H contain the continuous severity coordinates. Set $r^{dyn}_{H,k} = x_{k+1} - F_{H,k}(x_k;\vartheta_H)$ and $r^{pmu}_{H,k} = y_k - h_{H,k}(x_k;\vartheta_H)$. The pilot minimizes the conditional Gaussian MAP objective

$$\begin{aligned} J_H(X,\vartheta_H) = {} & \tfrac{1}{2}\|x_0 - \bar{x}_0\|^2_{P_0^{-1}} + \tfrac{1}{2}\sum_k \|r^{\mathrm{dyn}}_{H,k}\|^2_{Q^{-1}} \\ & + \tfrac{1}{2}\sum_k \|r^{\mathrm{pmu}}_{H,k}\|^2_{R^{-1}} - \log p(\vartheta_H \mid H). \end{aligned} \tag{27}$$

The DAE transition $F_{H,k}$ changes with the physical operator: a fault adds a temporary shunt, an outage removes a branch primitive, a load event changes withdrawal, and a generator event perturbs a governor reference. Algebraic sensitivity alone is blind to that last pathway; its PMU effect appears after differential propagation. The objective is conditional on the candidate and the declared Gaussian approximation. A minimum of J_H is not a normalized model probability.

The numerical update uses structured Gauss–Newton smoothing, whose relationship to the iterated extended Kalman smoother is established [60], [61]. Partition its local normal equations into trajectory and event-parameter blocks. If H_{XX} is nonsingular, eliminating the trajectory step gives

$$\begin{aligned} S_H &= H_{\vartheta\vartheta} - H_{\vartheta X} H_{XX}^{-1} H_{X\vartheta}, \\ b_H &= g_\vartheta - H_{\vartheta X} H_{XX}^{-1} g_X, \\ \Delta\vartheta_H &= -S_H^{-1} b_H, \quad \Delta X = -H_{XX}^{-1}(g_X + H_{X\vartheta}\Delta\vartheta_H). \end{aligned} \tag{28}$$

One Armijo factor is applied to both increments and tested against the nonlinear objective. Equation (28) is algebraically the full joint step; calling b_H a gradient of the *profiled* objective additionally requires the trajectory to be conditionally stationary. The appendix derives both statements and explains the numerical guards.

The pilot separates proposal recall from final selection. An H_0 innovation detector identifies an alarm window; the onset is refined separately, and a common pre-event state is fitted without event-contaminated samples. Smoothly nested load and generator candidates use the nuisance-profiled screening strength $T_H = b_H^{\mathsf{T}} S_H^{-1} b_H$ when $S_H \succ 0$. Fault and outage candidates instead use short-horizon physical signatures and nuisance-projected matched scores, because a topology switch or finite fault pulse is not an ordinary infinitesimal perturbation of H_0. Scores are ranked *within* families; a frozen quota retains representatives of every family because their cheap scores are not cross-family calibrated. A sequence of increasing nonlinear budgets then refines the shortlist. These steps are search heuristics, and neither a high Stage-A rank nor survival at ten iterations is a converged localization decision.

For valid final fits, a Laplace approximation would add prior, covariance, curvature, and measure normalizers to the optimized log density. Boundary severities require a cone correction, and failed or singular fits cannot be assigned a comparable regular Laplace score. The present pilot has not closed that final comparison. It also has not passed the $\mathcal{R}_0$ reduction test in (25); therefore its four-family experiments are kept separate from the prospective 137-support load result.

E. A SEPARATE CAUSAL LOAD–LINE COMPONENT

The batch estimator $\mathcal{E}_{\mathrm{B}}$ above uses a known onset and integrates load amplitudes with its original quadrature. We retain that implementation. A separate executable component, denoted R11, tests unknown-onset load and line interventions in a continuous stream. It is not a numerical realization of the complete joint estimator $\mathcal{E}_{\mathrm{J}}$: its source catalog, startup conditions, and uncertainty approximation are narrower.

R11 uses the frozen Julia/PowerDynamics IEEE-39 candidate and eight PMUs. Its 47 input coordinates comprise eight voltage magnitudes, eight voltage angles, eight terminal-current magnitudes, seven terminal-current angles, eight absolute frequencies, and eight ROCOFs. The current angle at BUS39 is excluded because its terminal interpretation is only partially validated. The BUS5 terminal retains the provisional historical 5–6 hypothesis. Currents are oriented terminal measurements, not nodal injections; bus frequency is not rotor speed. No waveform harmonics or THD are inferred from these phasor reports.

The physical bank propagates conditional state histories with local DAE transition and observation operators. Load interventions enter affine operators with an uncertain continuous amplitude; line interventions change the operator. Retrospective screening compares source/onset candidates over a 10-s history. A proposed suffix is replayed from its stored no-event state estimate; its screening score is not multiplied into the posterior as another observation. Conditional Gaussian state updates and truncated amplitude integration therefore have a different contract from the batch quadrature in $\mathcal{E}_{\mathrm{B}}$. Physical restoration changes the intervention operator while retaining the dynamic state. Restored histories continue to predict electrical recovery, which may outlast the intervention. The startup prefix ends before 10 s and the inherited monitoring warmup ends at 12 s; neither is supplied with event times.

Five fixed network hypotheses represent nominal, global $\pm 2\%$, and selected-branch $\pm 3\%$ paired R/X perturbations. They are constant over a trajectory. For each hypothesis θ_ℓ, a 300-report startup prefix estimates an observation intercept,

$$\widehat{b}_\ell = \bar{y}_{0:299} - y_\ell^{\mathrm{ref}}, \qquad \widehat{b}_{\ell,k} = \widehat{b}_\ell \quad (k \geq 300). \tag{29}$$

Angular differences are wrapped consistently. Frequency and ROCOF already use empirical startup centers and receive no additional intercept. This is an observational correction, not an identified network impedance or an AC-feasible state recentering. Its uncertainty is treated by substitution rather than fully marginalized. A stationary intervention-free prefix is required.

Let $\mathcal{F}_{k-1}$ contain the reports available before report k. The outer weights update once using each conditional model's one-step predictive density:

$$w_{\ell,k} = \frac{w_{\ell,k-1}\, p_\ell(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_\ell)}{\sum_j w_{j,k-1}\, p_j(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_j)}. \tag{30}$$

Startup reports are not scored again as independent evidence. The density includes the retained normal, active, restored, and newly born histories before current-report screening and state updates. Adaptive proposals, pruning, and shared covariance approximations make this an approximate prequential mixture; neither optimality nor calibrated event probabilities are claimed.

The source/family array has a computational role: it marginalizes the conditional history masses $M_{\ell,c,k}$ over static parameter hypotheses. Here c indexes a bus-load or line cell, rather than a dense array of physically admissible events at all bus pairs. The implementation checks the exact decomposition

$$\begin{aligned} M_{c,k} &= \sum_\ell w_{\ell,k} M_{\ell,c,k}, \\ \Delta M_{c,k} &= \sum_\ell w_{\ell,k}(M_{\ell,c,k} - M_{\ell,c,k-1}) \\ &\quad + \sum_\ell (w_{\ell,k} - w_{\ell,k-1}) M_{\ell,c,k-1}. \end{aligned} \tag{31}$$

The terms distinguish conditional event-evidence evolution from reweighting network hypotheses. They do not constitute a new tensor-factorization theorem. The nominal ablation retains one physical hypothesis and its own episode publication path. Alarm, diagnosis, restoration, and electrical recovery remain distinct outputs; publication does not reset the conditional observers.

F. PUBLISHED PHYSICAL COMPARATOR AND ITS ELIGIBILITY

We implement the persistent single-line quickest-change detector of Chen *et al.* [32]. Its reference-reduced incremental network matrix H_i defines $M_i = H_i^{-1}$. With Gaussian injection increments of covariance Σ and PMU selection matrix C, the observed covariance is $CM_i\Sigma M_i^\top C^\top$. Selection occurs after inversion. A parallel positive-part log-likelihood recursion stops when its maximum exceeds $\log(L\beta)$, where L is the number of admissible outages and β is expressed in observation intervals. The line at the maximum is the stopping diagnosis.

The F01 adapter uses flat-voltage DC branch slopes $1/(Xa)$ with transformer tap a, known nominal topology, and a scalar injection covariance estimated only from the startup prefix. Nonoverlapping PMU angle pairs give 15 increments/s from the common 30-report/s input; circular angle differences prevent wrapping jumps. Prefix mean increments compensate the observation-frame drift. These are declared adapter assumptions, not a reproduction of the article's original test system or an average-run-length guarantee on colored instrumental noise.

The primary eligible comparison is one persistent line outage or a normal record. The comparator has no load-family or restoration output. We do not manufacture reclosing decisions, restart it at true event times, or count unsupported load labels as a failure of its published scope. Its monitoring starts after its 10-s prefix; normal exposure is reported for each method's actual monitoring interval. Mixed continuous streams provide only an explicitly out-of-contract stress test.

VI. EXPERIMENTAL QUESTIONS AND FROZEN EVIDENCE

The experiments follow the logic of the paper rather than the order in which the software was developed. Each research question has its own independent unit, frozen selection boundary, and claim limit (Table 3). The complete manifests, seeds, acceptance gates, simulator versions, and secondary campaigns appear in Supplementary Appendix F.

A. RQ1 AND RQ2 STATE-RECONSTRUCTION CONTRACTS

The primary plant is the package-defined IEEE 39-bus PowerDynamics benchmark [62]. Before scoring estimator error, we verified the power-flow translation, terminal-current orientation, equilibrium, local modes, nonlinear trajectories, and first-order response. These checks establish numerical consistency with the implemented benchmark; they do not make the tutorial model an authoritative external grid.

TABLE 3. Four research questions and their evidence contracts. Results from different rows are not pooled.

RQ	Scientific question	Primary evidence and independent unit	Claim boundary
RQ1	What hidden voltage information do eight PMUs recover?	100 frozen nonlinear PowerDynamics TEST trajectories; complete trajectory is the unit	Matched model and fixed PMU set
RQ2	What fails under physical mismatch, and can a constrained recentering repair it?	980 independently rebuilt plants for diagnosis; 60 held-out equilibria for physical recentering	Point recovery under tested shifts, not sequential adaptation
RQ3	When can simultaneous load sources be resolved?	3,488 fresh nonlinear paths and 10,564 complete-record decisions over all 137 hypotheses	One event family, known onset, fixed topology and sensing
RQ4	Does supervised localization survive physical-source holdout?	303 decisions from 101 held-out event parents in ANDES	Historical baseline; true event family supplied

RQ1 compares B0–B3 on the same 3-s trajectories. Model parameters and covariance choices are fixed before TEST, and the 31 requested buses exclude all eight observed buses. Separate calibration trajectories fit uncertainty corrections. RQ2 then rebuilds the plant under seven physical mismatch families while keeping the nominal estimator frozen. A distinct held-out operating-point campaign tests the constrained recentering in (21). Hidden voltages are revealed only after each solve, and rejected solves remain in the planned denominator.

B. RQ3 SIMULTANEOUS LOAD-SOURCE CONTRACT

Development data select the normal covariance and likelihood before the prospective support test. The final bank freezes D_i, Q_i, Q_{ij}, the priors, onset, and quadrature, then generates new nonlinear trajectories for H_0, every single source, and every source pair. Event identifiers do not overlap the historical development manifests. Repeated noise realizations produce separate decisions but are not counted as new physical-system trajectories.

The primary outcome is exact support, evaluated only after cardinality. A retrospective weak-regime audit asks which geometric distance explains errors without refitting the likelihood. A separate 120-frame study checks physical-manifold fidelity, information accumulation, and runtime. Because that study uses only one operating point and 32 validation paths, it is a diagnostic rather than a prospective recovery result.

C. RQ4 HISTORICAL BASELINE AND EVENT SEMANTICS

The ANDES campaign [63] preserves the earlier physics-guided hierarchy as a baseline. Its compact representation uses local voltage, angle, frequency, and availability information; its expanded representation adds multiscale dynamics, adaptive residuals, and simultaneous PMU contrasts. A deterministic availability gate handles observed dropout. The ensemble is not DAE constrained, and its leaf votes are not treated as calibrated probabilities.

The ordinary split places new replicas of known sources in TEST. RQ4 imposes the harder contract: it removes each physical source from every source-labeled fitting and tuning step and supplies the true event family at evaluation. This isolates physical-source generalization from detection and family-classification errors.

Figure 4 grounds the event ontology before any score is reported. A fault produces a short voltage sag, a line outage redistributes current, and injection changes evolve more slowly. Missing or corrupted measurements require different evidence. The clock trace in panel (e) makes the distinction explicit: time advances while packets from PMU 29 are absent.

All intervals and tests use the experiment's independent unit: a complete trajectory, event parent, or rebuilt plant. Generated tables and figures read only the hashed frozen snapshot. Sample-level PMU frames are never promoted to independent replicates.

D. STATUS OF THE SINGLE-EVENT PILOT

The multi-family pilot in Section V-D has a different provenance tier. Two author-supplied technical dossiers dated September 24, 2026 summarize its campaign packages, but the underlying solver logs, frozen manifests, and final-survivor archive are not present in this repository. Its four development cases, four blind mini-holdout cases, and four quiet controls are independent *cases*, not a population performance sample; no interval or false-alarms-per-hour estimate is computed from them. We report this line as an exploratory engineering audit in Section VIII-D and Supplementary Appendix H. None of its numerical values is pooled with RQ1–RQ4 or promoted to the abstract's primary evidence.

VII. STATE EVIDENCE FROM RECOVERY TO MODEL FAILURE

A. RQ1 MEASUREMENT CONDITIONING RECOVERS THE TARGET

The estimator sequence reveals where the nominal accuracy comes from. Conditioning the equilibrium on the current PMU frame (B1) reduces hidden-voltage error by more than one order of magnitude. Adding causal propagation (B2) improves most trajectories, but by a much smaller amount. Thus the network-constrained measurement map supplies the main gain; temporal propagation refines it.

Figure 5(a) shows the scale change directly, and panel (c) uses paired trajectory differences rather than overlapping marginal intervals. The comparison is deliberately internal: B0, B1, and B2 share the same local physical model. It does not claim superiority over every learned or nonlinear estimator.

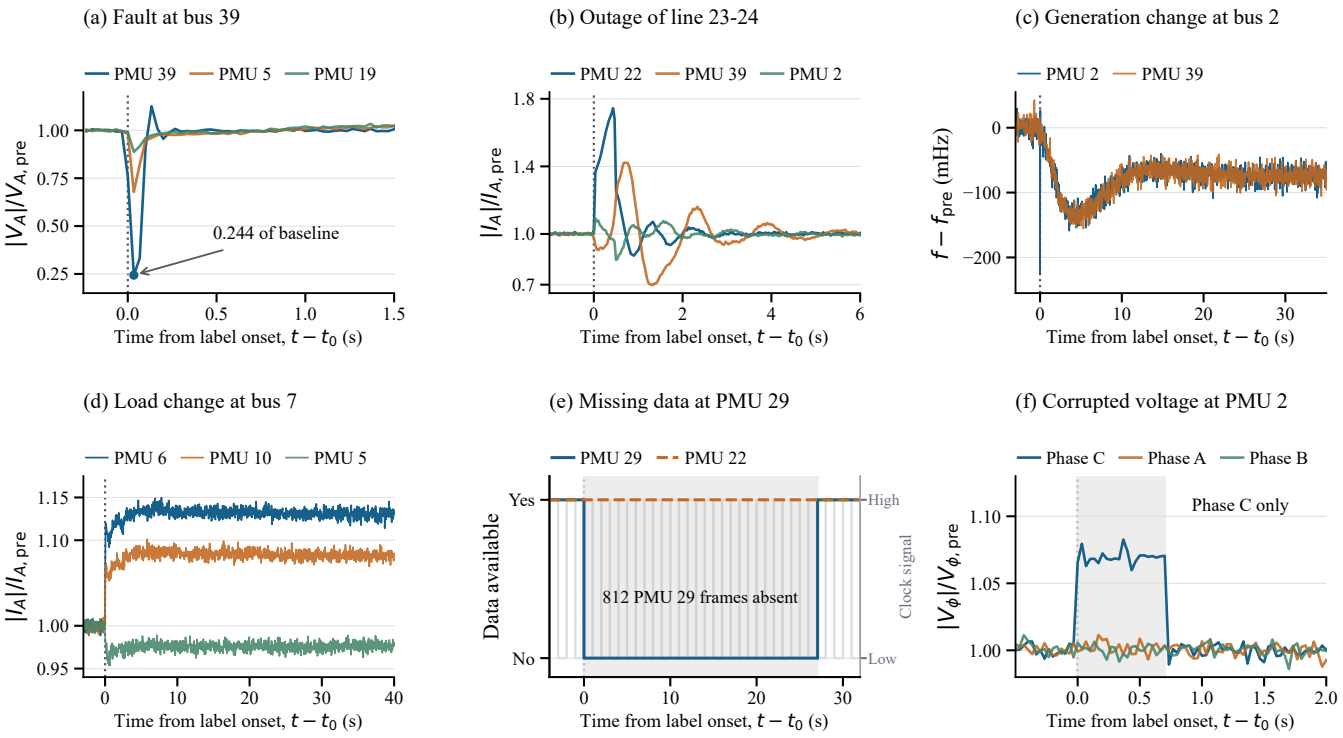


FIGURE 4. **Representative measured signatures from the RAW0001 reference simulation. Physical events alter voltage, current, or frequency through the network; integrity events alter availability or channel consistency. In panel (e), the gray reference clock advances while PMU 29 packets are absent. Curves are unsmoothed, and each panel uses the time scale needed to show its physical response.**

TABLE 4. **Nominal reconstruction on 100 frozen nonlinear TEST trajectories. HVRE is the trajectory-and-bus mean relative complex-voltage error over the 31 unobserved buses; intervals bootstrap independent trajectories.**

Method	Complex RMSE	$\|V\|$ RMSE	Angle RMSE (°)	Mean HVRE (%)	95% HVRE interval (%)
B0 equilibrium	3.486e-03	2.496e-05	0.19497	0.340200	[0.293231, 0.387428]
B1 snapshot LMMSE	1.529e-04	5.091e-05	0.00821	0.009904	[0.008342, 0.011514]
B2 causal Kalman	1.338e-04	4.235e-05	0.00722	0.009634	[0.008006, 0.011302]

The structural residual in Fig. 5(b) correlates with bus-level B2 error, but the relationship is incomplete. The buses share trajectories and network coupling, and the residual comes from the threshold-sensitive projector discussed in Section IV. It is therefore a difficulty indicator, not an observability certificate.

Two negative checks sharpen this result. The fixed-lag smoother does not improve the nonlinear TEST trajectories, even though a correctly specified linear-Gaussian smoother cannot increase expected squared error. Innovation autocorrelation identifies the missing premise: the local model treats deterministic governor and load excitation as white process noise. Marginal temperature scaling repairs coverage under nominal operation but does not whiten those innovations. More data cannot compensate for the wrong temporal model.

B. RQ2 PHYSICAL SHIFT EXPOSES THE MODEL BOUNDARY

The mismatch campaign asks whether nominal accuracy survives a changed plant. It does not fail uniformly. Network-parameter changes and operating-point shifts dominate the state error, while the tested machine, governor, AVR, and load-model perturbations remain close to the nominal level over the short excitation [Fig. 6(a)]. Those near-null responses show that the experiment did not expose the corresponding parameters; they are not evidence of general robustness.

This pattern suggests a mechanism. A stale equilibrium moves the center of both the measured and hidden voltage fields. The offset-only ablation estimates that movement from measured buses but ignores AC balance, and it closes almost none of the oracle gap. The constrained recentering in (21) instead rebuilds a physically admissible center. Across all three held-out operating shifts, it removes nearly all of the point-error gap [Fig. 7(a)–(b)].

The recovery does not identify every operating-point coordinate. The nuisance information is rank deficient, yet the requested voltage field changes little along much of that nullspace. This is the empirical counterpart of Proposition 1: a target can be recoverable even when the full latent vector is not. The remaining limitation is uncertainty. Marginal inter-

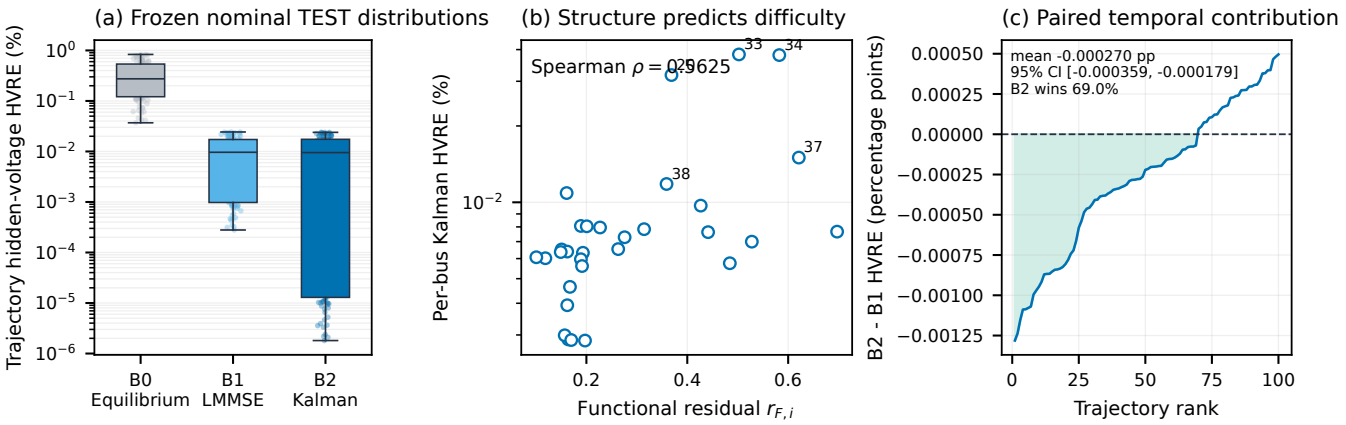


FIGURE 5. **Where nominal state-reconstruction accuracy comes from. (a) PMU conditioning produces the dominant reduction in trajectory-level hidden-voltage error. (b) The preregistered structural residual predicts some, but not all, difficult buses. (c) Paired B2-minus-B1 differences isolate the smaller contribution of causal propagation.**

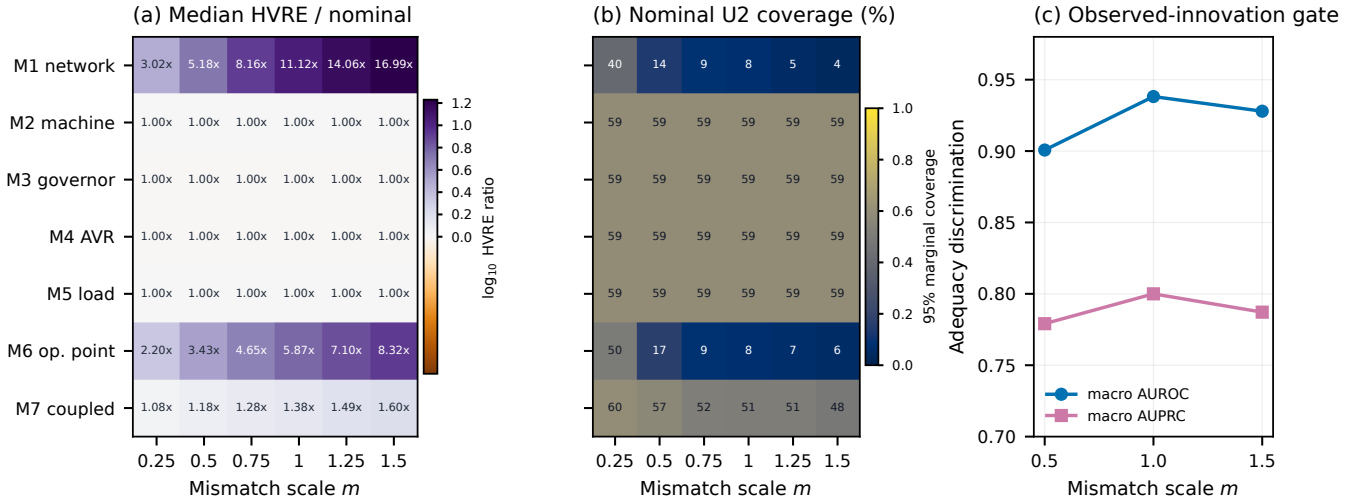


FIGURE 6. **How the frozen state estimator fails under physical mismatch. (a) Network and operating-point changes dominate the hidden-voltage error. (b) A covariance correction fitted under nominal operation loses coverage after physical shift. (c) Observed innovations often flag unacceptable hidden error but do not identify its physical cause. Each family-scale cell contains 20 rebuilt plants.**

TABLE 5. **Corrected physical static recentering on 60 held-out M6 operating-point cases. Gap-closure intervals bootstrap the 20 independent TEST cases at each scale. Rank and nullity refer to the 43-dimensional nuisance basis; runtime is per solve.**

m	n	Frozen HVRE (%)	Corrected HVRE (%)	Gap closure [95% CI] (%)	Rank	Nullity	Median NEES	Runtime (ms)
0.5	20	0.19885	0.00525	97.4 [94.9, 98.8]	25/43	18	111.6	57.6
1.0	20	0.38834	0.00749	98.1 [96.7, 98.8]	25/43	18	183.3	61.1
1.5	20	0.57795	0.00974	98.3 [97.4, 98.8]	25/43	18	294.2	69.2

vals cover, but the joint NEES-like statistic is severely underdispersed, and the independent constrained solver converges on only part of its small diagnostic subset. The evidence supports point reconstruction for the tested operating shifts, not joint parameter recovery or sequential adaptation.

VIII. EVENT EVIDENCE FROM DETECTION TO SOURCE AMBIGUITY

A. RQ3 CARDINALITY RESOLVES BEFORE EXACT SUPPORT

The prospective test opens the full 137-hypothesis bank only after the physical dictionary, covariance, priors, onset, and numerical integration are frozen. Every source pair competes in one normalized posterior. The resulting pattern is monotone with event strength, but the two decisions do not improve at the same rate (Table 6). Finite pairs are recovered exactly, whereas weak–weak Top-1 remains 24.4%. Weak sources are therefore the decisive regime, not a small perturbation around an otherwise uniform accuracy claim.

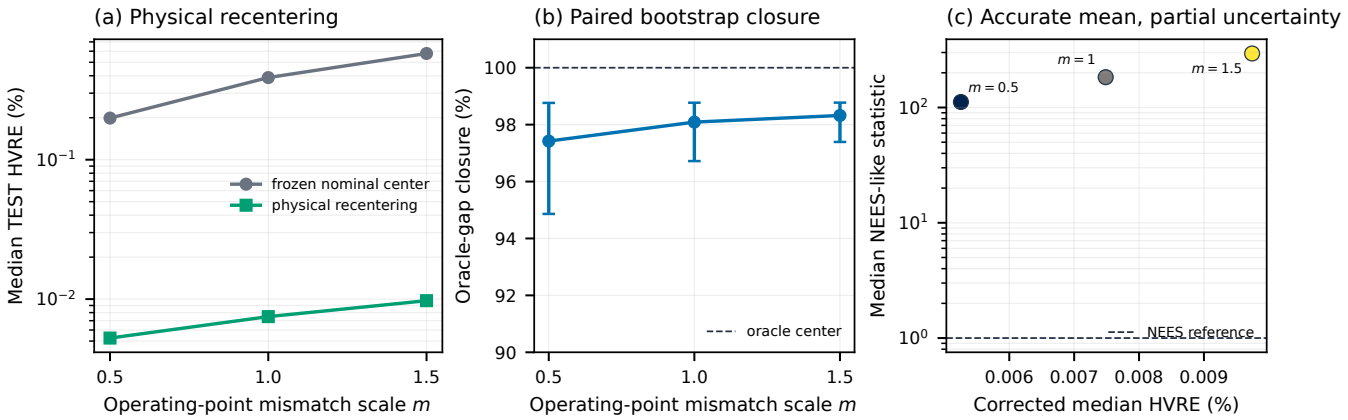


FIGURE 7. **Physical recentering separates point recovery from uncertainty. (a) Enforcing the AC map sharply reduces hidden-voltage error under operating-point shift. (b) The paired oracle-gap closure is consistent across held-out cases. (c) The joint NEES-like diagnostic remains too large even when marginal intervals cover, so the point estimate is supported but its joint covariance is not.**

TABLE 6. **Fresh prospective 137-hypothesis confirmation. The bank contains H_0, 16 single-source hypotheses, and all 120 two-source supports. Repeated noise realizations are decisions, not independent physical trajectories. Top-k is exact support recovery and is undefined for H_0 and single-source rows in this exported support table.**

Regime	Physical trajectories	Decisions	Cardinality (%)	Top-1 (%)	Top-3 (%)	Mean $p(K=2)$
No event	—	100	100.0	—	—	0.000
Single source	128	384	93.8	—	—	0.051
Weak–weak pair	1920	5760	37.4	24.4	59.8	0.427
Weak–strong pair	480	1440	77.6	66.2	95.8	0.808
Moderate pair	480	1440	99.7	98.8	100.0	0.996
Finite pair	480	1440	100.0	100.0	100.0	1.000

Figure 8 follows the inference failure in the same order as the method. Panel (a) establishes the strength-dependent recovery boundary. Panel (b) then shows that evidence for $K = 2$ grows while exact pair recovery remains nearly flat. The model has detected multiplicity before it has localized the support. Reporting only event detection would hide this failure.

Event presence is easier than source recovery. The frozen detector separates most event and no-event records, but its no-event exposure is too short for an operational false-alarm claim. Per-bus inclusion probabilities also rank useful candidates, yet their missed-source rate remains too high to repair weak exact-support decisions. Supplementary Appendix G reports these secondary metrics and denominators.

B. PROFILED GEOMETRY EXPLAINS THE WEAK CASES

The retrospective audit asks why the weak bank fails. A support-only score compares candidate signatures while holding the rest of the explanation fixed. The profiled distance instead allows the competing support to choose its best amplitude and thereby measures the actual overlap between response manifolds. The latter tracks exact weak-case recovery far more closely [Fig. 8(c)].

This result changes the interpretation of the weak regime. Every tested direction is first-order resolvable, so a curvature-dominated a^{-4} delay law is not the observed mechanism. The errors arise because several finite-horizon candidate responses lie close after amplitude profiling. Detectability and source identity therefore require different geometric quantities.

The first-order theory separates two reasons for that proximity. A source may have little response outside the nuisance span, or it may remain visible but nearly collinear with another source after nuisance removal. Shared-source competitors make the second mechanism especially severe because their common component cancels before the exclusive source is compared. The frozen audit uses the full nonlinear profiled distance and supports this explanation retrospectively; it does not yet validate the analytic small-amplitude coefficient in (13).

The 120-frame diagnostic adds a second piece of the explanation. Extending the window increases the separation proxy by more than eightfold, and no candidate pair is exactly degenerate in the frozen equilibrium model. Yet the hardest signatures remain almost collinear [Fig. 8(e)–(f)]. More time adds information without guaranteeing a well-conditioned decision.

The physical-manifold check also shows why the nonlinear interaction in (23) cannot be dropped. Adding only single-source curvature does not improve the simultaneous response; adding Q_{ij} reduces the median whitened error by roughly two orders of magnitude [Fig. 8(d)]. This check covers 32 nominal-operating-point paths. It validates the representation locally, not population recovery at 120 frames or robustness

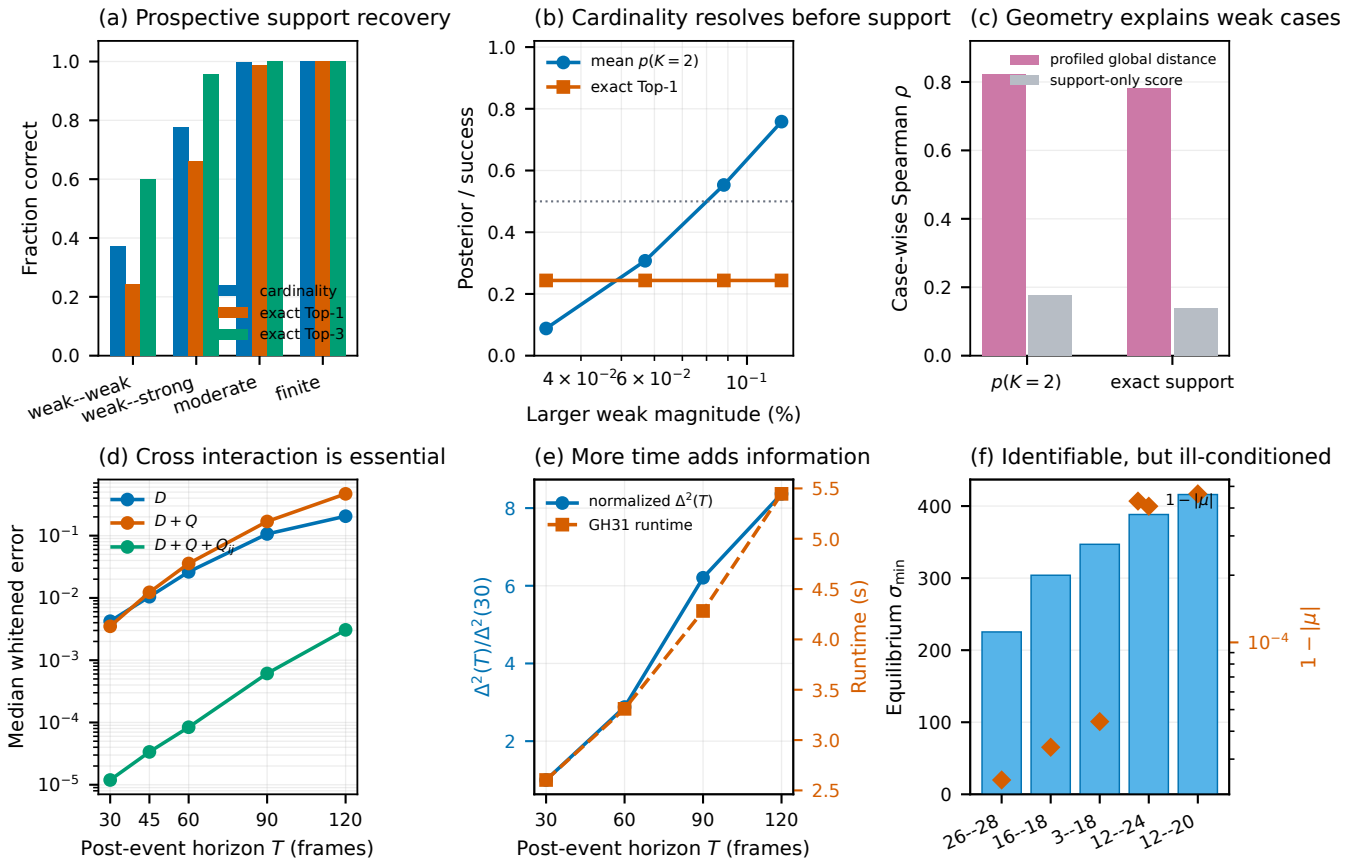


FIGURE 8. Why simultaneous-source recovery fails in the weak regime. Prospective results first separate cardinality from exact support (a–b). Retrospective geometry then links weak-case errors to profiled response distance (c). The longer-horizon diagnostic shows that the pairwise interaction is needed (d), information accumulates with time (e), and distinct equilibrium signatures can remain nearly collinear (f). Panels (c)–(f) explain the result but do not replace a prospective recovery test.

to operating shift.

C. RQ4 SOURCE LABELS DO NOT GENERALIZE TO NEW ASSETS

The supervised hierarchy provides a useful contrast. On the ordinary split it detects events and ranks known sources well, confirming that multiscale dynamics, adaptive residuals, and PMU contrasts carry event information. That result answers interpolation at represented assets.

The source-holdout result answers a different question. With the physical asset removed from all source-labeled fitting and the true event family supplied, only 3 of 303 decisions are exact. The failure remains after detection and family confusion have been removed. The location heads and asset descriptors therefore memorize source-specific associations that do not transfer to an unseen physical source.

The physical load bank avoids that shortcut by generating a response for every candidate support. Its success is narrower in another direction: it covers only load interventions under fixed topology, known onset, matched operation, and synthetic noise. Together, the two experiments motivate the next model rather than a claim of general superiority: learned residuals should correct a physical candidate likelihood and must be tested on event parents and source assets excluded from training.

D. EXPLORATORY MULTI-FAMILY PILOT: SEARCH SURVIVES, SELECTION DOES NOT

The single-event pilot tests a different bottleneck from the load-support bank: can the true physical explanation remain in a tractable four-family shortlist when onset and severity are fitted? The author-supplied campaign summaries report that a development-selected, frozen screening and racing policy retained the true candidate in all four blind mini-holdout cases (one fault, load change, generator change, and line outage). The generator case reached the six-survivor cutoff with no spare place. Four quiet windows provided only a pilot check of proposal scores; they cannot establish operational false-alarm control. The solver diagnostics also report full-block/Schur agreement on 30 synthetic systems and fault-resistance estimates of 0.02006–0.02133 from four initializations around a synthetic truth of 0.021. These checks establish numerical behavior of components on the reported cases, not a validated source-localization rate.

The same dossiers report a negative strict-final outcome: none of the four truth candidates won the final comparison.

TABLE 7. Executed continuous-stream development evidence (H04). Timely means correct family/source emitted before physical restoration; extras are unmatched alarms. OUT denotes out-of-bank R/X; F1 is report-level detection F1. These are paired realizations of 18 events from three event parents, not 54 independent events.

Method	Timely	Extras	OUT timely	OUT F1
R11 mixture	53/54	4	17/18	0.818
R11 nominal	54/54	7	18/18	0.828
Admission ablation	52/54	0	16/18	0.854
ExtraTrees reference	27/54	3	9/18	0.619
ExtraTrees short	27/54	3	9/18	0.626

That 0/4 is currently a terminal-reported result without its matching final-survivor archive in the repository. Its mechanisms must remain distinct. Fault and line truth/competitor fits did not all reach strict convergence, so their ranking is numerically unresolved. In the load and generator cases, converged neighboring same-family candidates reportedly received better final scores. That pattern motivates an oracle sequence for event-operator parity, pre-event state, onset and duration, severity, noise, and counterfactual PMU coverage. It does not yet prove an intrinsic identifiability limit. The pilot thus supports continued work on the joint numerical route but supplies no positive multi-family end-to-end claim. Supplementary Appendix H gives the case ledger, reported diagnostics, and the exact artifact checks needed to turn this audit into primary evidence.

E. UNKNOWN-ONSET CONTINUOUS EVIDENCE: SCOPE AND FAILURE MODES

The continuous component is evaluated separately from the prospective batch bank. Two completed development campaigns, H03 and H04, each contain three 80-s event parents with six successive load/line interventions and one normal parent. The same schedules are paired across nominal, in-bank R/X, and out-of-bank mixed R/X conditions. Thus each campaign contains 18 unique interventions and 54 paired event realizations. All use F01 operation, one instrumental-noise level, eight fixed PMUs, and the known nine-load/nine-line catalog. Neither campaign demonstrates unseen-source generalization. They are now development evidence and cannot be reused as a new blind confirmation.

In H03, R11 obtains 54/54 correct family/source decisions before physical restoration, with 6 unmatched alarms. Table 7 reports the subsequent H04 outcomes. R11 retains 53/54 timely decisions, whereas its nominal ablation retains 54/54. The mixture has fewer extras (4 versus 7), but loses one timely attribution. Parameter uncertainty therefore yields a trade-off, not uniform dominance. Under out-of-bank R/X, timely BUS31 attribution is 2/3 for the mixture and 3/3 for the nominal ablation. Eventual correct attribution would conceal that delay.

An additional admission ablation retains 52/54 timely decisions with 0 extras. Its zero-extra result is insufficient for promotion: it rejects a true BUS16 load intervention whose retrospective onset lacks the required early witness. The change reduces public alarms while leaving the inherited physical states and hypothesis masses unchanged. We retain it as negative evidence rather than silently replacing R11. Both R11 references are now frozen; no bus-specific correction follows these observations.

The two frozen ExtraTrees adaptations obtain 27/54 timely decisions here. This is an identical-stream causal comparison under the restricted load/line contract, not a recomputation of the hackathon leaderboard or an assertion about all event classes. Report-level F1 and event-level timely attribution measure different targets and are shown separately. The short normal exposure and three event parents cannot support a precise operational false-alarm rate or generalization confidence interval.

The published physical comparator passes equation-level tests for the reduced network, partial-observation marginalization, Gaussian log-likelihood, suffix-max recursion, angle wrapping, and stopping localization. On independently generated three-bus sequences under its own Gaussian law, 299/300 post-change trials stop after onset at the correct line; 1 trial stops before the change. No alarm occurs over 100,000 normal increments, with every normal sequence right-censored at its observation horizon. This checks an implementation under matched assumptions, not the numerical results of the original article. On three previously opened F01 normal records, the adapter alarms in all three; colored noise and frame dynamics violate its assumed law. We preserve that mismatch result and make no general superiority claim from it.

The confirmatory C01 configuration is sealed before its new draw. It specifies independent physical parents, longer normal records, paired R/X conditions, two instrumental-noise scales, persistent-line eligibility for the published comparator, and paired parent-cluster intervals. Both R11 references and both ExtraTrees artifacts remain fixed. Solver rejects, inference failures, late decisions, and right-censoring remain in the ledger. C01 has not been executed; it contributes no performance number to this manuscript. Source-disjoint inference, changing PMUs, faults, generation interventions, and the complete joint estimator remain outside this confirmatory contract.

IX. DISCUSSION AND LIMITATIONS

The experiments support one central distinction: recovering a requested state function is not the same as identifying the intervention that produced it. The state estimator can reconstruct hidden voltages because many unobservable state directions have little effect on that target. Source inference is stricter. It must separate every admissible candidate after amplitude and other nuisance directions are allowed to adjust.

The state results make this distinction concrete. Instantaneous PMU conditioning supplies most of the nominal improvement, while causal propagation adds a smaller correction. When the equilibrium becomes stale, an unconstrained offset does not solve the problem; rebuilding an AC-feasible center does. The resulting point estimate is accurate even though the operating-point coordinates remain non-unique.

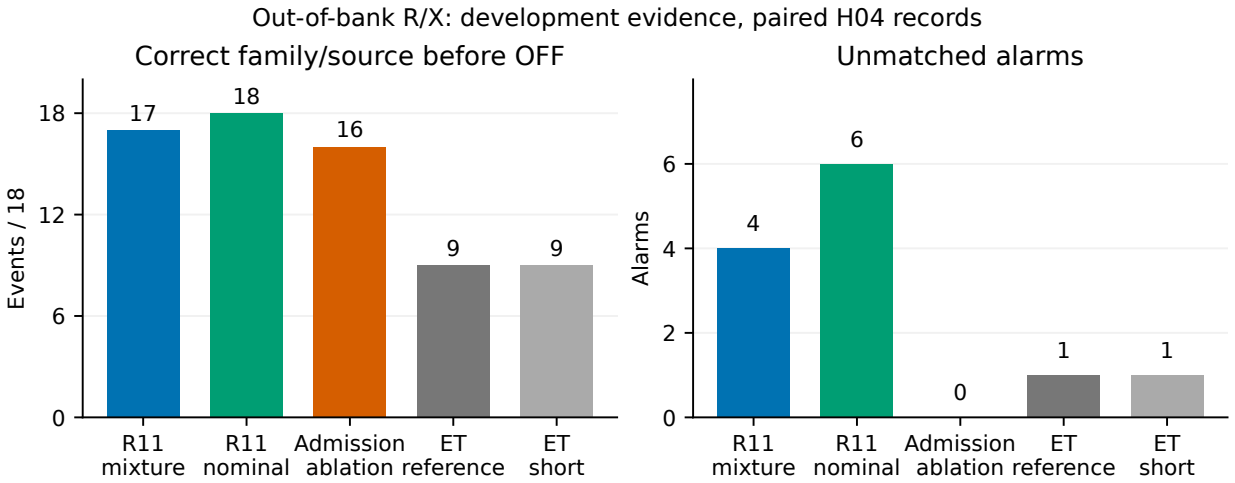


FIGURE 9. Failure trade-off under out-of-bank network error in H04. An additional causal-admission ablation eliminates unmatched alarms but rejects a real BUS16 intervention; the nominal R11 retains every timely event but has more extra alarms. These paired development results do not establish a universally superior variant.

Its joint covariance is still unreliable. Target recovery and parameter recovery must therefore be evaluated separately.

An archived ANDES calibration checkpoint makes the parameter distinction more concrete. Individual screening retained 10 of 18 directions, but their joint condition number was 722.9. Rank-revealing selection reduced the set to eight directions with condition number 7.83, and a final 6.15σ displacement gate left seven in the released local posterior. The rejected directions did not share one explanation: six were information-poor, two violated the local curvature gate, two added little first-order direction, and one left the local model domain. Supplementary Appendix D traces these counts to the frozen gate ledger. Because the checkpoint lacks the required higher derivatives, it does not establish which rejected directions are curvature-resolved or structurally redundant.

The event results expose a parallel hierarchy. Weak simultaneous events often provide enough evidence for two active sources before they identify the pair. Profiled manifold distance explains this behavior because it measures what remains after a competing support chooses its best amplitude. Longer observation increases that distance, but nearly collinear signatures still amplify noise. A positive structural margin is therefore a uniqueness statement, not a reliability guarantee. This information–delay trade-off has a measurement-level analogue: our previous benchmark showed disturbance-dependent trade-offs among transient error, settling behavior, observation characteristics, and computational cost across dynamic frequency estimators [41]. In the present problem, the corresponding trade-off occurs at the inference level: longer PMU histories can increase physical source separation while delaying the localization decision.

The supervised baseline adds a warning about learning. Engineered temporal and spatial features work when the source appears in training. Under complete source holdout, the same representation loses source identity even when the event family is supplied. A learned correction should therefore operate on candidate residuals, not on asset labels alone, and its value must be measured against the physics-only likelihood on the same event parents.

The evidence remains bounded in four ways. First, the state and event paths are evaluated separately rather than as one fitted posterior. Second, the Bayesian support result covers one event family, fixed topology, known onset, fixed PMUs, and matched operation. Third, the 120-frame study is diagnostic and lacks independent operating-point validation. Fourth, the normal exposure is too short to establish false alarms per hour, and no field or cross-simulator validation is claimed.

The separate continuous component supplies unknown-onset load/line evidence under a fixed catalog, but leaves those broader gaps open. Its frozen observation intercept corrects the measured residual center; it does not contradict the hidden-state study's finding that an arbitrary offset cannot replace AC-feasible recentering. Likewise, its finite network mixture does not identify true R/X or establish calibrated probabilities. The nominal component's better timely attribution and worse alarm burden make both references necessary.

The next decision is a finite confirmatory comparison, not another bus-specific repair. The sealed C01 protocol compares those fixed references and the frozen ExtraTrees adaptations on continuous records, and compares the published physical detector only on persistent-line and normal records. It retains failed and censored cases. That campaign can validate the stated restricted causal contract; it cannot establish source-disjoint transfer, a learned-discrepancy benefit, changing-PMU robustness, or the complete joint estimator. Those claims remain blocked rather than inferred from improved development scores.

X. CONCLUSION

Sparse PMUs impose different information limits on state reconstruction and event-source inference. The requested

hidden-voltage function can be estimated accurately even when the full local state is not observable. Event support requires a stronger condition: candidate responses must remain separated after amplitude and nuisance directions are profiled.

The experiments follow that logic. Eight PMUs reconstruct the 31 unmeasured bus voltages with 0.009634% mean relative error under the matched model, and physical recentering repairs most of the tested operating-point shift. In the prospective load bank, finite two-source supports are recovered exactly, whereas weak-pair Top-1 remains 24.4%. The gap is explained by finite-horizon response overlap and near-collinearity, not by event absence alone.

The present evidence supports matched-model state reconstruction and simultaneous load-source inference as separate components. A separate causal load/line component obtains 53/54 timely decisions with 4 unmatched alarms in development streams, while its nominal ablation retains one more timely event at a higher alarm burden. That trade-off and the negative admission ablation are part of the result. R11 and its nominal ablation are frozen for a preregistered comparison with ExtraTrees and an implemented published line-outage detector. No confirmatory outcome is claimed before execution. The complete multi-family state/event/integrity posterior, source holdout, changing sensing, and field transfer remain open.

SUPPLEMENTARY MATERIAL

The companion supplement contains the paper-by-paper literature matrix, electrical and PMU conventions, proofs, complete estimator equations, frozen experimental contracts, and secondary result tables. These details preserve reproducibility without interrupting the main scientific argument.

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Digital Object Identifier

Supplementary Material for Physics-Informed State Reconstruction and Event-Source Inference From Sparse PMUs Under Limited Observability

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ABSTRACT This supplement records the detailed literature matrix, electrical conventions, mathematical proofs, Fisher-information analysis, Bayesian estimator construction, frozen experimental contracts, and secondary results supporting the main article. It distinguishes the independently evaluated state component and implemented batch event estimator $\mathcal{E}_{\mathrm{B}}$ from the proposed joint Hybrid-DAE estimator $\mathcal{E}_{\mathrm{J}}$. The latter remains unexecuted. This supplement does not add empirical claims or change any test denominator. Every empirical table and figure is generated from the same hashed evidence snapshot used by the main manuscript; separate deterministic tables audit stated algebraic identities.

INDEX TERMS Bayesian inference, Fisher information, functional observability, phasor measurement units, reproducibility, state estimation, event-source inference.

APPENDIX A DETAILED LITERATURE CAPABILITY MATRIX

Table 1 expands the family-level synthesis in the main paper. "Held source" means that the physical asset is absent from source-labeled training and tuning. The entries describe published tasks and assumptions; they are not a cross-paper performance ranking.

The direct competitors solve different contracts. State and unknown-input estimators define the latent reconstruction problem but not the event taxonomy. Learned localizers rank represented sources but rarely remove the physical asset from every source-labeled fitting operation. Model-based outage methods provide the closest likelihood precedent, while functional observability and placement studies provide the information tests. These distinctions determine the baselines and prevent cross-paper accuracy comparisons with incompatible targets.

APPENDIX B ELECTRICAL MODEL AND EVENT OPERATORS

Within a smooth network mode, the DAE is locally index one after fixing the reference angle. Differentiating the algebraic constraint gives

$$\dot{z} = -g_z^{-1}(g_x f + g_u \dot{u} + g_\theta \dot{\theta}). \tag{1}$$

This reduction is not applied across a topology switch or fault insertion, where algebraic variables may jump. Differential states remain continuous unless a simulator-specific reset is declared.

With $Y_{\mathrm{bus}} = G + \mathrm{j}B$ and $V_i = |V_i|e^{\mathrm{j}\vartheta_i}$, the network injections are

$$P_i = |V_i| \sum_j |V_j|(G_{ij}\cos\vartheta_{ij} + B_{ij}\sin\vartheta_{ij}), \tag{2}$$

$$Q_i = |V_i| \sum_j |V_j|(G_{ij}\sin\vartheta_{ij} - B_{ij}\cos\vartheta_{ij}). \tag{3}$$

For branch $e = (i,j)$ with complex tap t_e, series admittance y_e, and terminal shunts $y_e^{f,sh}$ and $y_e^{t,sh}$,

$$\begin{bmatrix} I_{e,f} \\ I_{e,t} \end{bmatrix} = \begin{bmatrix} (y_e + y_e^{f,sh})/|t_e|^2 & -y_e/t_e^* \\ -y_e/t_e & y_e + y_e^{t,sh} \end{bmatrix} \begin{bmatrix} V_i \\ V_j \end{bmatrix}. \tag{4}$$

The implemented PMU operator selects the actual terminal and orientation. Replacing this current with complete bus-injection current changes the likelihood.

Positive-sequence phasors define the balanced model. Negative- and zero-sequence components remain integrity

TABLE 1. Detailed capability comparison for the closest methodological works.

Work	State target	Event/source target	Physics mechanism	Sparse sensing	Held source	Integrity treatment
Katanic *et al.* [1]	Differential and algebraic DAE states	Unknown injectors	Incomplete nonlinear DAE and least squares	Yes	N/A	No
Nugroho *et al.* [2]	Differential and algebraic DAE states	Unknown input and input-sensor failure	Outlier-resistant DAE observer	PMU voltage/current	N/A	Failure tolerance
Qu *et al.* [3]	Machine state	Unknown field voltage and torque	Event-triggered nonlinear state/input estimator	Event-triggered PMUs	N/A	Covariance bounds
Ngo *et al.* [4]	Power-system states	None	Branch-current physics plus graph learning	Yes	N/A	No
Yue *et al.* [5]	Distribution-system state	Line outage	Grid-model-informed graph learning	Yes	Not stated	Outlier-resistant SE
Moshtagh *et al.* [6]	Transmission-system state	Topology variation	Topology-aware graph learning	PMU-unobservable systems	Not a source task	PMU failure and bad data
Pandey *et al.* [7]	None	Detection, class, and source	Topology/equipment information	PMU network	Not stated	No
Yildiz and Abur [8]	None	Fault detection and location	OLS/CNN and PMU placement	Yes	Not stated	No
Li *et al.* [9]	None	Multi-family disturbance source	Topology/type-constrained STGCN loss	Yes	Not stated	No
Kummerow *et al.* [10]	None	Known/unknown event type and location	Siamese distance and learned rejection	Limited/full sets	Open-set rejection	Rejection boundary
Hu and Cheng [11]	None	Open-set event detection/classification	Generative window evidence and fusion	Streaming PMUs	Unknown-event category	No source posterior
Rimkus *et al.* [12]	Missing PMU trajectories	Downstream fault localization	Graph reconstruction before diagnosis	Yes	Imputer only	Corruption via imputation
Zhai *et al.* [13]	Full dynamical state	Line-attack location	Learned temporal dynamics	Yes	Across benchmarks	Attack target only
Main article	Hidden voltage field	All load supports with $K \leq 2$; historical eight-label ontology	DAE hypotheses and quadratic physical likelihood	Eight PMU buses	Physical bank has no source-label head	Physical/integrity variables separate

or unbalance evidence. Ideal continuous-frequency channels satisfy

$$f_i^{\mathrm{phys}} = f_0 + \frac{1}{2\pi}\frac{d\angle V_i}{dt}, \qquad \mathrm{ROCOF}_i^{\mathrm{phys}} = \frac{1}{2\pi}\frac{d^2\angle V_i}{dt^2}, \tag{5}$$

but an operational likelihood must reproduce the causal filtering of the PMU backend.

The preceding equations define the ideal physical quantities f_i^{phys} and $\mathrm{ROCOF}_i^{\mathrm{phys}}$ implied by the network trajectory. Reported quantities f_i^{meas} and $\mathrm{ROCOF}_i^{\mathrm{meas}}$, however, are outputs of a dynamic frequency-estimation process and can exhibit disturbance- and estimator-dependent transient error and latency. A prior benchmark by the authors compared 16 dynamic frequency estimators spanning six practical groups over 34 disturbance scenarios and quantified these trade-offs under ramps, phase jumps, harmonics, ringdown, and composite IBR disturbances [14]. We therefore distinguish

$$\begin{bmatrix} f_i^{\mathrm{meas}} \\ \mathrm{ROCOF}_i^{\mathrm{meas}} \end{bmatrix} = \begin{bmatrix} f_i^{\mathrm{phys}} \\ \mathrm{ROCOF}_i^{\mathrm{phys}} \end{bmatrix} + \boldsymbol{\varepsilon}_{fr,i}, \qquad \mathrm{Cov}(\boldsymbol{\varepsilon}_{fr,i}) = \boldsymbol{\Omega}_{fr}. \tag{6}$$

Because phasor, frequency, and ROCOF outputs may be estimated from overlapping underlying waveform or phasor information, their errors need not be conditionally independent. An augmented likelihood therefore permits cross-covariance

$$\boldsymbol{\Omega}_{\mathrm{aug}} = \begin{bmatrix} \boldsymbol{\Omega}_{VI} & \boldsymbol{\Omega}_{VI,fr} \\ \boldsymbol{\Omega}_{fr,VI} & \boldsymbol{\Omega}_{fr} \end{bmatrix}; \tag{7}$$

neither the magnitude of $\boldsymbol{\Omega}_{VI,fr}$ nor a block-diagonal approximation is assumed without calibration. Frequency and ROCOF are retained as candidate channels for an augmented sparse-PMU likelihood. Their incremental source-discrimination value, covariance structure, and effect on localization delay are not assumed here and require the dedicated VI, $VI + f$, $VI + \mathrm{ROCOF}$, and $VI + f + \mathrm{ROCOF}$ ablation.

APPENDIX C METRICS, CAUSALITY, AND EVALUATION UNITS

For trajectory r, hidden-voltage relative error is

$$\mathrm{HVRE}^{(r)} = \frac{1}{T_r|\mathcal{U}|}\sum_{k=1}^{T_r}\sum_{i\in\mathcal{U}}\frac{|\widehat{V}_{i,k} - V_{i,k}|}{\max(|V_{i,k}|, \epsilon)}. \tag{8}$$

It is a trajectory-and-bus mean reported as a percentage; it is not the standard per-frame synchrophasor TVE conformity statistic. Detection, event-family macro-F1, exact physical Top-1/Top-3, cardinality, candidate-set coverage, and integrity-source Top-1 retain their own populations. A joint outcome is correct only when every applicable component is correct.

A causal estimate at k uses $y_{0:k}$. A fixed-lag smoother, complete-window score, or centered temporal feature is retrospective. For source generalization, the physical asset is absent from source-labeled training, feature fitting, threshold selection, and calibration. Intervals and significance tests use the complete trajectory, event parent, or rebuilt plant as the in-

TABLE 2. Physical intervention operators represented by the common DAE formulation.

Family	DAE action at source ℓ	Candidate support	Severity coordinates	State continuity
Bus fault	Add a shunt primitive for the clearing interval	admissible buses	complex admittance or resistance/reactance	x continuous; z may jump
Line outage	Remove branch tap, series, charging, and shunt primitives	in-service branches	status and switching time	x continuous; z may jump
Generation change	Modify scheduled injection or a declared governor reference	generator buses	ΔP_G, ΔQ_G, duration	continuous input-driven response
Load change	Modify the active/reactive withdrawal model	load buses	ΔP_L, ΔQ_L, ZIP convention	continuous input-driven response

dependent unit; samples within a 30-Hz stream are correlated observations, not replicates.

APPENDIX D PROOFS AND ADDITIONAL DIAGNOSABILITY RESULTS

This appendix develops the mathematical chain used by the paper. It begins with the sampled hybrid event map, separates state-functional recovery from event-source separation, derives the relevant Fisher information after physical nuisance is removed, and then states the finite-horizon and temporal limits. The results are local unless a statement explicitly says otherwise.

A. REGULARITY AND STATISTICAL CONTRACT

Let $u = \mathrm{col}(x, z)$ collect differential and algebraic coordinates and consider the descriptor model

$$E\dot{u} = F(u, p), \qquad y = h(u, p). \tag{9}$$

The analysis uses four assumptions.

1) The reference angle has been fixed and the algebraic Jacobian is nonsingular on the local mode under study, so a regular index-one chart exists between discrete events.
2) The scheduled event time is fixed. In a neighborhood of zero amplitude, the complete callback–solver–sampling map is twice continuously differentiable.
3) Every covariance used as a metric is positive definite on the retained measurement subspace. Candidate-dependent covariances retain their log determinants in likelihood comparisons.
4) The support set is finite and all Bayesian priors are proper. Exact signed-amplitude equivalences use a neighborhood of zero within each support subspace; constrained physical amplitudes instead use their actual difference set.

These assumptions distinguish the local theory from general descriptor left invertibility, state-triggered saltation, and global nonlinear identifiability.

B. SIMULATOR-CONSISTENT EVENT SENSITIVITIES

For a fixed-time, impulse-free event, the differential state is continuous, but the algebraic state must satisfy the post-event constraint. This physical right limit must be distinguished from the values stored inside a simulator callback and from the first sampled frame. The following result makes that distinction explicit.

Theorem 1 (Local consistency-jump expansion): Let $x_0 = x(\tau^-)$ and suppose $g_+(x_0, z_0, 0) = \mathbf{0}$. If g_+ is three times continuously differentiable and g_z is nonsingular at $(x_0, z_0, 0)$, there is a unique local consistent branch

$$z_+(a) = z_0 + az_1 + \tfrac{1}{2}a^2 z_2 + O(a^3), \tag{10}$$

where all derivatives below are evaluated at the nominal right-limit mode and

$$z_1 = -g_z^{-1} g_a, \tag{11}$$

$$z_2 = -g_z^{-1}\left(g_{zz}[z_1, z_1] + 2g_{za}z_1 + g_{aa}\right). \tag{12}$$

The corresponding differential state satisfies $x(\tau^+) = x(\tau^-)$, and its first event-amplitude sensitivity starts with

$$\dot{x}_a(\tau^+) = f_a - f_z g_z^{-1} g_a \equiv B_{r,a}. \tag{13}$$

Its second-order initial forcing is

$$\dot{x}_{aa}(\tau^+) \equiv B_{r,aa} = f_{zz}[z_1, z_1] + 2f_{za}z_1 + f_{aa} - f_z g_z^{-1}\left(g_{zz}[z_1, z_1] + 2g_{za}z_1 + g_{aa}\right). \tag{14}$$

a: Proof.

The implicit-function theorem applied to $g_+(x_0, z_+(a), a) = \mathbf{0}$ gives a unique smooth local branch. Differentiating once gives $g_z z_1 + g_a = \mathbf{0}$, which yields (11). Differentiating a second time gives $g_z z_2 + g_{zz}[z_1, z_1] + 2g_{za}z_1 + g_{aa} = \mathbf{0}$, which yields (12). An impulse-free fixed-time switch does not reset the energy-storage coordinates, so the differential state is continuous. Differentiating $\dot{x} = f_+(x, z_+(a), a)$ at the right limit, with $x_a(\tau^+) = \mathbf{0}$, gives (13). A second differentiation, using $x_{aa}(\tau^+) = \mathbf{0}$ and (12), gives (14). □

The theorem also defines an implementation-independent consistency diagnostic. If a callback changes the parameter but temporarily retains z_0, its algebraic residual is

$$r_{\mathrm{cb}}(a) = g_+(x_0, z_0, a) = ag_a + \tfrac{1}{2}a^2 g_{aa} + O(a^3). \tag{15}$$

Substituting the zeroth-, first-, and second-order approximations to (10) produces constraint residuals of orders $O(a)$, $O(a^2)$, and $O(a^3)$, respectively. Central finite differences of an independently solved algebraic branch can therefore test both derivatives and the expected convergence orders without relying on undocumented callback storage.

For a differentiable PMU map $y = h(x, z, a)$, the direct right-limit signature is

$$y_a(\tau^+) = h_a - h_z g_z^{-1} g_a, \qquad \Delta y_{\mathcal{P}} = aM_{\mathcal{P}} y_a(\tau^+) + O(a^2). \tag{16}$$

This direct algebraic visibility is not the entire event signature. Even when (16) is nearly zero at the retained PMUs, (13) can generate a distinguishable dynamic response over later frames. Instantaneous invisibility therefore does not imply finite-horizon nonidentifiability.

The frozen artifacts begin at the first retained 30-Hz frame and do not record the callback microstate or an independently solved algebraic right limit. Accordingly, the paper treats the right-limit equations above as the physical contract, not as an observed fact about the current ANDES callback implementation. The relevant differentiable object for the existing response dictionary is the sampled map

$$\mathbf{\Psi}_{T_s}^{\mathrm{sim}} : (u^-, p(a)) \mapsto u(\tau + T_s; a), \qquad T_s = 1/30\ \mathrm{s}. \tag{17}$$

For event coordinates a_i, define

$$L_i = \left.\frac{\partial \mathbf{\Psi}_{T_s}^{\mathrm{sim}}}{\partial a_i}\right|_{a=\mathbf{0}}, \qquad L_{ij}^{(2)} = \left.\frac{\partial^2 \mathbf{\Psi}_{T_s}^{\mathrm{sim}}}{\partial a_i \partial a_j}\right|_{a=\mathbf{0}}. \tag{18}$$

For $t \geq \tau + T_s$, the first and second sensitivities obey

$$E\dot{s}_i = F_u s_i + F_p p_i, \tag{19}$$
$$E\dot{s}_{ij} = F_u s_{ij} + r_{ij}, \tag{20}$$

where

$$r_{ij} = F_{uu}[s_i, s_j] + F_{up}[s_i, p_j] + F_{pu}[p_i, s_j] + F_{pp}[p_i, p_j] + F_p p_{ij}. \tag{21}$$

The initial conditions are $s_i(\tau + T_s) = L_i$ and $s_{ij}(\tau + T_s) = L_{ij}^{(2)}$. For the affine event parameterization, $p_{ij} = \mathbf{0}$.

Theorem 2 (Propagation of event-map mismatch): Suppose two second-order models have the same nominal trajectory, first-order sensitivities, parameter derivatives, and forcing in (20), but different second-order event-map initial conditions. Their difference $e_{ij} = s_{ij}^A - s_{ij}^B$ satisfies

$$E\dot{e}_{ij} = F_u e_{ij}, \qquad e_{ij}(t) = \mathbf{\Phi}_E(t, t_1) e_{ij}(t_1) \tag{22}$$

on the consistent solution subspace, where $t_1 = \tau + T_s$.

b: Proof.

Subtract (20) for the two models. Every forcing term cancels by assumption, leaving the homogeneous descriptor variational equation. Applying its transition operator gives (22). □

When the retained PMU operator is locally affine, $y = H_y u + d$, the sampled response coefficients are

$$D_i = H_y s_i, \quad Q_i = \tfrac{1}{2} H_y s_{ii}, \quad Q_{ij} = H_y s_{ij}. \tag{23}$$

Thus all quadratic curvature comes from the physical solution and event map, not from PMU post-processing. This is why the dictionary must differentiate the same event semantics used to generate the trajectories.

C. EXACT FUNCTIONAL RECONSTRUCTION

Consider the finite-horizon linear experiment

$$Y = \mathcal{O}x + \varepsilon, \qquad \psi = Lx, \qquad \varepsilon \sim \mathcal{N}(\mathbf{0}, R), \quad R \succ \mathbf{0}. \tag{24}$$

Theorem 3 (Functional reconstruction equivalence): The following statements are equivalent: (i) $\mathcal{O}x_1 = \mathcal{O}x_2$ implies $Lx_1 = Lx_2$; (ii) $\ker \mathcal{O} \subseteq \ker L$; and (iii) there is a linear map K such that $L = K\mathcal{O}$.

a: Proof.

If (i) holds and $v \in \ker \mathcal{O}$, the observations of v and $\mathbf{0}$ coincide, hence $Lv = \mathbf{0}$ and (ii) follows. If (ii) holds, define $K(\mathcal{O}x) = Lx$ on $\mathrm{range}\,\mathcal{O}$. The definition is independent of the representative because two representatives differ by a vector in $\ker \mathcal{O} \subseteq \ker L$. Extend K linearly to the observation space to obtain (iii). Finally, (iii) immediately implies (i). □

Let

$$G = \mathcal{O}^{\mathsf{T}} R^{-1} \mathcal{O}. \tag{25}$$

Theorem 4 (Best linear unbiased estimate of an estimable function): If the equivalent conditions of Theorem 3 hold, then

$$\widehat{\psi}_\star = L G^\dagger \mathcal{O}^{\mathsf{T}} R^{-1} Y, \tag{26}$$
$$\mathrm{Cov}(\widehat{\psi}_\star) = L G^\dagger L^{\mathsf{T}}. \tag{27}$$

It has minimum covariance, in the positive-semidefinite order, among all linear unbiased estimators of Lx [15].

b: Proof.

Because $v^{\mathsf{T}} G v = \|\mathcal{O}v\|_{R^{-1}}^2$, $\ker G = \ker \mathcal{O}$. Estimability gives $L = LG^\dagger G$, so taking expectation in (26) yields Lx. Direct covariance propagation gives (27). Let AY be any other linear unbiased estimator and set $T = A - LG^\dagger \mathcal{O}^{\mathsf{T}} R^{-1}$. Then $T\mathcal{O} = \mathbf{0}$ and the cross covariance vanishes because $LG^\dagger \mathcal{O}^{\mathsf{T}} T^{\mathsf{T}} = \mathbf{0}$. Consequently,

$$\mathrm{Cov}(AY) = \mathrm{Cov}(\widehat{\psi}_\star) + TRT^{\mathsf{T}} \succeq \mathrm{Cov}(\widehat{\psi}_\star). \tag{28}$$

If the kernel inclusion fails, two states induce the same data law but different targets, so no estimator can be unbiased for every state. □

The pseudoinverse has this interpretation only for estimable targets. In the nonlinear DAE, a pointwise kernel test is not sufficient: local factorization requires the condition to hold on a regular constant-rank neighborhood with connected local fibers [16].

D. FISHER INFORMATION WITH PHYSICAL NUISANCE

For a differentiable Gaussian model

$$r \mid \boldsymbol{\theta} \sim \mathcal{N}(\boldsymbol{\mu}(\boldsymbol{\theta}), \boldsymbol{\Sigma}(\boldsymbol{\theta})), \tag{29}$$

the Fisher information contains both mean and covariance contributions. For scalar a,

$$I(a) = \boldsymbol{\mu}_a^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_a + \tfrac{1}{2} \mathrm{tr}(\boldsymbol{\Sigma}^{-1} \boldsymbol{\Sigma}_a \boldsymbol{\Sigma}^{-1} \boldsymbol{\Sigma}_a). \tag{30}$$

The second term must be retained when event amplitude changes the covariance. The evaluated load bank freezes $\boldsymbol{\Sigma}$, so only the first term is used.

To expose nuisance geometry, consider

$$r = ad + N\boldsymbol{\nu} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}), \tag{31}$$

where $\boldsymbol{\nu}$ may contain initial-state, operating-point, parameter, or integrity directions. Let $W^{\mathsf{T}}W = \boldsymbol{\Sigma}^{-1}$, $\overline{d} = Wd$, $\overline{N} = WN$, and

$$P_{\perp} = I - \overline{N}\overline{N}^{\dagger}. \tag{32}$$

Theorem 5 (Efficient Fisher information after nuisance profiling): For deterministic unknown $\boldsymbol{\nu}$, the efficient information for a is

$$I_{a|\nu}^{\mathrm{det}} = \overline{d}^{\mathsf{T}} P_{\perp} \overline{d} \tag{33}$$

$$= d^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}d - d^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}N(N^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}N)^{\dagger}N^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}d. \tag{34}$$

It is zero exactly when the event tangent lies in the observable nuisance span.

a: Proof.

After whitening, the score for a is $\overline{d}^{\mathsf{T}}\overline{\boldsymbol{\varepsilon}}$. Nuisance scores span $\{v^{\mathsf{T}}\overline{\boldsymbol{\varepsilon}} : v \in \mathrm{range}\,\overline{N}\}$. Their Fisher inner product is Euclidean because $\mathbb{E}[\overline{\boldsymbol{\varepsilon}}\overline{\boldsymbol{\varepsilon}}^{\mathsf{T}}] = I$. Removing the nuisance-score projection leaves $(P_{\perp}\overline{d})^{\mathsf{T}}\overline{\boldsymbol{\varepsilon}}$, whose variance is (33). Expanding the projector by an SVD gives (34). □

If instead $\boldsymbol{\nu} \sim \mathcal{N}(m_{\nu}, \boldsymbol{\Gamma})$ with a proper prior, marginalization yields

$$r \mid a \sim \mathcal{N}(ad + Nm_{\nu}, \boldsymbol{\Sigma} + N\boldsymbol{\Gamma}N^{\mathsf{T}}), \tag{35}$$

$$I_{a|\nu}^{\mathrm{Bayes}} = d^{\mathsf{T}}(\boldsymbol{\Sigma} + N\boldsymbol{\Gamma}N^{\mathsf{T}})^{-1}d. \tag{36}$$

Woodbury's identity shows that (36) tends to the no-nuisance information as $\boldsymbol{\Gamma} \to \mathbf{0}$ and to (33) as a full-rank nuisance prior becomes diffuse. Thus zero deterministic efficient information is not a universal impossibility statement: a restrictive proper prior can still supply separation.

Corollary 1 (Pre-event information cannot reduce local event information): If pre-event data add nuisance precision $\boldsymbol{\Lambda}_{\mathrm{pre}} \succeq \mathbf{0}$, then

$$I_{a|\nu,\mathrm{pre}} = I_{aa} - I_{a\nu}(I_{\nu\nu} + \boldsymbol{\Lambda}_{\mathrm{pre}})^{-1}I_{\nu a} \tag{37}$$

is nondecreasing in $\boldsymbol{\Lambda}_{\mathrm{pre}}$.

b: Justification.

If $\boldsymbol{\Lambda}_2 \succeq \boldsymbol{\Lambda}_1 \succeq \mathbf{0}$, matrix inversion reverses the order. Premultiplication and postmultiplication by the cross-information blocks preserve the positive-semidefinite order, and subtraction reverses it once more. □

For a known source, known positive sign, and fixed covariance, the standardized matched statistic

$$T = \frac{d^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}r}{\sqrt{d^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}d}} \tag{38}$$

is standard normal under H_0 and has mean $a\sqrt{I}$ under amplitude a. A level-α one-sided test has power $1-\Phi(z_{1-\alpha}-a\sqrt{I})$. Hence the amplitude required for power $1-\beta$ is

$$a_{\alpha,\beta} = \frac{z_{1-\alpha} + z_{1-\beta}}{\sqrt{I}}. \tag{39}$$

This is a conditional known-source benchmark, not the threshold of the 137-support search. Its empirical use in the paper is limited to the exploratory source-wise detectability association.

E. FIRST-ORDER MULTI-EVENT SUPPORT GEOMETRY AND ONSET INFORMATION

Let two first-order source signatures be d_g and d_j under fixed precision $G \succeq \mathbf{0}$. Allow the competing source to refit its amplitude. The local profiled distance satisfies

$$b_{\star}(a) = a\frac{d_j^{\mathsf{T}}Gd_g}{d_j^{\mathsf{T}}Gd_j} + O(a^2), \tag{40}$$

$$\Delta^2_{g\to j}(a) = a^2 J_{g\to j} + O(a^3), \tag{41}$$

$$J_{g\to j} = d_g^{\mathsf{T}}Gd_g - \frac{(d_g^{\mathsf{T}}Gd_j)^2}{d_j^{\mathsf{T}}Gd_j} \geq 0. \tag{42}$$

Equation (42) is a pairwise local separation quantity, not the Fisher information for amplitudes conditional on the true support.

The pairwise formula extends to arbitrary supports. Let N_c be the unwhitened nuisance Jacobian, let $G_e = [g_1, \ldots, g_M]$ collect the event tangents, and define

$$A = WN_c, \quad F = WG_e, \quad R_N = I - AA^{\dagger}, \quad \overline{F} = R_N F. \tag{43}$$

Theorem 6 (Profile-likelihood diagnostic quotient): Consider the whitened local experiment

$$y = A\boldsymbol{\nu} + F\boldsymbol{\alpha} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I), \tag{44}$$

with deterministic unrestricted nuisance $\boldsymbol{\nu}$. Let $U_{\mathcal{D}}$ have orthonormal columns spanning $\mathcal{D} = \mathrm{range}(\overline{F})$, and define

$$z = U_{\mathcal{D}}^{\mathsf{T}}R_N y, \qquad d_{\mathcal{D}} = \dim\mathcal{D} = \mathrm{rank}(\overline{F}). \tag{45}$$

Then every profile-likelihood ratio over $\boldsymbol{\alpha}$ depends on y only through z:

$$\min_{\boldsymbol{\nu}} \|y - A\boldsymbol{\nu} - F\boldsymbol{\alpha}\|_2^2 = \|z - U_{\mathcal{D}}^{\mathsf{T}}\overline{F}\boldsymbol{\alpha}\|_2^2 + C(y), \tag{46}$$

where $C(y)$ is independent of $\boldsymbol{\alpha}$. The projector, quotient statistic up to an orthogonal change of coordinates, and $d_{\mathcal{D}}$ depend only on the nuisance span, not its parameterization.

a: Proof.

Profiling nuisance gives $\|R_N(y - F\boldsymbol{\alpha})\|_2^2$. Decompose this residual orthogonally into $\mathcal{D}$ and $\mathrm{range}(R_N) \cap \mathcal{D}^{\perp}$. The first component is the squared norm in (46); because $\overline{F}\boldsymbol{\alpha} \in \mathcal{D}$, the second is $C(y) = \|(I - U_{\mathcal{D}}U_{\mathcal{D}}^{\mathsf{T}})R_N y\|_2^2$ and contains no $\boldsymbol{\alpha}$. Reparameterizing nuisance without changing $\mathrm{range}(A)$ leaves $AA^{\dagger}$ and therefore every stated object unchanged. □

Thus $d_{\mathcal{D}}$ is the intrinsic dimension of the *linear profiled* event experiment. It is not a nonlinear global observability count and need not equal the number of PMUs. If candidates use different nuisance spans, or if nuisance has an informative proper prior, the corresponding candidate-specific profile or marginal likelihood must be retained; the common projector above is then insufficient.

Theorem 7 (Signed first-order support-to-support separation): For true support S with fixed severity direction $\boldsymbol{\alpha}_S$ and competing support C with freely fitted local amplitudes, define

$$\gamma^2_{S\to C}(\boldsymbol{\alpha}_S) = \min_{u,\boldsymbol{\beta}_C} \|F_S\boldsymbol{\alpha}_S - Au - F_C\boldsymbol{\beta}_C\|_2^2. \tag{47}$$

Then

$$\gamma^2_{S\to C}(\boldsymbol{\alpha}_S) = \| \left(I - P_{\overline{F}_C}\right) \overline{F}_S\boldsymbol{\alpha}_S\|_2^2, \tag{48}$$

$$= \boldsymbol{\alpha}_S^{\mathsf{T}} K_{S|C}\boldsymbol{\alpha}_S, \tag{49}$$

$$K_{S|C} = \overline{F}_S^{\mathsf{T}} \left(I - P_{\overline{F}_C}\right) \overline{F}_S \succeq \mathbf{0}. \tag{50}$$

Here $P_{\overline{F}_C} = \overline{F}_C\overline{F}_C^{\dagger}$, so rank deficiency in either nuisance or event coordinates does not change the projected trajectory.

b: Proof.
For fixed $\boldsymbol{\beta}_C$, minimizing (47) over u removes the component in $\operatorname{range} A$ and leaves $\|\overline{F}_S\boldsymbol{\alpha}_S - \overline{F}_C\boldsymbol{\beta}_C\|_2^2$. A second least-squares projection gives (48). Symmetry and idempotence of the orthogonal projector give (49) and its positive-semidefinite order. □

The theorem exposes two distinct weak-event mechanisms. For nonzero residualized single-source signatures $\overline{f}_i$ and $\overline{f}_j$,

$$\gamma^2_{i\to j}(a_i) = a_i^2\|\overline{f}_i\|_2^2(1-\rho_{ij}^2), \qquad \rho_{ij} = \frac{\overline{f}_i^{\mathsf{T}}\overline{f}_j}{\|\overline{f}_i\|_2\|\overline{f}_j\|_2}. \tag{51}$$

Small $\|\overline{f}_i\|$ is event–nuisance invisibility; large $|\rho_{ij}|$ is event–event aliasing after nuisance removal.

For general supports, write $H = S \cap C$, $B = S \setminus C$, and $D = C \setminus S$. The shared amplitudes can be refitted, so

$$\gamma^2_{S\to C} = \| \left(I - P_{[A,F_H,F_D]}\right) F_B\boldsymbol{\alpha}_B\|_2^2. \tag{52}$$

Thus the shared true component cancels exactly at first order for signed amplitudes or an interior parameter point. In particular, for $S = \{i,j\}$ and $C = \{i,k\}$ the relaxed margin is independent of the amplitude at i. Enlarging a competitor can only reduce the margin because its explanatory subspace grows.

The exclusive-support quotient removes the zero directions introduced by a literal support intersection. Define

$$\begin{aligned} Z_H &= [A, F_H], \quad R_H = I - Z_H Z_H^{\dagger}, \\ \widetilde{F}_B &= R_H F_B, \quad \widetilde{F}_D = R_H F_D. \end{aligned} \tag{53}$$

Then

$$\begin{aligned} \gamma^2_{S\to C}(\boldsymbol{\alpha}_B) &= \min_{\boldsymbol{\beta}_D} \|\widetilde{F}_B\boldsymbol{\alpha}_B - \widetilde{F}_D\boldsymbol{\beta}_D\|_2^2 \\ &= \boldsymbol{\alpha}_B^{\mathsf{T}} K_{B|D;H}\boldsymbol{\alpha}_B, \\ K_{B|D;H} &= \widetilde{F}_B^{\mathsf{T}}(I - P_{\widetilde{F}_D})\widetilde{F}_B \succeq \mathbf{0}. \end{aligned} \tag{54}$$

The first equality follows by minimizing first over the common span $\operatorname{range} Z_H$ and then over $\boldsymbol{\beta}_D$. Moreover, $K_{B|D;H} \succ \mathbf{0}$ if and only if

$$\ker \widetilde{F}_B = \{\mathbf{0}\}, \qquad \operatorname{range} \widetilde{F}_B \cap \operatorname{range} \widetilde{F}_D = \{\mathbf{0}\}. \tag{55}$$

Indeed, a zero quadratic form means that $\widetilde{F}_B\boldsymbol{\alpha}_B$ lies in the competing range; the two conditions force $\boldsymbol{\alpha}_B = \mathbf{0}$, and the converse is immediate. Under the coefficient normalization $\|\boldsymbol{\alpha}_B\|_2 = 1$, Rayleigh–Ritz gives the worst relaxed direction and margin as

$$\min_{\|\boldsymbol{\alpha}_B\|_2=1} \gamma^2_{S\to C} = \lambda_{\min}(K_{B|D;H}). \tag{56}$$

This value depends on the physical scaling of the amplitude coordinates.

For a scale-free comparison, let Q_B, Q_D be orthonormal bases of the two conditional ranges. If $\theta_{\min}$ is their smallest principal angle,

$$\lambda_{\min}\left[Q_B^{\mathsf{T}}(I - Q_DQ_D^{\mathsf{T}})Q_B\right] = \sin^2\theta_{\min}. \tag{57}$$

The angle must be computed after removing nuisance and shared directions; otherwise every shared event creates a trivial zero angle. If the residual ambient dimension is $q_H = m - \operatorname{rank} Z_H$, then $r_B + r_D > q_H$ forces a nonzero intersection between the two conditional *spans*. This is a span-relaxed dimensional obstruction, not yet a one-sided physical ambiguity.

If $S \subset C$ and active amplitudes may equal zero, the competitor contains the true response and $\gamma_{S\to C} = 0$. If zero is excluded but belongs to the amplitude-domain closure, the infimum remains zero. Pure best-fit geometry therefore cannot select cardinality among nested support models. The marginal likelihood in Supplementary Appendix E supplies the required amplitude-volume and support-prior terms.

F. PHYSICAL TANGENT-CONE SEPARATION

Let $\mathcal{A}_C$ be the physical amplitude domain of a competing event model at the nominal point, and let $T_C = T_{\mathcal{A}_C}(\mathbf{0})$ denote its Bouligand tangent cone. Signed load changes give $T_C = \mathbb{R}^{|C|}$; one-sided trip, shedding, outage-continuation, and fault-severity coordinates give nonnegative rays. Missing PMU data are a discrete measurement-mode change and are not forced into this continuous construction.

Theorem 8 (Constrained first-order separation): Suppose a competing trajectory family has a local C^2 effective chart ϕ with $\phi(\mathbf{0}) = y_0$, full-column-rank derivative J, and local parameter cone $K = \mathbb{R}^p \times T_C$, where T_C is closed and polyhedral. Assume no distant branch approaches y_0. For $y_\star(\epsilon) = y_0 + \epsilon b + o(\epsilon)$,

$$d_C^2(\epsilon) = \epsilon^2 \operatorname{dist}^2(b, JK) + o(\epsilon^2). \tag{58}$$

a: Proof.

For every $v \in K$, conicity gives $\phi(\epsilon v) = y_0 + \epsilon J v + o(\epsilon)$, which yields the upper bound after taking the infimum. Conversely, an $o(\epsilon)$-optimal local coordinate u_ϵ satisfies $\|\phi(u_\epsilon) - y_0\| = O(\epsilon)$. Full column rank and differentiability imply $\|u_\epsilon\| = O(\epsilon)$. Every convergent subsequence of u_ϵ/ϵ has a limit in the closed cone K, and division of the residual by ϵ gives the matching lower bound. The isolated-branch assumption prevents a nonlocal chart from attaining a smaller leading distance. □

With A the whitened unconstrained nuisance tangent and F_C the event tangent, nuisance profiling is still exact:

$$\gamma_{\text{phys}}^2 = \min_{\beta_C \in T_C} \|R_N(b - F_C\beta_C)\|_2^2 = \text{dist}^2(R_N b, R_N F_C T_C). \tag{59}$$

The observable projection is unique even when the coefficient vector is not. Let $\mathcal{C} = R_N F_C T_C$ and $\mathcal{L} = \text{span}\,\mathcal{C}$. Since $\mathcal{C} \subseteq \mathcal{L}$,

$$\gamma_{\text{phys}} \geq \gamma_{\text{span}}, \quad \gamma_{\text{phys}} = \gamma_{\text{span}} \iff \Pi_{\mathcal{L}}(R_N b) \in \mathcal{C}. \tag{60}$$

Thus a positive span margin certifies physical separation, whereas a zero span margin does not prove one-sided ambiguity.

For $T_C = \mathbb{R}_+^{|C|}$, the optimum is a nonnegative least-squares projection. With residual $r = R_N(b - F_C\beta_C)$, the KKT conditions are

$$\beta_C \geq 0, \quad (R_N F_C)^\top r \leq 0, \quad \beta_j (R_N f_j)^\top r = 0. \tag{61}$$

For one true signature x and one admissible competing ray d,

$$\gamma_{\text{ray}}^2 = \|x\|_2^2 - \frac{[d^\top x]_+^2}{\|d\|_2^2}, \qquad [v]_+ = \max(0, v). \tag{62}$$

An antiparallel signature can therefore have zero span margin and a strictly positive physical margin.

For a shared one-sided support H, the exact difference coordinate satisfies $\delta_H \in T_H - \alpha_H$, which need not be a subspace. Hence the physical margin can depend on the shared true amplitude; the unconditional cancellation statement applies only when the relaxed solution is feasible. Competitor enrichment still cannot increase the margin whenever the old feasible set is contained in the new one. Likewise, a nested superset still has zero margin when it can match the shared amplitudes and set every added amplitude to zero.

For nontrivial closed cones $\mathcal{C}_S, \mathcal{C}_C$, the directed scale-free separation

$$\delta(\mathcal{C}_S \to \mathcal{C}_C) = \min_{\substack{x \in \mathcal{C}_S \\ \|x\|_2 = 1}} \text{dist}(x, \mathcal{C}_C) \tag{63}$$

vanishes exactly when the cones share a nonzero vector. It reduces to $\sin\theta_{\min}$ for subspaces, but need not be symmetric. Consequently, the classical signed-sparse condition $\text{spark}(\overline{F}) > 2K$ remains sufficient under one-sided constraints, while its failure need not imply a physically feasible ambiguity.

TABLE 3. Deterministic checks of conditional and one-sided geometry. These toy systems are not IEEE 39-bus evidence.

Check	Recomputed outcome
Conditional quotient	full–quotient error 0.0e+00 (pass).
Worst coefficient direction	eigenvalue–grid error 1.9e-08 (pass).
Conditional angle	naive 0.0e+00 deg; conditional 18.689 deg (pass).
Ray versus span	squared margins 0.0 versus 1.0 (pass).
One-sided shared source	squared-margin sequence 5.0, 2.0, 1.0, 1.0 (pass).
Cone enrichment	2000 trials; max violation 3.6e-15 (pass).
Nested cone	squared margin 2.4e-31 (pass).

For a multi-source support S, the Fisher information conditional on that known support is

$$I_S(a) = J_S(a)^\top G J_S(a), \quad J_S = \frac{\partial \mu_S}{\partial a_S}. \tag{64}$$

Positive definiteness of I_S permits local amplitude estimation conditional on S; it does not exclude another support with the same mean.

Unknown onset has a separate degeneracy. For $\mu(a, \tau) = a d(\tau)$ and fixed covariance,

$$I(a, \tau) = \begin{bmatrix} d^\top \Sigma^{-1} d & a d^\top \Sigma^{-1} \dot{d} \\ a \dot{d}^\top \Sigma^{-1} d & a^2 \dot{d}^\top \Sigma^{-1} \dot{d} \end{bmatrix}. \tag{65}$$

Profiling amplitude by a Schur complement gives

$$I_{\tau|a} = a^2 \left[\dot{d}^\top \Sigma^{-1} \dot{d} - \frac{(d^\top \Sigma^{-1} \dot{d})^2}{d^\top \Sigma^{-1} d} \right]. \tag{66}$$

At $a = 0$, onset is unidentifiable. If $\dot{d}$ is proportional to d, delay and gain are locally confounded even for nonzero amplitude. The evaluated bank avoids this unresolved problem by fixing onset.

G. CANDIDATE INDISTINGUISHABILITY AND SPARSE INJECTIVITY

For candidate h and fixed integrity support S, define the affine mean set

$$\mathcal{M}_h(S) = s_{h,H} + \text{range}[\mathcal{O}_{h,H}, \mathcal{E}_{S,H}]. \tag{67}$$

The profiled margin in the main paper is half the squared distance between two such sets under R_H^{-1}.

Theorem 9 (Local candidate indistinguishability): If $D_{ij,S}(H) = 0$, the candidate manifolds intersect. At any intersection, the two candidates induce the same Gaussian law; no decision rule can distinguish them uniformly over the declared nuisance set.

a: Proof.

Zero distance between closed affine sets means that there exist nuisance vectors for which their means coincide. The covariance is common by the stated contract, so the two conditional probability measures are identical. For any decision region, the probabilities of choosing i and j under this common law sum to one; at least one conditional error is therefore at least one half. □

For two fixed simple candidates with squared Mahalanobis distance $\Delta^2 = 2D_{ij,S}$ and equal priors, projection onto their mean difference reduces the optimal test to one dimension, giving

$$P_e^\star = \Phi(-\Delta/2) = \Phi(-\sqrt{D_{ij,S}/2}). \tag{68}$$

This formula does not apply directly to amplitude-marginalized support mixtures.

For the local persistent-event model $Y_T = \mathcal{H}_T a + \varepsilon_T$, define the physically admissible difference set

$$\mathcal{D}_K = \{a - b : a, b \text{ admissible and } K\text{-sparse}\}. \tag{69}$$

Theorem 10 (Finite-horizon sparse event uniqueness): Every admissible K-sparse event is uniquely determined in the noise-free local model if and only if

$$\ker \mathcal{H}_T \cap \mathcal{D}_K = \{\mathbf{0}\}. \tag{70}$$

For a signed neighborhood in each support subspace, $\mathcal{D}_K = \mathcal{A}_{2K}$ locally, recovering the condition in the main paper.

b: Proof.

If two admissible events produce the same output, their nonzero difference belongs to both sets in (70). Conversely, any nonzero vector in the intersection is, by definition of $\mathcal{D}_K$, the difference of two admissible K-sparse events that produce the same output. □

Unknown initial-state nuisance can be removed locally by whitening and projection:

$$\widetilde{\mathcal{H}}_T = \left(I - P_{\mathrm{range}(W_T\mathcal{O}_T)}\right) W_T \mathcal{H}_T, \qquad W_T^\mathsf{T} W_T = R_T^{-1}. \tag{71}$$

Theorem 10 then applies with $\widetilde{\mathcal{H}}_T$. Define the restricted margin

$$\gamma_{2K}(T) = \min_{1 \le |U| \le 2K} \sigma_{\min}(\widetilde{\mathcal{H}}_{T,U}). \tag{72}$$

If two K-sparse explanations both fit data within whitened residual ϵ, then

$$\|a - b\|_2 \le \frac{2\epsilon}{\gamma_{2K}(T)}. \tag{73}$$

Indeed, the triangle inequality bounds $\|\widetilde{\mathcal{H}}_T(a-b)\|_2$ by 2ϵ, while the definition of γ_{2K} gives the reverse lower bound. For the evaluated $K \le 2$ bank, the relevant union size is four.

H. INFORMATION CANNOT DECREASE WHEN VALID MEASUREMENTS ARE ADDED

For a true trajectory at severity a and a competing physical family with parameter $\boldsymbol{\eta} \in \Theta_C$, define the exact profiled prefix distance

$$d_T^2(a) = \inf_{\boldsymbol{\eta} \in \Theta_C} r_T(a, \boldsymbol{\eta})^\mathsf{T} \boldsymbol{\Sigma}_T^{-1} r_T(a, \boldsymbol{\eta}), \tag{74}$$

where every $\boldsymbol{\Sigma}_T$ is a principal prefix of one positive-definite process covariance and the physical competitor domain does not change with T. Partition the difference between candidate means into old and added blocks, $\boldsymbol{\delta} = \mathrm{col}(\boldsymbol{\delta}_1, \boldsymbol{\delta}_2)$, with joint covariance $\boldsymbol{\Sigma} \succ \mathbf{0}$. Block Gaussian elimination gives

$$\boldsymbol{\delta}^\mathsf{T} \boldsymbol{\Sigma}^{-1} \boldsymbol{\delta} = \boldsymbol{\delta}_1^\mathsf{T} \boldsymbol{\Sigma}_{11}^{-1} \boldsymbol{\delta}_1 + \widetilde{\boldsymbol{\delta}}_2^\mathsf{T} \boldsymbol{\Sigma}_{2|1}^{-1} \widetilde{\boldsymbol{\delta}}_2, \tag{75}$$

$$\widetilde{\boldsymbol{\delta}}_2 = \boldsymbol{\delta}_2 - \boldsymbol{\Sigma}_{21} \boldsymbol{\Sigma}_{11}^{-1} \boldsymbol{\delta}_1, \tag{76}$$

$$\boldsymbol{\Sigma}_{2|1} = \boldsymbol{\Sigma}_{22} - \boldsymbol{\Sigma}_{21} \boldsymbol{\Sigma}_{11}^{-1} \boldsymbol{\Sigma}_{12}. \tag{77}$$

Theorem 11 (Prefix monotonicity and temporal rescue): Under the prefix and fixed-domain contract above,

$$d_{T+1}^2(a) \ge d_T^2(a). \tag{78}$$

If both minima are attained, equality holds if and only if an element of $\operatorname{Argmin} d_T^2$ also has zero conditional innovation residual in the added block. Suppose further that, locally,

$$d_T^2(a) = c_T a^{2k_T} + o(a^{2k_T}), \qquad c_T > 0. \tag{79}$$

Then $k_{T+1} \le k_T$; if the order remains k, then $c_{T+1} \ge c_T$.

a: Proof.

The conditional covariance in (75) is positive definite, so its second quadratic form is nonnegative for every fixed $\boldsymbol{\eta}$. Taking the infimum over the same Θ_C proves (78). If the minima are attained, equality of the infima forces an extended-horizon minimizer to minimize the prefix term and to make the innovation term zero; the converse is immediate. If $k_{T+1} > k_T$, the ratio $d_{T+1}^2(a)/d_T^2(a)$ tends to zero as $a \to 0$, contradicting (78). When both orders equal k, division by a^{2k} and passage to the limit yield $c_{T+1} \ge c_T$. □

The theorem permits a longer prefix to reveal a first-order direction that was absorbed at an earlier horizon, or to reveal a higher-order obstruction that was previously invisible. It does not say that a concrete estimator improves: changing its normalization, nuisance set, covariance model, or optimization breaks the prefix contract. In particular, the archived RAWSIM39 calibration renormalized each event–signal–horizon block to unit weight and is not a prefix likelihood for this theorem.

If whitened model errors under two supports satisfy $\|We_S\| \le \epsilon_S$ and $\|We_R\| \le \epsilon_R$, the reverse triangle inequality gives the robust separation bound

$$d_{\mathrm{true}} \ge [d_{\mathrm{nom}} - \epsilon_S - \epsilon_R]_+, \qquad \mathcal{J}_{\mathrm{true}} \ge \tfrac{1}{2}[d_{\mathrm{nom}} - \epsilon_S - \epsilon_R]_+^2. \tag{80}$$

This is the required bridge for future simulator-shift claims; it is not evaluated by the current nominal load bank.

I. TEMPORAL INFORMATION UNDER THE FROZEN AR(1) CONTRACT

For true support S, competing support R, true amplitude a, and competitor amplitude b, let $\boldsymbol{\delta}_t(b) = \boldsymbol{\mu}_{S,t}(a) - \boldsymbol{\mu}_{R,t}(b)$. The frozen temporal covariance is

$$\boldsymbol{\Sigma}_T = R_T(\rho) \otimes \boldsymbol{\Omega}, \qquad [R_T]_{ij} = \rho^{|i-j|}, \quad |\rho| < 1. \tag{81}$$

For fixed b, its squared Mahalanobis separation has the innovation form

$$q_T(b) = \|\boldsymbol{\delta}_1(b)\|^2_{\boldsymbol{\Omega}^{-1}} + \frac{1}{1-\rho^2}\sum_{t=2}^{T}\|\boldsymbol{\delta}_t(b) - \rho\boldsymbol{\delta}_{t-1}(b)\|^2_{\boldsymbol{\Omega}^{-1}}. \quad (82)$$

Define the profiled simple-model separation

$$\mathcal{J}_{S\to R}(T) = \tfrac{1}{2}\inf_{b\in\mathcal{A}_R} q_T(b). \quad (83)$$

It equals the minimum KL divergence over fixed-amplitude simple Gaussian competitors. It is not the KL divergence or Bayes factor of the amplitude-mixture support model.

Theorem 12 (Persistent AR(1) information rate): If the amplitude domain is compact and fixed and $\boldsymbol{\delta}_t(b)$ converges uniformly to $\boldsymbol{\delta}_\infty(b)$, then

$$\lim_{T\to\infty}\frac{2\mathcal{J}_{S\to R}(T)}{T} = \frac{1-\rho}{1+\rho}\inf_{b\in\mathcal{A}_R}\|\boldsymbol{\delta}_\infty(b)\|^2_{\boldsymbol{\Omega}^{-1}}. \quad (84)$$

a: Proof.

Under uniform convergence, $\boldsymbol{\delta}_t - \rho\boldsymbol{\delta}_{t-1}$ converges uniformly to $(1-\rho)\boldsymbol{\delta}_\infty$. Divide (82) by T; the first-frame term vanishes and the Cesàro mean converges uniformly to $(1-\rho)^2(1-\rho^2)^{-1}\|\boldsymbol{\delta}_\infty\|^2_{\boldsymbol{\Omega}^{-1}}$. Compactness and uniform convergence permit interchange of infimum and limit, giving (84). □

Monotonicity of $\mathcal{J}_{S\to R}(T)$ follows independently from Theorem 11; the AR(1) calculation supplies the explicit limiting rate.

If the profiled equilibrium distance is positive, separation grows as $\Theta(T)$. If an admissible competitor matches equilibrium and its innovation differences are square summable, evaluating (83) at that competitor gives a finite upper bound; monotonicity then implies convergence to a finite ceiling. Exact trajectory overlap gives zero separation for every horizon.

For a true pair $\{i,j\}$, a competing pair $\{i,k\}$ can refit the shared source. With whitened equilibrium columns $\bar{h}_i, \bar{h}_j, \bar{h}_k$,

$$a_j^2\|(I - P_{i,k})\bar{h}_j\|_2^2 \le a_j^2\|(I-P_i)\bar{h}_j\|_2^2. \quad (85)$$

The inequality follows from $\mathrm{span}(\bar{h}_i) \subseteq \mathrm{span}(\bar{h}_i, \bar{h}_k)$. It formalizes why identifying which weak second source occurred cannot be easier than rejecting the shared single-source explanation.

J. HIGHER-ORDER CONTACT AND LOCAL DETECTION SCALE

First-order Fisher information can vanish while nonlinear separation remains. In whitened coordinates, let the no-event nuisance manifold and event curve be

$$F_0(b) = Nb + \tfrac{1}{2}H_0[b,b] + O(\|b\|^3), \quad (86)$$
$$c(a) = ad + a^2 q + O(a^3). \quad (87)$$

Theorem 13 (Relative first- and second-order separation): Assume the nuisance manifold has constant rank and use a regular chart after removing redundant coordinates. Define the profiled first-order coefficient

$$\gamma_1^2 = \|P_\perp d\|_2^2, \qquad P_\perp = I - (WJ_c)(WJ_c)^\dagger, \quad (88)$$

where J_c is the nuisance tangent before whitening and d is the already whitened event tangent in (87). If $\gamma_1 > 0$, the local squared distance to the nuisance manifold is

$$\Delta^2(a) = a^2\gamma_1^2 + O(a^3). \quad (89)$$

If $\gamma_1 = 0$, so the whitened event tangent lies in the nuisance tangent space, choose the regular-chart vector v satisfying $Nv = d$ and define

$$\boldsymbol{\kappa} = P_\perp\left(q - \tfrac{1}{2}H_0[v,v]\right). \quad (90)$$

Then

$$\Delta^2(a) = a^4\gamma_2^2 + O(a^5), \qquad \gamma_2^2 = \|\boldsymbol{\kappa}\|_2^2. \quad (91)$$

a: Proof.

The local minimizer satisfies $DF_0(b)^\mathsf{T}[F_0(b) - c(a)] = \mathbf{0}$. At the origin, its derivative with respect to the regular chart coordinate is $N^\mathsf{T}N \succ \mathbf{0}$, so the implicit-function theorem gives a smooth minimizing branch. Write $b_\star(a) = av + a^2 w + O(a^3)$. At first order, $N^\mathsf{T}(Nv - d) = \mathbf{0}$, yielding residual $aP_\perp d + O(a^2)$ and (89). If $d = Nv$, the first-order residual vanishes. At second order, minimization projects $q - \frac{1}{2}H_0[v,v]$ onto $\mathrm{range}\,N$; the remaining normal component is (90). Squaring the residual gives (91). The projector is unchanged by any full-rank reparameterization of the same nuisance tangent space, so γ_1 is a property of the local manifolds and covariance metric rather than of a particular nuisance coordinate basis. □

The same constant-rank qualification makes γ_2 intrinsic. Redundant original parameters may give several representatives of the same observable tangent, but their normal acceleration equals the second fundamental form of the same manifold branch. If $\gamma_2 = 0$, the conclusion is only $\Delta^2(a) = o(a^4)$ under the corresponding little-o expansion; an a^6 law or exact overlap requires a separate third-order analysis.

K. OPERATIONAL ERROR AND SUPPORT-RECOVERY LIMITS

For two simple hypotheses with a shared covariance, $P_i = \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma})$, define

$$d^2 = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0). \quad (92)$$

The equal-prior log likelihood ratio is Gaussian with mean $\mp d^2/2$ and variance d^2 under P_0 and P_1, respectively. Hence

$$P_e^\star = \Phi(-d/2), \quad D_{\mathrm{KL}}(P_1\|P_0) = d^2/2, \quad C_{\mathrm{Ch}} = d^2/8. \quad (93)$$

The last identity gives $P_e^\star \le \frac{1}{2}\exp(-d^2/8)$. If a common affine nuisance is $A\boldsymbol{\eta}$, orthogonal projection after whitening removes it exactly and replaces d by the quotient margin.

Thus $P_e^\star = \Phi(-\gamma/2)$ is exact for that quotient Gaussian experiment. It is not an exact formula for two nonlinear amplitude mixtures with different priors or dimensions.

Suppose the first nonzero contact order at horizon T is k and

$$d_T^2(a) = T\mathcal{I}_k|a|^{2k}[1+o(1)], \qquad \mathcal{I}_k > 0. \tag{94}$$

For target simple-Gaussian error $\alpha < 1/2$, inversion of (93) yields

$$|a|_{\min}(\alpha, T) \sim \left[\frac{2\Phi^{-1}(1-\alpha)}{\sqrt{T\mathcal{I}_k}}\right]^{1/k}. \tag{95}$$

Equivalently, if $a_T = cT^{-\beta}$, the optimal binary error tends to one half for $\beta > 1/(2k)$, to a nontrivial constant at equality, and to zero for $\beta < 1/(2k)$. This gives the $T^{-1/2}$ and $T^{-1/4}$ boundaries for first- and second-order separation.

Now let $\Theta = \bigcup_{S\in\mathfrak{S}} \Theta_S$ and define the minimax exact-support risk

$$R^\star = \inf_{\widehat{S}} \sup_{\theta\in\Theta} \Pr_\theta\{\widehat{S} \neq S(\theta)\}. \tag{96}$$

Theorem 14 (Binary and minimum-severity limits): For every pair $\theta_i \in \Theta_{S_i}$ with $S_0 \neq S_1$,

$$\begin{aligned} R^\star &\geq \frac{1}{2}[1 - \mathrm{TV}(P_{\theta_0}, P_{\theta_1})] \\ &\geq \frac{1}{2}\left[1 - \sqrt{\tfrac{1}{2}D_{\mathrm{KL}}(P_{\theta_0}\|P_{\theta_1})}\right]_+. \end{aligned} \tag{97}$$

If $S^+ = S \cup \{j\}$ and $P_{S^+,a_j} \to P_S$ in total variation as $a_j \downarrow 0$, then uniform exact-support recovery over all $a_j > 0$ has minimax risk at least one half.

a: Proof.

Restriction of any support rule to a two-point subset cannot reduce its worst-case risk; the optimal equal-prior binary risk is $[1 - \mathrm{TV}(P,Q)]/2$. Pinsker's inequality gives the second bound. For the nested pair, the binary risk tends to one half as the active amplitude tends to zero, so the supremum over positive amplitudes is at least that limit. □

The theorem requires either a declared $|a_j| \geq a_{\min} > 0$ contract or a decision that may abstain near absence. It also distinguishes a source-ranking error from an impossible exact-cardinality decision.

For M equiprobable supports, Fano's inequality gives

$$P_e \geq 1 - \frac{I(V;Y) + \log 2}{\log M}. \tag{98}$$

With $\overline{P} = M^{-1}\sum_j P_j$, convexity of KL in its second argument implies

$$\begin{aligned} I(V;Y) &= \frac{1}{M}\sum_i D_{\mathrm{KL}}(P_i\|\overline{P}) \\ &\leq \frac{1}{M^2}\sum_{i,j} D_{\mathrm{KL}}(P_i\|P_j) = \overline{D}_{\mathrm{KL}}. \end{aligned} \tag{99}$$

Therefore a necessary condition for $P_e \leq \alpha$ is

$$\overline{D}_{\mathrm{KL}} \geq (1-\alpha)\log M - \log 2. \tag{100}$$

If $M = \binom{N}{K}$ and $\overline{D}_{\mathrm{KL}}(T) \leq T\overline{\mathcal{I}}_k a^{2k}/2$, then necessarily

$$T \geq \frac{2[(1-\alpha)\log\binom{N}{K} - \log 2]}{\overline{\mathcal{I}}_k a^{2k}}. \tag{101}$$

This is an impossibility certificate, not a sufficient sample-size guarantee [17], [18]. For sparse $K \ll N$, it has the familiar scaling $T \gtrsim K\log(N/K)/(\overline{\mathcal{I}}_k a^{2k})$, now expressed through the physical Hybrid-DAE information rate.

L. DIAGNOSABILITY-PRESERVING DISCREPANCY COVARIANCE

A residual correction can improve fit while making physical events harder to distinguish. The following calculation makes that tradeoff explicit. Work after whitening and removal of common nuisance. Let $Q_U \in \mathbb{R}^{m\times r}$ be an orthonormal basis for the discriminant directions that must be protected, and let $Q_D \in \mathbb{R}^{m\times q}$ be an orthonormal discrepancy basis. The former should be frozen from critical candidate pairs; the latter must be estimated from mismatch-only records so that unlabeled events are not absorbed as model error.

Theorem 15 (Discrepancy retention and finite intensity): Let $P_D = Q_D Q_D^\mathsf{T}$ and model the whitened residual with covariance

$$C_s = I + sP_D, \qquad s = \tau^2 \geq 0. \tag{102}$$

Then $C_s^{-1} = I - \alpha P_D$, where $\alpha = s/(1+s)$, and the worst information fraction retained over the protected subspace is

$$\rho(s, Q_D) = 1 - \alpha\,\sigma_{\max}^2(Q_U^\mathsf{T} Q_D). \tag{103}$$

Set $c^2 = \sigma_{\max}^2(Q_U^\mathsf{T}Q_D)$ and $\delta = 1 - \rho_{\min}$. The constraint $\rho \geq \rho_{\min}$ imposes no finite upper bound on s when $c^2 \leq \delta$; otherwise it requires

$$s \leq s_{\max} = \frac{\delta}{c^2 - \delta}. \tag{104}$$

If the true residual covariance is $I + C_m$ and $E_D = \mathrm{tr}(Q_D^\mathsf{T} C_m Q_D)$, the expected Gaussian negative twice-log-likelihood relative to $s = 0$, up to constants, is

$$\Delta\mathcal{L}(s) = q\log(1+s) - \frac{s}{1+s}(q + E_D). \tag{105}$$

For fixed Q_D, its constrained minimizer is finite:

$$s^\star = \min\{E_D/q, s_{\max}\}, \tag{106}$$

with $s_{\max} = +\infty$ in the first geometric case.

a: Proof.

The inverse follows from $P_D^2 = P_D$. For $x = Q_U a$, the information ratio is the Rayleigh quotient of $I - \alpha Q_U^\mathsf{T} P_D Q_U$; its smallest eigenvalue is (103) because the singular values of $Q_U^\mathsf{T}Q_D$ are the principal-angle cosines. Solving $\alpha c^2 \leq \delta$ gives (104). Finally, $\log|C_s| = q\log(1+s)$ and taking the expectation of $y^\mathsf{T}C_s^{-1}y$ under $\mathrm{Cov}(y) = I + C_m$ gives (105). Its derivative is $(qs - E_D)/(1+s)^2$, which gives (106). □

Under strict no-loss protection, $\rho_{\min} = 1$ and $s > 0$ require $\mathrm{range}(Q_D) \subseteq \mathrm{range}(Q_U)^\perp$. For fixed rank, the basis

TABLE 4. Deterministic checks for the discrepancy-retention identities. These are algebraic tests, not power-system experiments.

Check	Error	Tolerance	Status
Low-rank covariance inverse	4.53e-16	1.0e-12	PASS
Principal-angle retention	2.22e-16	1.0e-12	PASS
Retention variance cap	0.00e+00	1.0e-12	PASS
Finite full-likelihood intensity	5.50e-09	1.0e-07	PASS
Safe projected PCA	1.78e-15	1.0e-10	PASS
Protected-distance floor	1.11e-16	1.0e-12	PASS

that captures the most mismatch energy is therefore the top-q eigenspace of $P_{U^\perp} C_m P_{U^\perp}$, provided that the orthogonal complement has dimension at least q. This is a safe projected-PCA baseline, not a learned-estimator result.

Maximizing only αE_D omits both the covariance determinant and the whitened measurement-noise contribution. It therefore drives s toward an infinite-variance limit whenever geometry permits. That utility is a valid description of reduced quadratic mismatch penalty, but it is not a normalized Bayesian tuning objective. The general joint orientation problem for $q > 1$ is an offline nonconvex design; spectral, semidefinite, or moment relaxations belong in a separate optimization study unless they are executed on the frozen event and mismatch dictionaries.

M. APPLIED GATE TAXONOMY FROM AN ARCHIVED ANDES CHECKPOINT

The frozen RAWSIM39 V1.2 calibration ledger provides an applied boundary check on the theory. It contains complete first-order screening and rank-selection decisions for 18 plant/event directions, but not the Jacobian and higher-order arrays needed to evaluate γ_2 or γ_3. Table 5 therefore classifies only what the recorded gates establish. In particular, failure of the curvature gate is not relabeled as second-order identifiability, and a small orthogonal fraction is not relabeled as exact redundancy.

N. THEORY-TO-EVIDENCE BOUNDARY

Equation (89) supplies a direct falsification test: for independently generated nonlinear trajectories with decreasing $|a|$, the profiled squared distance divided by a^2 should converge to γ_1^2. That convergence study has not been run for the present ANDES bank; the retrospective distance association and the gate-only RAWSIM39 taxonomy are not substitutes for it.

If n independent Gaussian replications have local distance $\Delta(a) = |a|^r \|k_r\| + O(|a|^{r+1})$, their KL divergence to the closest null is $n\Delta^2(a)/2$. Pinsker's inequality bounds the power of any level-α test by

$$\mathrm{Power}_a \le \alpha + \min\{1, \sqrt{n}\Delta(a)/2\}. \tag{107}$$

Thus $n|a_n|^{2r} \to 0$ prevents power from separating from size. The reference scales are $n^{-1/2}$ for first-order contact and $n^{-1/4}$ for second-order contact; singular-information asymptotics are established more broadly [19]. Correlated frames in one PMU window are not independent replications and must be handled through their joint covariance.

All audited load directions in the present bank have nonzero first-order separation. The second-order result is therefore a boundary theorem, not the empirical explanation for weak-pair failure.

TABLE 5. Conservative taxonomy of the frozen RAWSIM39 V1.2 calibration checkpoint. The gate sequence was 18 candidates, 10 individually eligible ($\kappa = 722.91$), eight rank-selected ($\kappa = 7.83$), and seven released after one 6.15σ local-domain rejection.

Category	Count	Directions	Evidence boundary
Regular local	7	plant_inertia_scale, plant_governor_droop_scale, generation_delta_p_pu, generation_delta_q_pu, generation_amplitude, generation_peak_s, load_delta_p_fraction	Released only under the local restricted posterior
Weak first-order	6	plant_damping, plant_governor_lag_scale, plant_turbine_time_scale, plant_avr_gain_scale, plant_avr_time_scale, plant_subtransient_x_scale	Low information; order and temporal rescue unresolved
Curvature unresolved	2	fault_duration_s, line_duration_s	Linear surrogate failed; not evidence of order 2
Near-redundant first-order	2	generation_tau_s, generation_residual	Small new orthogonal fraction; exact redundancy unproved
Local-domain failure	1	fault_reactance_pu	MAP left the declared local prior domain

TABLE 6. Recomputed checks of higher-order evidence and information limits. Controlled systems are not IEEE 39-bus performance evidence.

Check	Recomputed outcome
Evidence classes	slopes 0.9994, 0.4951, and 0.0003 for regular, order-2, and redundant additions (pass).
Prefix monotonicity	500 correlated-noise trials; max violation 0.0e+00 (pass).
Temporal rescue	first-order $\gamma_1^2(2) = 0.452724$; second-order relative error 1.0e-04 (pass).
Gaussian error map	quadrature error 5.6e-17 (pass).
Fano lower bound	$M = 8$ bounds 0.5993/0.3974; no Monte Carlo violation (pass).

TABLE 7. Deterministic checks of the local separation identities. These toy systems are not IEEE 39-bus evidence.

Check	Prediction and recomputed outcome
Regular first-order DAE	slope 2 and $\gamma_1^2 = 0.4527244641$; 2.000000; relative error 3.8e-12 (pass).
Shared-tangent second order	slope 4 and $\gamma_2^2 = 2.89$; 4.000000; coefficient 2.890000 (pass).
Rank-deficient nuisance	unique projected fit and $\gamma_1^2 = 0.5$; rank 2/3; max variation 1.2e-16 (pass).
Signed shared-source cancellation	margin independent of shared amplitude; squared-margin range 0.0e+00 (pass).
Competitor enrichment	$\gamma(S \to C_2) \leq \gamma(S \to C_1)$; 0.0196 versus 0.5096 (pass).
Nested support	geometric margin 0; 6.1e-35 (pass).
Principal angle	unit margin $\sin\theta$; absolute error 0.0e+00 (pass).
Nuisance enlargement	profiled margin cannot increase; sequence 2.0, 1.0, 0.0 (pass).
Diagnostic quotient	profiled costs factor through $d_{\mathcal{D}}$ coordinates; rank 4/6; max error 1.1e-15 (pass).
Nuisance reparameterization	unchanged nuisance span gives the same projector; spectral error 9.5e-16 (pass).

TABLE 8. Open checks for the simulator event semantics.

Check	Evidence required for closure
Right limit	Solver-scale $\|g_+(x(\tau+\epsilon), z(\tau+\epsilon), a)\|$ after re-equilibration; the callback residual may remain $O(a)$.
Algebraic derivatives	Agreement of analytic z_1, z_2 with central differences on a decreasing symmetric amplitude grid.
Taylor order	Constraint and state-approximation slopes $1, 2, 3$, or amplitude-halving ratios $2, 4, 8$.
Event families	One declared semantics and tolerance set for load, generation, and continuous branch-admittance perturbations.
Dynamic first order	Convergence of $\Delta y/a$ to $B_{r,a}$ over the declared early-time window.
Dynamic second order	Analytic second variation within a preregistered tolerance of nonlinear trajectories; the current discrepancy is unresolved.

TABLE 9. Status of the mathematical results relative to the frozen evidence.

Result	Present evidence boundary
Consistency jump	DAE contract only; callback/right-limit logging and derivative convergence remain open.
Functional equivalence and BLUE	Exact in the local linear model; nonlinear TEST accuracy remains empirical.
Fisher, support, and cone geometry	Signed and one-sided toy identities pass; IEEE 39-bus convergence and physical cone margins remain untested.
Sparse injectivity and γ_{2K}	T120 diagnostics concern the frozen load bank, not general descriptor left-invertibility.
Temporal order and rate	Prefix and controlled correlated-noise identities pass; the frozen long-horizon load study remains diagnostic and lacks independent operating-point validation.
Relative curvature	The controlled parabola passes; curvature-dominated weak-event behavior was not observed in the frozen atlas.
Error and minimax limits	Gaussian quotient, minimum-severity, and Fano formulas pass controlled checks; no IEEE 39 pairwise oracle-error atlas has been frozen.
RAWSIM39 gate taxonomy	The 18-direction ledger supports five conservative gate categories; derivative arrays needed to assign higher-order classes are unavailable.
Discrepancy retention	Exact for the whitened low-rank covariance model and checked algebraically; the protected dictionary, mismatch covariance, learned correction, and shifted IEEE 39-bus test remain unexecuted.

APPENDIX E ESTIMATOR AND LIKELIHOOD DETAILS

This appendix distinguishes a state-reconstruction component from two event estimator contracts. Sections E-A–E-C specify the state recursion and the implemented batch physical-manifold Bayes estimator $\mathcal{E}_{\mathrm{B}}$. Section E-F specifies the proposed joint Hybrid-DAE estimator $\mathcal{E}_{\mathrm{J}}$. Filtering, smoothing, quadrature, and optimization are numerical operations within those contracts, not additional estimator proposals.

A. BAYESIAN RECURSION AND THE STATE HIERARCHY

For a Markov state model, Bayesian filtering alternates prediction and correction [20]. More broadly, this construction treats the PMU task as a Bayesian inverse problem in which physical candidate models and latent states are compared through predictive evidence [21]:

$$p(x_k \mid y_{0:k-1}) = \int p(x_k \mid x_{k-1}) p(x_{k-1} \mid y_{0:k-1})\, dx_{k-1}, \tag{108}$$

$$p(x_k \mid y_{0:k}) = \frac{p(y_k \mid x_k) p(x_k \mid y_{0:k-1})}{\int p(y_k \mid x_k) p(x_k \mid y_{0:k-1})\, dx_k}. \tag{109}$$

The denominator in (109) is the predictive evidence. Omitting it is harmless for a state mean under one fixed model, but not when physical candidates are compared.

B0 returns the equilibrium. B1 conditions the equilibrium prior on one PMU frame:

$$K_{\mathrm{LMMSE}} = P_0 C^{\mathsf{T}} (C P_0 C^{\mathsf{T}} + R)^{-1}, \tag{110}$$

$$\widehat{\delta x}_k^{\mathrm{B1}} = K_{\mathrm{LMMSE}} (y_k - y_0). \tag{111}$$

B2 retains state and covariance between frames. For the linear Gaussian model, substituting Gaussian densities into (108)–(109) yields

$$m_k^- = A_d m_{k-1}^+, \qquad P_k^- = A_d P_{k-1}^+ A_d^{\mathsf{T}} + Q, \tag{112}$$

$$S_k = C P_k^- C^{\mathsf{T}} + R, \qquad K_k = P_k^- C^{\mathsf{T}} S_k^{-1}, \tag{113}$$

$$m_k^+ = m_k^- + K_k (y_k - C m_k^-), \qquad P_k^+ = (I - K_k C) P_k^-. \tag{114}$$

Theorem 16 (Linear-Gaussian reduction): Equations (112)–(114) are the exact posterior mean and covariance generated by the Bayesian recursion for the stated linear Gaussian model.

a: Proof.

An affine transformation of the Gaussian posterior at $k-1$, followed by addition of independent Gaussian process noise, gives (112). The pair (x_k, y_k) is jointly Gaussian with cross covariance $P_k^- C^{\mathsf{T}}$ and measurement covariance S_k. The standard conditional-Gaussian formula gives (113)–(114). Equivalently, multiplying the predictive density by the likelihood and completing the square gives posterior precision $(P_k^-)^{-1} + C^{\mathsf{T}} R^{-1} C$; Woodbury's identity produces the same covariance and mean. □

The implementation may use the Joseph covariance form

$$P_k^+ = (I - K_k C) P_k^- (I - K_k C)^{\mathsf{T}} + K_k R K_k^{\mathsf{T}} \tag{115}$$

to preserve symmetry and positive semidefiniteness numerically.

B3 uses the Rauch–Tung–Striebel backward recursion [22]:

$$J_t = P_t^+ A_d^{\mathsf{T}} (P_{t+1}^-)^{-1}, \tag{116}$$

$$m_{t|k} = m_t^+ + J_t (m_{t+1|k} - m_{t+1}^-), \tag{117}$$

$$P_{t|k} = P_t^+ + J_t (P_{t+1|k} - P_{t+1}^-) J_t^{\mathsf{T}}. \tag{118}$$

It estimates a past state using later measurements. The negative B3 result therefore diagnoses model mismatch; it does not imply that correct future information generally lacks value.

The nominal uncertainty correction rescales the rectangular voltage covariance as

$$\boldsymbol{\Sigma}_{i,k}^{\mathrm{U2}} = T \boldsymbol{\Sigma}_{i,k} T^{\mathsf{T}}, \qquad T = \mathrm{diag}(\sqrt{\tau_{\Re}}, \sqrt{\tau_{\Im}}), \tag{119}$$

with temperatures fitted on independent calibration trajectories. The observed-only adequacy statistic is a five-frame trailing mean of

$$a_k = (\boldsymbol{\nu}_k - \boldsymbol{\mu}_\nu)^{\mathsf{T}} \boldsymbol{\Sigma}_\nu^{-1} (\boldsymbol{\nu}_k - \boldsymbol{\mu}_\nu). \tag{120}$$

Neither operation changes the physical state estimate.

B. CONSTRAINED PHYSICAL RECENTERING

The static recentering estimator treats active/reactive load changes and active generator changes as nuisance coordinates d. It solves

$$(\widehat{V}, \widehat{d}) = \arg\min_{V,d} \|e_y(V)\|_2^2 + \gamma^2 \|g_{\mathrm{AC}}(V,d)\|_2^2 + \lambda q_d^{-2} \|d\|_2^2. \tag{121}$$

Development data fix $\gamma = 10^3$, $q_d = 1$, and $\lambda = 0.01$. Because the AC constraint appears through a finite penalty rather than a normalized stochastic density, the implemented objective is a regularized physical estimator, not a fully normalized posterior.

For an exactly constrained local MAP, let $r(\boldsymbol{\chi})$ be the stacked residual, $c(\boldsymbol{\chi}) = \mathbf{0}$ the equality constraints, and $\boldsymbol{\Omega}$ the residual precision. A Gauss–Newton step solves

$$\begin{bmatrix} H & G_c^{\mathsf{T}} \\ G_c & \mathbf{0} \end{bmatrix} \begin{bmatrix} \Delta\boldsymbol{\chi} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} -\nabla J \\ -c \end{bmatrix}, \quad H = J_r^{\mathsf{T}} \boldsymbol{\Omega} J_r + H_{\mathrm{prior}}. \tag{122}$$

Theorem 17 (Feasible local covariance): If G_c has full row rank and Z spans $\ker G_c$, then $Z^\mathsf{T} H Z \succ \mathbf{0}$ makes the KKT matrix nonsingular. The local constrained Gaussian covariance is

$$P_\chi = Z(Z^\mathsf{T} H Z)^{-1} Z^\mathsf{T}. \tag{123}$$

a: Proof.

Any vector in the KKT nullspace satisfies $G_c v = \mathbf{0}$, hence $v = Zq$. Premultiplying the first block equation by v^T gives $q^\mathsf{T} Z^\mathsf{T} H Z q = 0$, so $q = \mathbf{0}$ and $v = \mathbf{0}$. Full row rank then forces the multiplier to zero. Parameterizing feasible perturbations by $\delta\boldsymbol{\chi} = Zq$ gives Gaussian precision $Z^\mathsf{T} H Z$ in q; its linear pushforward is (123). □

This covariance is local and unimodal. The reported joint NEES-like failure shows that it is not calibrated for the full shifted-plant experiment.

C. IMPLEMENTED BATCH PHYSICAL-MANIFOLD BAYES $\mathcal{E}_\mathsf{B}$

Let r be the centered T-frame PMU window. For support S with amplitudes a_S, the evaluated mean is

$$\boldsymbol{\mu}_S(a_S) = \sum_{i\in S}(a_i D_i + a_i^2 Q_i) + \sum_{\substack{i,j\in S\\ i<j}} a_i a_j Q_{ij}. \tag{124}$$

The W2 likelihood is

$$q_S(a_S) = \|r - \boldsymbol{\mu}_S(a_S)\|^2_{\boldsymbol{\Sigma}_T^{-1}}, \tag{125}$$

$$p(r \mid S, a_S) = \frac{\exp[-q_S(a_S)/2]}{(2\pi)^{n_y T/2}|\boldsymbol{\Sigma}_T|^{1/2}}, \tag{126}$$

$$\boldsymbol{\Sigma}_T = R_T(\rho) \otimes \boldsymbol{\Omega}. \tag{127}$$

The covariance is fitted only on normal calibration records and remains fixed across supports. The selected priors are

$$p(K = 0, 1, 2) = (0.20, 0.50, 0.30), \tag{128}$$

$$p(S \mid K) = \binom{16}{K}^{-1}, \tag{129}$$

$$a_i \mid i \in S \sim \mathcal{N}(0, \sigma_a^2), \qquad \sigma_a = 0.05. \tag{130}$$

For reference, the linear single-source model has a closed form.

Proposition 1 (Conjugate amplitude posterior and Bayes factor): Let $r = ad + \varepsilon$, $\varepsilon \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, and $a \sim \mathcal{N}(0, \sigma_a^2)$. Define $I = d^\mathsf{T}\boldsymbol{\Sigma}^{-1} d$ and $s = d^\mathsf{T}\boldsymbol{\Sigma}^{-1} r$. Then

$$v = (I + \sigma_a^{-2})^{-1}, \qquad m = vs, \tag{131}$$

$$a \mid r, S \sim \mathcal{N}(m, v), \tag{132}$$

$$\log \frac{p(r \mid S)}{p(r \mid H_0)} = \tfrac{1}{2}\frac{s^2}{I + \sigma_a^{-2}} - \tfrac{1}{2}\log(1 + \sigma_a^2 I). \tag{133}$$

a: Proof.

Terms in the joint log density that depend on a are $-\frac{1}{2}[(I + \sigma_a^{-2})a^2 - 2sa]$. Completing the square gives mean m and variance v. Integrating the resulting Gaussian contributes $\sqrt{v/\sigma_a^2}\exp(vs^2/2)$ relative to H_0; taking its logarithm gives (133). The logarithmic term is the amplitude-volume or Occam penalty. □

Corollary 2 (One-sided amplitude evidence): Under Proposition 1, replace the signed Gaussian slab by the proper half-normal prior $p(a) = 2\mathcal{N}(a; 0, \sigma_a^2)\mathbb{I}(a \geq 0)$. If $BF_\pm$ denotes the signed Bayes factor in (133), then

$$BF_+ = 2\Phi(z) BF_\pm, \qquad z = \frac{\sigma_a s}{\sqrt{1 + \sigma_a^2 I}}. \tag{134}$$

Conditional on the active model, the severity posterior is $\mathcal{N}(m, v)$ truncated to $[0, \infty)$.

b: Proof.

The fixed-amplitude likelihood ratio is $\exp(as - Ia^2/2)$. Multiplication by the half-normal density and completion of the square give twice the signed Gaussian integral restricted to $a \geq 0$. The unrestricted posterior has standardized probability $\Phi(m/\sqrt{v}) = \Phi(z)$ on that half-line, which yields (134). □

The factor $2\Phi(z)$ preserves direction: a large score with the prohibited sign is evidence against the one-sided event even though the signed factor depends on s^2. Proper slab scales are essential because Bayes factors across different support dimensions are undefined up to an arbitrary constant under an improper amplitude prior. Under the one-sided null $a = 0$, the likelihood ratio statistic satisfies

$$2\Delta\ell = [Z]_+^2, \qquad Z \sim \mathcal{N}(0, 1), \tag{135}$$

and therefore has the boundary mixture $\frac{1}{2}\delta_0 + \frac{1}{2}\chi_1^2$, rather than the regular χ_1^2 law.

The quadratic multi-source mean is nonconjugate. Its evidence is

$$Z_S = p(r \mid S) = \int p(r \mid S, a_S) \prod_{i\in S} p(a_i)\, da_S. \tag{136}$$

With Q-node Gauss–Hermite abscissas x_q and weights w_q, the Gaussian-prior transformation $a_i = \sqrt{2}\sigma_a x_{q_i}$ gives

$$Z_S \approx \pi^{-|S|/2} \sum_{q_1=1}^{Q} \cdots \sum_{q_{|S|}=1}^{Q} \left(\prod_{j=1}^{|S|} w_{q_j}\right) p(r \mid S, \sqrt{2}\sigma_a x_q). \tag{137}$$

The implementation uses $Q = 31$ and evaluates (137) in the log domain using log-sum-exp. Independent adaptive Gauss–Kronrod integration checks representative two-dimensional cases. These checks support numerical integration under the frozen model; they do not validate the physical model itself.

D. POSTERIOR OUTPUTS AND THE CARDINALITY–LOCALIZATION DISTINCTION

The normalized support posterior is

$$p(S \mid r) = \frac{Z_S p(S)}{\sum_{R\in\mathcal{H}} Z_R p(R)}. \tag{138}$$

It induces cardinality, source-inclusion, and amplitude distributions:

$$p(K = k \mid r) = \sum_{|S|=k} p(S \mid r), \tag{139}$$

$$p(i \in S \mid r) = \sum_{S:\, i \in S} p(S \mid r), \tag{140}$$

$$p(a_i \mid r) = p(i \notin S \mid r)\delta_0 + \sum_{S:\, i \in S} p(S \mid r) p(a_i \mid S, r). \tag{141}$$

Equation (141) is a model-averaged spike-and-slab output: source absence contributes a point mass at zero.

The experimentally observed gap between cardinality and support has an exact algebraic decomposition. Let $N_k = \binom{16}{k}$, $\ell_S = \log Z_S$, and $\ell_k^\star = \max_{|S|=k} \ell_S$. Define the effective multiplicity

$$m_{\mathrm{eff},k} = \sum_{|S|=k} \exp(\ell_S - \ell_k^\star). \tag{142}$$

Proposition 2 (Exact effective-multiplicity identity): Under a uniform prior over supports of cardinality k,

$$\log p(r \mid K = k) = \ell_k^\star - \log N_k + \log m_{\mathrm{eff},k}, \tag{143}$$

$$\max_{|S|=k} p(S \mid K = k, r) = m_{\mathrm{eff},k}^{-1}. \tag{144}$$

a: Proof.

Factor $\exp(\ell_k^\star)$ from $N_k^{-1} \sum_{|S|=k} \exp(\ell_S)$ and take logarithms to obtain (143). Conditional normalization within cardinality k divides the largest evidence by the same sum, giving (144). □

Many similar pair supports can therefore increase evidence for $K = 2$ while preventing any one pair from dominating. Entropy separates the same two questions exactly because K is a deterministic function of S:

$$H(S \mid r) = H(K \mid r) + \sum_k p(K = k \mid r) H(S \mid K = k, r). \tag{145}$$

The first term measures cardinality uncertainty; the second measures localization ambiguity within each cardinality.

For set-valued output, sort supports by posterior mass and define the smallest credible set

$$\mathcal{C}_q(r) = \arg\min_{\mathcal{C}} |\mathcal{C}| \quad \text{subject to} \quad \sum_{S \in \mathcal{C}} p(S \mid r) \geq q. \tag{146}$$

Candidate-set size and coverage are more informative than an arbitrary Top-m when the posterior is multimodal. The present paper has not yet calibrated an abstention threshold or credible-set coverage prospectively.

E. STEP-BY-STEP EVALUATED BAYESIAN ALGORITHM

Table 10 records the exact order of the load-support computation. Steps marked "evaluated" produced the 137-hypothesis results; optional outputs are mathematically defined but are not promoted to validated claims.

The direct quadrature cost per record is

$$1 + GQ + \binom{G}{2} Q^2 \tag{147}$$

likelihood-node evaluations for $K \leq 2$. With $G = 16$ and $Q = 31$, this is 115,817 nodes before vectorization or reuse. The count explains why the support bank, integration order, and horizon must be frozen together when runtime is compared.

An incremental implementation of $\mathcal{E}_{\mathrm{B}}$ is computational, not a new estimator. If the exact innovations factorization includes the same initial AR(1) density, then at every frozen prefix it must reproduce the batch quadratic form, evidence, and posterior to numerical tolerance. This is a regression test, not evidence for the joint model below.

F. PROPOSED JOINT HYBRID-DAE BAYES $\mathcal{E}_{\mathrm{J}}$

The proposed extension would infer state trajectory X, algebraic variables Z, event family c, onset τ, support S, amplitudes a_S, operating nuisance $\boldsymbol{\eta}$, and integrity corruption b. Its posterior factorization is

$$\begin{aligned} p(X, Z, \boldsymbol{\eta}, b, c, \tau, S, a_S \mid Y) &\propto p(\boldsymbol{\eta}) p(c) p(\tau \mid c) p(S \mid c) \\ &\times p(a_S \mid S, c) p(\boldsymbol{\chi}_0 \mid \boldsymbol{\eta}) \\ &\times \prod_{t=0}^{T-1} p(\boldsymbol{\chi}_{t+1} \mid \boldsymbol{\chi}_t, \boldsymbol{\eta}, c, \tau, S, a_S) \\ &\times \prod_{t=0}^{T} p(y_t \mid \boldsymbol{\chi}_t, b_t, c, \tau, S, a_S) p(b_t), \end{aligned} \tag{148}$$

subject to the candidate DAE constraints. Here $\boldsymbol{\chi}_t$ denotes a consistent state/nuisance chart. A constrained negative twice-log density, excluding the discrete candidate priors, is

$$\begin{aligned} \mathcal{J}_S^\circ = \|\boldsymbol{\chi}_0 - \overline{\boldsymbol{\chi}}_0\|^2_{P_0^{-1}} + \sum_{t=0}^{T} \|r_{S,t}^{\mathrm{pmu}}\|^2_{R_t^{-1}} \\ + \sum_{t=0}^{T-1} \|r_{S,t}^{\mathrm{dyn}}\|^2_{Q_t^{-1}} + 2\lambda_b \sum_{t=0}^{T} \|b_t\|_1. \end{aligned} \tag{149}$$

Candidate evidence must be evaluated in a common feasible measure. The boundary correction is exposed by an explicit concentration parameter rather than by informally assuming a concentrated posterior.

Let $\mathcal{R}_0$ fix operating point, onset, and load family and replace the candidate prediction with the same quadratic manifold, W2 covariance, support prior, and amplitude prior used by $\mathcal{E}_{\mathrm{B}}$. Then the defining regression condition is

$$p_{\mathrm{J}}(S, a_S \mid Y, \mathcal{R}_0) = p_{\mathrm{B}}(S, a_S \mid r, \boldsymbol{\eta}_0, \tau_0, c_{\mathrm{load}}). \tag{150}$$

The equality is not claimed for an unrestricted nonlinear DAE likelihood; it holds only when the complete statistical contract is shared.

Theorem 18 (Local cone–Laplace evidence): Let $L_{S,\epsilon}$ be the log likelihood plus the log of a proper prior in identifiable

TABLE 10. Evaluated 137-hypothesis Bayesian load-support algorithm.

Step	Operation	Computation and frozen choice	Status
0	Declare contract	Fix 16 candidate load buses, $K \leq 2$, known onset, 30-frame window, eight PMUs, nominal topology, and no corrupted channels	Evaluated
1	Validate plant	Check equilibrium, terminal-current orientation, sampling, nonlinear solve acceptance, and first-order response before building the bank	Evaluated
2	Build physical dictionary	Generate centered D_i, self-curvature Q_i, and cross-interaction Q_{ij} from simulator-consistent interventions	Evaluated
3	Fit noise on CAL	Estimate channel covariance and AR(1) dependence from normal calibration records; freeze W2 before event TEST	Evaluated
4	Freeze priors and integrator	Set $p(K)$, uniform support priors, Gaussian amplitude slab, GH31 nodes/weights, and independent GK checks	Evaluated
5	Form the residual	Align the known onset, subtract the frozen normal mean, retain the fixed PMU/channel order, and apply the declared temporal covariance	Evaluated
6	Enumerate supports	Construct H_0, 16 single-source supports, and 120 two-source supports; no learned source label prunes the bank	Evaluated
7	Integrate each support	Evaluate (136) with (137); retain quadrature-weighted amplitude moments	Evaluated
8	Normalize once	Add the support prior in log space, subtract the global log-sum-exp, and obtain (138)	Evaluated
9	Read decisions in order	Report event presence, $p(K \mid r)$, conditional support ranking, source inclusion, and conditional amplitude intervals separately	Evaluated
10	Quantify ambiguity	Compute entropy, effective multiplicity, or $\mathcal{C}_q$ without relabeling them as validated abstention guarantees	Defined; partial evidence
11	Record provenance	Store the plant identifier, support, amplitude, noise seed, likelihood version, and frozen artifact hash for every independent record	Evaluated

local coordinates of dimension d_S. Suppose the rescaled feasible domains converge locally to a closed polyhedral cone K_S and

$$L_{S,\epsilon}(\boldsymbol{\xi}_0 + \epsilon u) - L_{S,\epsilon}(\boldsymbol{\xi}_0) = h_S^{\mathsf{T}} u - \tfrac{1}{2} u^{\mathsf{T}} H_S u + r_\epsilon(u), \quad (151)$$

where $H_S \succ \mathbf{0}$ and $r_\epsilon \to 0$ uniformly on compact sets. Assume an integrable dominating envelope and negligible evidence from distant modes. Then

$$Z_{S,\epsilon} = e^{L_{S,\epsilon}(\boldsymbol{\xi}_0)} \epsilon^{d_S} (2\pi)^{d_S/2} |H_S|^{-1/2} \times \exp\left(\tfrac{1}{2} h_S^{\mathsf{T}} H_S^{-1} h_S\right) \Pr(U_S \in K_S)[1 + o(1)], \quad (152)$$

$$U_S \sim \mathcal{N}(H_S^{-1} h_S, H_S^{-1}).$$

a: Proof.

Set $\boldsymbol{\xi} = \boldsymbol{\xi}_0 + \epsilon u$, which contributes the Jacobian ϵ^{d_S}. The compact-uniform expansion gives pointwise convergence of the rescaled integrand to $\exp(h_S^{\mathsf{T}} u - u^{\mathsf{T}} H_S u/2)$. The assumed envelope permits dominated convergence, while the distant-mode condition localizes the original integral. Completing the square in the limiting quadratic integral gives the Gaussian normalizer and the cone probability in (152). □

For an interior model $K_S = \mathbb{R}^{d_S}$ and the probability factor is one. For one or more boundary amplitudes it is a Gaussian orthant probability, not generally 2^{-q}. Half-normal prior normalizers supply their own factors in addition to this cone probability. Chart Jacobians, prior normalizers, and candidate-dependent covariance determinants belong in $L_{S,\epsilon}$ or the measure term. Comparing only optimized residuals drops both volume and boundary terms and does not produce a posterior.

For a small-noise Gaussian Hybrid-DAE trajectory map $\boldsymbol{\mu}_S(\boldsymbol{\xi}_0 + \epsilon u) = \boldsymbol{\mu}_0 + \epsilon J_S u + O(\epsilon^2)$,

$$h_S = J_S^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} z, \qquad H_S = J_S^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} J_S, \quad (153)$$

so positive curvature is equivalent to first-order identifiability in the reduced coordinates. This is a theoretical local specialization, not a validation of the unevaluated joint estimator. If H_S is singular, merely replacing its inverse by a pseudoinverse is invalid: redundant coordinates must first be quotiented out, while genuine higher-order identification changes the evidence scale. For example,

$$\int_0^\infty e^{-na^4/4}\, da = \frac{\Gamma(1/4)}{4^{3/4}} n^{-1/4}, \quad (154)$$

not the $n^{-1/2}$ rate of regular quadratic Laplace theory.

Proposition 3 (Exact reduction along redundant fibers): Suppose a parameter chart $\boldsymbol{\theta} \leftrightarrow (\boldsymbol{\xi}, \boldsymbol{\zeta})$ is one-to-one on the prior support and the observable map satisfies $\boldsymbol{\mu}(\boldsymbol{\theta}) = \overline{\boldsymbol{\mu}}(\boldsymbol{\xi})$ there. If $\widetilde{\pi}(\boldsymbol{\xi}, \boldsymbol{\zeta})$ is the transformed proper prior, then the evidence equals the reduced integral with induced prior

$$\overline{\pi}(\boldsymbol{\xi}) = \int \widetilde{\pi}(\boldsymbol{\xi}, \boldsymbol{\zeta})\, d\boldsymbol{\zeta}. \quad (155)$$

Thus an exactly redundant coordinate contributes no vanishing small-noise power; its effect is the pushforward prior measure, not a pseudodeterminant.

b: Proof.

The likelihood is independent of $\boldsymbol{\zeta}$. Fubini's theorem integrates the proper transformed prior along each fiber and gives $Z = \int p(y \mid \boldsymbol{\xi}) \overline{\pi}(\boldsymbol{\xi}) d\boldsymbol{\xi}$ exactly. □

For a merely local chart, the same reduction is asymptotic only after a localization condition excludes evidence from distant branches.

Theorem 19 (Mixed regular and higher-order evidence): In whitened coordinates, suppose the first observable expansion after redundancy removal is

$$\boldsymbol{\mu}(\boldsymbol{\xi}, a) = \boldsymbol{\mu}_0 + A\boldsymbol{\xi} + \frac{c_k}{k!} a^k + o(\|\boldsymbol{\xi}\| + |a|^k), \quad (156)$$

where A has rank r, the proper prior density is continuous and positive at the origin, and

$$c_{k,\perp} = (I - P_A) c_k \neq \mathbf{0}. \quad (157)$$

TABLE 11. Deterministic checks of boundary-aware evidence identities. These toy systems are not evaluated estimator results.

Check	Recomputed outcome
Signed/one-sided Bayes factors	max quadrature error 2.7e-12 (pass).
Broad nuisance limit	relative information error 2.1e-09 (pass).
Boundary likelihood ratio	zero mass 0.4972; positive mean 0.9987 (pass).
Nonlinear cone–Laplace	relative-error slope 0.9497 (pass).
Singular quartic evidence	scaling slope -0.2500 (pass).

Assume a dominating envelope and negligible evidence away from this branch. For the Gaussian-normalizer-free evidence kernel with noise scale ϵ,

$$M_\epsilon \sim C\epsilon^{r+1/k}, \tag{158}$$

$$C = \pi(\mathbf{0},0)(2\pi)^{r/2}|A^{\mathsf{T}}A|^{-1/2} \tag{159}$$

$$\times \frac{\Gamma(1/(2k))}{k}\left[\frac{2(k!)^2}{\|c_{k,\perp}\|_2^2}\right]^{1/(2k)}. \tag{160}$$

c: Proof.

Use $\boldsymbol{\xi} = \epsilon u$ and $a = \epsilon^{1/k}t$. The Jacobian is $\epsilon^{r+1/k}$ and the limiting kernel is $\exp[-\|Au + c_k t^k/k!\|^2/2]$. The component of c_k in $\operatorname{range} A$ disappears after a translation of u. Gaussian integration over the remaining regular coordinates yields $(2\pi)^{r/2}|A^{\mathsf{T}}A|^{-1/2}$; integration of the normal component over $t \in \mathbb{R}$ gives the gamma-function factor. Dominated convergence supplies the prior density at the origin and the stated limit. □

For one additional amplitude, the theorem separates three cardinality cases. A regular direction supplies an $O(\epsilon)$ Occam factor, a direction first visible at order k supplies $O(\epsilon^{1/k})$, and an exactly redundant extension supplies an $O(1)$ factor determined by its induced prior. For $\epsilon \asymp n^{-1/2}$ these become $n^{-1/2}$, $n^{-1/(2k)}$, and n^0. These are local evidence rates; the RAWSIM39 gate ledger does not contain the higher-order derivatives required to assign them to its rejected parameters.

For a sliding window, eliminating old Gaussian variables creates an arrival prior. Write its canonical log kernel as $-\frac{1}{2}z^{\mathsf{T}}\Lambda z + h^{\mathsf{T}}z + c$. If the information matrix and vector are partitioned into variables a to eliminate and retained variables b, exact integration yields

$$\Lambda_{\text{new}} = \Lambda_{bb} - \Lambda_{ba}\Lambda_{aa}^{-1}\Lambda_{ab}, \tag{161}$$

$$h_{\text{new}} = h_b - \Lambda_{ba}\Lambda_{aa}^{-1}h_a, \tag{162}$$

$$c_{\text{new}} = c + \tfrac{1}{2}h_a^{\mathsf{T}}\Lambda_{aa}^{-1}h_a + \tfrac{n_a}{2}\log(2\pi) - \tfrac{1}{2}\log\det\Lambda_{aa}. \tag{163}$$

The first two lines preserve the next MAP problem; the last line is also required to preserve evidence between candidates.

Table 12 gives the intended sequence. It is a specification for the next experiment, not a claim about the results reported in the main paper.

For a posterior state $\boldsymbol{\pi}_t(S) = p(S \mid Y_{1:t})$, an idealized unit-cost support decision with per-frame delay cost c satisfies the Bellman equation

$$V(\boldsymbol{\pi}) = \min\left\{1 - \max_S \pi(S),\ c + \mathbb{E}[V(\boldsymbol{\pi}_{t+1}) \mid \boldsymbol{\pi}_t = \boldsymbol{\pi}]\right\}. \tag{164}$$

The one-step expected information gain is

$$I(S; y_{t+1} \mid Y_{1:t}) = H(S \mid Y_{1:t}) - \mathbb{E}[H(S \mid Y_{1:t+1})]. \tag{165}$$

Equations (164)–(165) motivate adaptive horizon selection. They are not an executed stopping rule for the current fixed-onset, fixed-window bank.

Finally, a hidden-voltage or derived-output posterior must average across support uncertainty. With $w_S = p(S \mid Y)$, conditional output mean m_S, and covariance P_S,

$$\overline{m} = \sum_S w_S m_S, \tag{166}$$

$$\operatorname{Cov}(\psi \mid Y) = \sum_S w_S P_S + \sum_S w_S (m_S - \overline{m})(m_S - \overline{m})^{\mathsf{T}}. \tag{167}$$

The second term is between-support ambiguity. Nonlinear output moments may be computed from posterior samples or a local delta method; phasors are averaged in rectangular coordinates before magnitude and angle are formed.

G. ESTIMATOR CLAIM BOUNDARY

The state recursion is an independently evaluated component. $\mathcal{E}_{\mathrm{B}}$ computes normalized Bayesian evidence under the frozen quadratic W2/L2 model and is the implemented reference event estimator. $\mathcal{E}_{\mathrm{J}}$, its reduction test, learned discrepancy, unknown-onset/family inference, changing-PMU operation, virtual outputs, and sequential stopping remain specified but unexecuted. The physical recentering is a regularized constrained estimator with separately assessed uncertainty; it is not relabeled as either event estimator.

APPENDIX F COMPLETE FROZEN EXPERIMENTAL CONTRACTS

The PowerDynamics validation ladder checks static translation, PMU terminal-current orientation, equilibrium residual, local modes, nonlinear acceptance, and first-order response before estimator scoring. All 980 main mismatch cases and 175 refinement cases pass the declared acceptance gates. The physical-recentering TEST includes 20 equilibria at each of three shift scales; a solve is accepted only when the optimizer succeeds, outputs are finite, and the maximum AC residual is at most 5×10^{-5}. Rejections remain in the denominator.

For GLOBAL-137, the physical dictionary, covariance, priors, onset, and quadrature are frozen before new trajectories are generated. Three disjoint noise realizations accompany each event path, and 100 additional H_0 records represent no event. The identifiers have zero overlap with 1,816 historical records. The T120 bank preserves exact prefixes at 30, 60, 90, and 120 frames; independent operating-point validation remains pending.

TABLE 12. Candidate-conditioned joint estimator design and current implementation boundary.

Step	Operation	Required calculation	Present status
1	Pre-event update	Use only pre-event measurements to update state and operating nuisance; retain cross covariance and normalized arrival evidence	State filter evaluated; joint nuisance pending
2	Integrity mask	Mark missing channels exactly and introduce sparse corruption variables for available but suspect channels	Ontology defined; common inference pending
3	Candidate generation	Enumerate physical family, support, onset, and severity domain without using source identity as a learned label	Load family evaluated; other families pending
4	Candidate prediction	Apply the candidate DAE/event map and the current PMU membership to generate a physical trajectory	Load bank evaluated
5	Candidate-conditioned solve	Fit state and nuisance on the feasible DAE manifold, conditional on the physical candidate and amplitude quadrature node	Not executed jointly
6	Learned discrepancy	If used, predict only candidate-residual discrepancy and covariance; do not replace the physical hypothesis or consume target source labels	Not implemented
7	Evidence normalization	Retain GH31 amplitude marginalization when $\lvert S\rvert \leq 2$; integrate or approximate the remaining coordinates with all determinants, priors, and Schur normalizers	GH31 load evidence evaluated; joint evidence pending
8	Posterior outputs	Return state-functional mixture, cardinality, support, integrity state, severity, and candidate-set uncertainty	Separate state/load outputs only
9	Sequential decision	Continue observing when expected information gain justifies delay; use distinct thresholds for detection, cardinality, and exact source	Not evaluated
10	Reconfigure and verify	After accepting a physical change, rebuild the operating center and check subsequent innovations before declaring a persistent model update	Static recentering only

TABLE 13. Complete frozen-contract inventory. The main paper groups these contracts by research question.

Study	Frozen contract	Independent evaluation unit
Structural audit	Preregistered voltage/current target residual, horizons 0–180	31 hidden buses
Nominal reconstruction	20 DEV / 100 TEST; 91 frames at 30 Hz; B0–B3 frozen	Complete nonlinear trajectory
Nominal calibration	30 independent CAL trajectories; U1/U2 and discrepancy only	Complete nonlinear trajectory
Physical mismatch	Seven families and seven scales; B1/B2/U2 frozen	980 rebuilt plants
Breakpoint refinement	New seeds in transition bands; B2 frozen	175 rebuilt plants
Causal recentering	M1/M2/M6/M7; 160 DEV / 320 TEST; tune on DEV	Rebuilt plant trajectory
Regularized physical recentering	M6 PQ/PV/Slack basis; 30 DEV / 60 TEST; one snapshot	Rebuilt equilibrium
Load tangent audit	16 load buses; centered $\pm 0.5\%$ nonlinear simulations	Candidate-bus intervention
Bayesian load inference V1	Eight amplitudes; 2,560 event / 320 no-event TEST records	Source-amplitude-noise record
Bayesian load inference V2	1,000 normal CAL; 500 normal + 640 event DEV; weak and finite TEST blocks	Source-amplitude-noise record
Prospective 137-hypothesis inference	Fresh H_0 + 16 singles +120 pairs; 3,488 paths; 10,564 decisions	Physical trajectory or complete record, as declared
Weak-regime mechanism audit	Frozen GH31 prefixes at 5–30 frames; 2,400 paths	Retrospective source-pair trajectory
Long-horizon diagnostic	Exact prefixes through 120 frames; 32 nominal pair paths; 137 hypotheses	Diagnostic trajectory, not population recovery
Known-source event diagnosis	Eight abnormal labels plus normal; 414 TRAIN / 138 DEV / 138 TEST	Complete ANDES trajectory
Source-transfer audit	Asset-disjoint folds; true family supplied	101 event parents, each evaluated across three seeds

The ANDES ordinary split uses three replicas for training, one for development, and one for testing at each source configuration. The source-holdout audit removes the physical asset from every source-labeled operation and supplies the true event family at evaluation. These contracts answer interpolation and source generalization separately.

APPENDIX G SECONDARY RESULTS

The main paper retains the figures needed for the scientific argument. This appendix records secondary diagnostics and exact values that would otherwise interrupt that argument. The two compact tables first preserve the exact mismatch and weak-signal values; the full-width panels then show the corresponding temporal, likelihood, and supervised-baseline diagnostics.

TABLE 14. Median B2 hidden-voltage HVRE at the largest mismatch scale, $m = 1.5$. Ratios use the 140-case main-grid nominal median of 0.014744%.

Mismatch family	HVRE (%)	Ratio
M1 network	0.250430	16.985×
M2 machine	0.014744	1.000×
M3 governor	0.014743	1.000×
M4 AVR	0.014744	1.000×
M5 load model	0.014761	1.001×
M6 operating point	0.122641	8.318×
M7 coupled	0.023615	1.602×

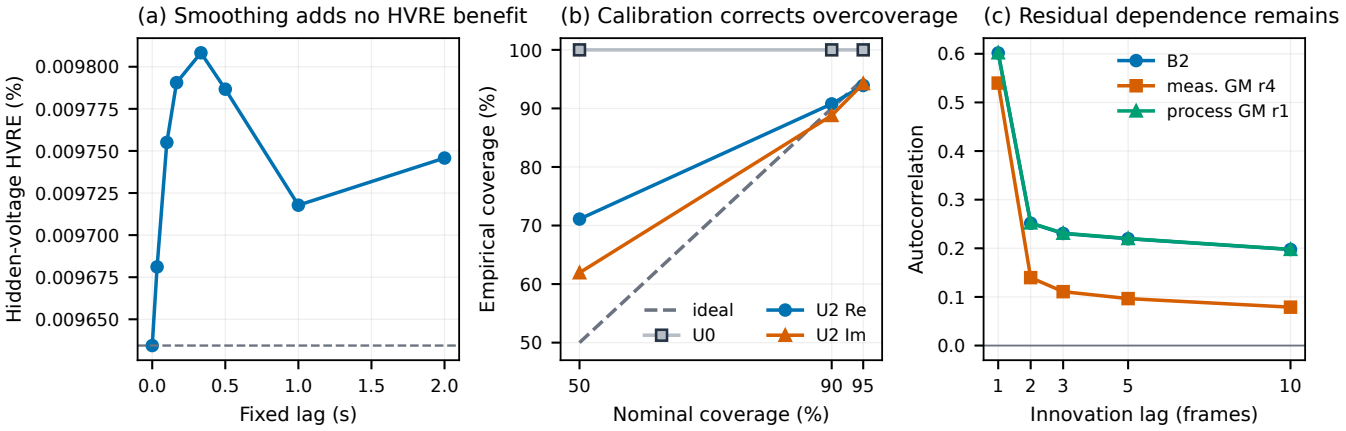


FIGURE 1. **Secondary temporal and uncertainty diagnostics for B2/B3 and calibration variants. These panels diagnose model adequacy; they do not upgrade nominal marginal coverage to a complete uncertainty claim.**

TABLE 15. **Sign-pooled weak load-event TEST results for the frozen W2/L2 model. Each row contains 1,600 event cases (16 source buses, two signs, 50 noise replicas); the false-positive rate uses 800 no-event records.**

$\lvert a \rvert$ (%)	AUROC	FPR (%)	FNR (%)	Top-1 (%)	Top-3 (%)
0.010	0.733	0.25	83.1	13.0	37.1
0.020	0.906	0.25	45.0	31.8	69.0
0.050	0.995	0.25	4.5	73.4	95.9
0.100	1.000	0.25	0.2	93.2	99.4
0.200	1.000	0.25	0.0	98.9	100.0
0.400	1.000	0.25	0.0	100.0	100.0

TABLE 16. Frozen TEST uncertainty and colored-discrepancy ablations. U2 is fitted on a disjoint 30-trajectory calibration set; Gauss–Markov (GM) ranks are selected on that calibration set.

Variant	Hidden NLL	95% cov. Re (%)	95% cov. Im (%)	Innovation ACF(1)	HVRE (%)
U0 original	-13.477	100.0	100.0	0.602	0.009634
U2 Re/Im temperature	-16.184	93.9	94.2	0.602	0.009634
D2-M measurement GM r4	14.991	66.6	48.1	0.540	0.009788
D2-P process GM r1	-13.475	100.0	100.0	0.602	0.009638

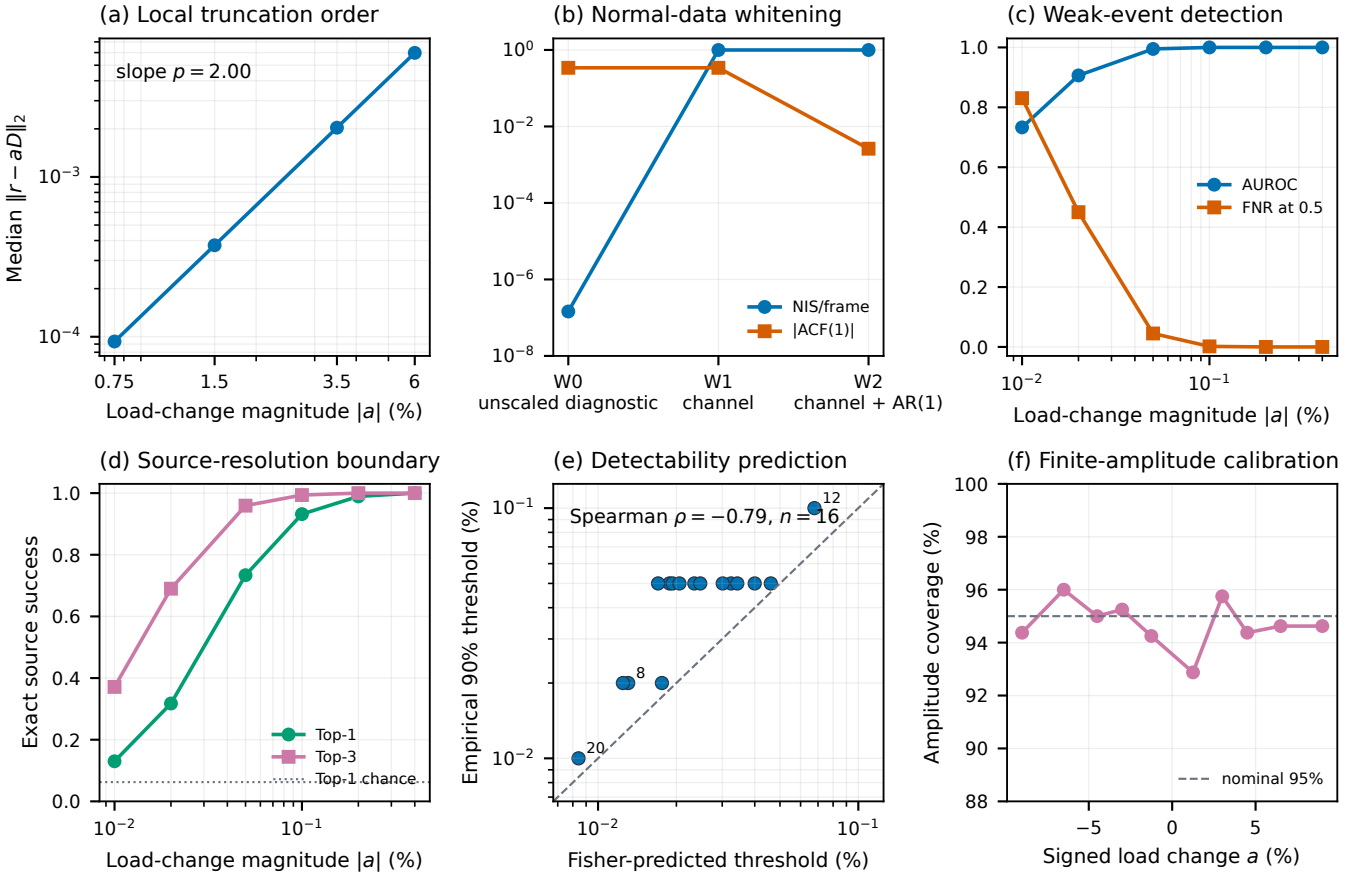


FIGURE 2. Single-source development evidence for the frozen load likelihood. The figure is retained as component validation; GLOBAL-137 in the main paper is the prospective support-recovery result.

TABLE 17. Audited sequential event-diagnosis baseline on the known-source TEST split, mean over three fitted seeds. The independent split unit is the complete trajectory; timestamp-level scores retain the denominators of the frozen evaluator.

Configuration	Detection F1	Event macro-F1	Physical Top-1	Physical Top-3	Integrity Top-1	False alarms/min
136-feature hierarchy	0.965	0.791	0.496	0.646	0.870	4.01
448-feature hierarchy	0.975	0.815	0.590	0.717	0.865	1.45
448 features + availability	0.989	0.905	0.592	0.719	0.974	1.45

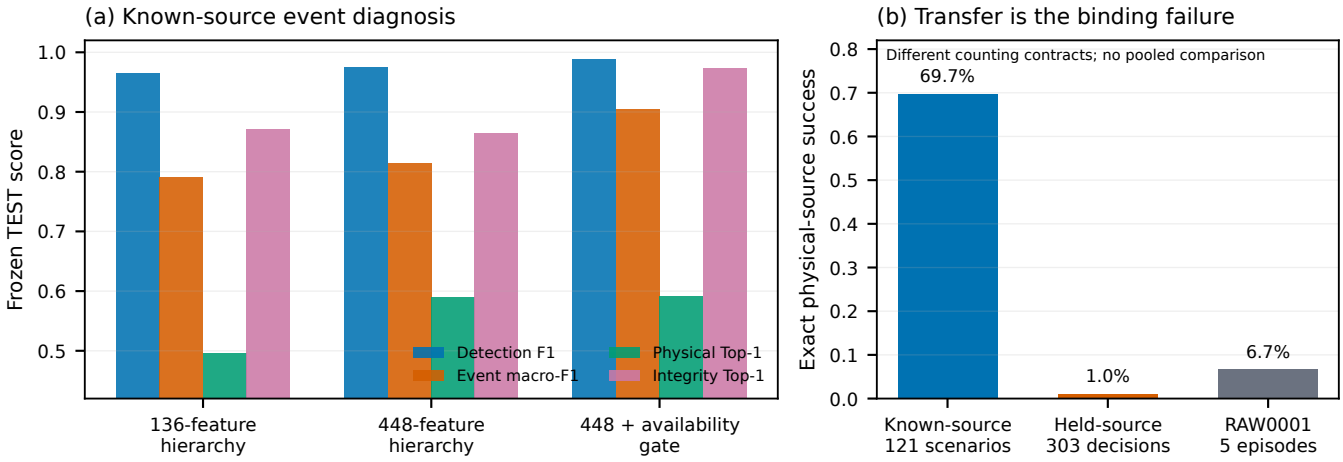


FIGURE 3. **Secondary evidence for the supervised ANDES baseline, including known-source performance and the source-holdout failure.**

APPENDIX H EXPLORATORY HIERARCHICAL SINGLE-EVENT PILOT

This appendix records the September 24, 2026 estimator handoff as a *provisional* numerical branch of the joint design $\mathcal{E}_{\mathrm{J}}$. Its source documents are the 30-page mathematical derivation book and 24-page technical integration dossier identified in the repository integration ledger. Their statements are technical source material, not independent instructions or substitutes for campaign outputs. The dossiers label many checks "artifact-verified" in packages supplied elsewhere; those packages were not supplied with this manuscript checkout. The latest strict-final comparison is explicitly tagged terminal-reported in both documents. All values below are therefore transcribed as *dossier-reported* diagnostics and excluded from the article's frozen primary tables.

A. STATE, MEASUREMENTS, AND EVENT OPERATORS

The pilot's declared model has 114 differential states, 78 algebraic rectangular-voltage coordinates, and 32 measured rectangular channels: voltage at eight PMUs and terminal current at eight monitored branches. Sampling is at $1/30$ s. Its default matrices are $P_0 = 10^{-2} I_{114}$, $Q = 10^{-6} I_{114}$, and $R = 10^{-6} I_{32}$. These numbers identify the pilot likelihood, not the separately frozen covariance W2 of $\mathcal{E}_{\mathrm{B}}$. Three-phase phasors, frequency, ROCOF, and data-present fields in the historical competition files do not silently become inputs to this 32-channel physical likelihood.

For a fixed candidate H and severity ϑ_H, the semi-explicit system is $\dot{x} = f_H(x, z; \vartheta_H)$, $\mathbf{0} = g_H(x, z; \vartheta_H)$, and $y = h_H(x, z) + v$. At a regular point with invertible $G_z = \partial g_H / \partial z$, the implicit-function theorem gives

$$Z_x = -G_z^{-1} G_x, \quad Z_\vartheta = -G_z^{-1} G_\vartheta, \quad A_c = F_x - F_z G_z^{-1} G_x. \tag{168}$$

The corresponding reduced event pathway is $B_\vartheta = F_\vartheta + F_z Z_\vartheta$ and the direct measured pathway is $D_\vartheta = H_\vartheta + H_z Z_\vartheta$. A generator-reference change can satisfy $D_\vartheta = \mathbf{0}$ while $B_\vartheta \neq \mathbf{0}$: it is invisible to a raw algebraic screen but visible after dynamic propagation. The event operator and its duration must be replayed consistently in truth generation and candidate fitting. The pilot dossier reports a five-frame fault pulse, persistent load steps unless recovery is declared, and finite generator and line events with recovery. Some earlier inference runs used persistent approximations for the latter two; those rankings have a known model-contract discrepancy.

The current single-event catalog has 39 fault buses, 19 load buses, nine controlled generators, 46 in-service branches, and a separate H_0 comparison. Thus it has 113 physical candidates and one null candidate; the screening rank denominators may be smaller when a candidate is numerically unscorable. Fault severity uses a log-resistance coordinate, load severity two active/reactive changes, generator severity a reference-power change, and an outage has a discrete branch and switching profile. Missing or bad PMU channels belong to the integrity variable of $\mathcal{E}_{\mathrm{J}}$ and were not jointly inferred in this pilot.

B. JOINT GAUSS–NEWTON STEP AND ITS INTERPRETATION

For one candidate, write the conditional MAP residual as $r_H(X, \vartheta)$ after whitening and appending proper prior residuals. Let $\Delta u = \mathrm{col}(\Delta X, \Delta\vartheta)$ and $g = \mathrm{col}(g_X, g_\vartheta)$. At an iterate, its local quadratic is

$$q(\Delta u) = g^{\mathsf{T}} \Delta u + \tfrac{1}{2} \Delta u^{\mathsf{T}} H \Delta u, \qquad H = \begin{bmatrix} H_{XX} & H_{X\vartheta} \\ H_{\vartheta X} & H_{\vartheta\vartheta} \end{bmatrix}. \tag{169}$$

Eliminating ΔX gives $q_{\mathrm{red}}(\Delta\vartheta) = \mathrm{const} + b_H^{\mathsf{T}} \Delta\vartheta + \tfrac{1}{2} \Delta\vartheta^{\mathsf{T}} S_H \Delta\vartheta$, where $S_H = H_{\vartheta\vartheta} - H_{\vartheta X} H_{XX}^{-1} H_{X\vartheta}$ and $b_H = g_\vartheta - H_{\vartheta X} H_{XX}^{-1} g_X$. Hence its unconstrained local decrease is $-\tfrac{1}{2} b_H^{\mathsf{T}} S_H^{-1} b_H$ when $S_H \succ 0$. This algebra needs no conditional convergence of X: it is the full-block Gauss–Newton step expressed with a smaller parameter solve. By contrast, if $\psi(\vartheta) = \min_X J_H(X, \vartheta)$, the identity $\nabla^2 \psi = H_{\vartheta\vartheta} - H_{\vartheta X} H_{XX}^{-1} H_{X\vartheta}$ is a local *profile-Hessian* statement only at a regular conditional optimum, and for the true Hessian rather than an arbitrary Gauss–Newton approximation. An unsettled state trajectory can change the sign of the reduced gradient without invalidating Schur elimination.

The numerical realization factors the temporally sparse state block and obtains its action on g_X and the few columns of $H_{X\vartheta}$ by solves; it does not form H_{XX}^{-1}. One Armijo step length updates both state and severity and evaluates the actual nonlinear DAE objective. The pilot selected Rodas4P for transition evaluation after the reported cross-integrator audit. That choice is a numerical guard, not a new inference theorem. Levenberg–Marquardt remains an independent finalizer candidate for the unresolved fault/line fits [23], [24].

C. DETECTION, PHYSICAL SCREENING, AND RACING

Let $\boldsymbol{\nu}_k = y_k - \widehat{y}_{k|k-1}^{H_0}$ be the observed innovation. An alarm tests the adequacy of H_0 on measured channels; it is not itself the physical onset. The pilot refines onset over a bounded interval and fits the common initial state $\widehat{X}_{\mathrm{pre}}$ on data before that onset. Every candidate starts from this same reference. This prevents a strong event from contaminating its own null-model baseline.

For a smooth candidate nested at H_0, the efficient local score is $T_H = b_H^{\mathsf{T}} S_H^{-1} b_H$ under positive definite S_H. If a severity is physically one-sided, maximizing the local quadratic over its tangent cone replaces the unconstrained inverse step and changes the null law; the ordinary signed score is then only a proposal proxy. For a fault or topology switch, let s_H be its short-horizon physical response from $\widehat{X}_{\mathrm{pre}}$ and r the corresponding null residual. With $W^{\mathsf{T}} W = R_H^{-1}$ and nuisance projector P_N, define $\widetilde{s}_H = (I - P_N) W s_H$ and $\widetilde{r} = (I - P_N) W r$. A signed matched score is $(\widetilde{r}^{\mathsf{T}} \widetilde{s}_H)^2 / \|\widetilde{s}_H\|^2$, with the allowed severity domain imposed before interpreting it as a physical fit. The two screening scores need not share a calibrated scale across families.

The frozen pilot consequently ranks within each family, retains a family quota, and allocates increasing nonlinear

TABLE 18. Single-event pilot ledger transcribed from the two author-supplied September 24 dossiers. All rows require the corresponding campaign archives for an independent paper audit.

Check or campaign	Dossier-reported result and precise interpretation	Source tier
Linear Gaussian oracle E0	RTS and batch MAP means agree near 10^{-16}; covariance/log-likelihood checks near 10^{-14} after an omitted evidence constant was repaired. This guards the numerical algebra only.	Dossier AV
Three-node and five-bus toys	E1: five development and 30 evaluation seeds, hidden algebraic two-sigma coverage 0.927 versus nominal 0.954. E2: first-order classification 0.2875 over 10 hypotheses × eight repetitions; not a nonlinear full-catalog result.	Dossier AV
Transition/derivative audit	Rodas4P/Rodas5P: 23 shared successful candidates with maximum endpoint difference 4.9×10^{-7}; Rodas4P evaluated three of three selected Rodas5P failures. Finite-difference coverage was 25/25; one random direction retained a moderate derivative discrepancy.	Dossier AV
Joint update and severity	Full-block and Schur steps agree in 30/30 synthetic block systems to numerical precision. Four fault-resistance initializations of 0.01, 0.03, 0.05, and 0.08 ended in 0.02006–0.02133 around truth 0.021; only the first two passed the stated strict stationarity gate.	Dossier AV
Small hidden-event bank	A hidden LOAD@16 and LINE(23,24) ranked first against selected decoys after joint severity and line-index corrections. These were not full-catalog tests.	Dossier AV
H_0 and onset pilots	Fifty-one disjoint normal calibration windows yielded no alarms at two reported thresholds on that same set; this is neither independent alarm testing nor false alarms per hour. Blind onset alternatives differed by 26–27 frames in two cases.	Dossier AV
Stage-A development	Four cases; 451/456 attempted physical scores were valid (98.9%); family-stratified shortlist retained 4/4 truths.	Dossier AV
Blind mini-holdout	FAULT@20, LOAD@28, GEN@38, and LINE(4,14): within-family truth ranks 3/38, 1/19, 1/9, and 1/46. The shortlist and subsequent racing retained 4/4 truths; GEN survived at the six-candidate cutoff. Four quiet windows were a proposal-level control only.	Dossier AV
Strict final comparison	The later terminal summary reports 0/4 exact winners after 24 final fits. Fault and line include unconverged truth/competitors; load and generator have better-scoring converged neighbors. This is a failure audit, not an identifiable physical error rate.	TR; final archive absent

budgets $B1, B2, B3, B5, B10$ before strict finalization. A candidate with a failed score is recorded as unscorable rather than inserted into a tie by catalog order. The dossier reports that ranking was nonmonotone with iteration budget and that the generator truth survived the blind racing cutoff at rank six of six. Thus survival is a search diagnostic; it says nothing by itself about a final converged winner or computational savings. In the reported racing run all 21 candidates were still evaluated to ten iterations, so a wall-clock advantage has not been demonstrated.

The operational order of this *pilot* is fixed: (1) process only the declared 32-channel PMU sequence and evaluate the H_0 innovation; (2) record alarm time, refine physical onset candidates, and fit a common pre-event state; (3) score every admissible physical family/location with its own event operator and explicit numerical status; (4) retain the frozen within-family quotas; (5) apply the same staged nonlinear budgets to all survivors while checkpointing objective and stationarity; (6) finalize every remaining candidate to the same strict gate; and (7) compare only validly converged modes with consistent priors, normalizers, and boundary treatment. Outputs before step (6) are candidate sets and diagnostics, not final source decisions. This route is retrospective whenever onset refinement or smoothing uses samples after the decision time; it has no measured end-to-end latency guarantee.

Several discarded results in the dossier were caused by implementation changes rather than by power-system information limits. A generator screen based only on the algebraic Jacobian missed its differential pathway; a previous joint update computed but did not apply the severity step; and an outage candidate was once switched at frame zero instead of the inferred onset. A full-window H_0 fit absorbed an event into the null reference, while independent caps on state and severity steps invalidated a quadratic proposal score. Unscored fault candidates inherited catalog order through a tied sentinel value. One detection-delay calculation used the window start rather than its end, and some failed integrator returns were encoded as nonfinite values despite finite unfinished states. The corrected pilot treats each as a separate numerical or semantic gate. Its four-case truth survival must not be merged with superseded rankings from those earlier variants.

D. EVIDENCE AND THE FINAL COMPARISON GATE

For converged, regular, identifiable interior modes only, the local Laplace approximation is

$$\log p(Y \mid H) \simeq -J_H(\widehat{u}_H) + \tfrac{d_H}{2}\log(2\pi) - \tfrac{1}{2}\log|\nabla^2 J_H(\widehat{u}_H)|, \tag{170}$$

where u_H uses a common feasible measure and J_H includes all model-dependent prior and likelihood normalizers. The normalizer, prior scale, curvature determinant, and any boundary or quotient correction must be retained when comparing families of different dimension. Section E-F derives the boundary-aware alternative. A candidate that stops at $B10$ or violates the strict stationarity/solver gate cannot be made comparable merely by evaluating (170). The pilot has not demonstrated exact normalization against the load-bank restriction $\mathcal{R}_0$.

E. DOSSIER-REPORTED NUMERICAL LEDGER

Table 18 preserves the numerical chronology without treating small pilot denominators as population rates. "Dossier AV" means the PDF claims an external campaign package supports the entry; it does not mean this repository has reopened that package. "TR" is the dossier's own terminal-report tier.

The case-specific terminal report places converged LOAD@27 and LOAD@26 ahead of true LOAD@28. It quotes approximate objectives -29986.9, -29985.9, and -29980.7, respectively, with evidence values around 4609.98, 4610.22, and 4607.17. Those values are deliberately absent from the main manuscript: their sign convention and normalizing terms must be checked against the final archive before quantitative model comparison. No exact generator

margin was supplied. The fault and line comparisons are not physically interpretable until both truth and nearest competitors pass a common strict convergence gate.

F. REQUIRED CLOSURE BEFORE A MULTI-FAMILY CLAIM

The next frozen audit must first replay each event operator, measurement conversion, onset, duration, and initial state under the same truth and inference model. It must then use an independent finalizer for the fault/line candidates, retaining every failed solve in the denominator. For converged load/generator mislocalizations, successive oracles should fix the true pre-event state, timing, severity, and trajectory, then remove noise and add a counterfactual nearby PMU. Only a persistent small whitened truth-to-competitor margin after those controls supports an information-limited explanation. The final-survivor archive must supply per-candidate status, objective, prior and likelihood constants, Hessian measure, and evidence normalization. A larger source-disjoint holdout, operational normal exposure, and the $\mathcal{R}_0$ reduction test are still needed. The current pilot cannot be used to upgrade the separate $\mathcal{E}_{\mathrm{B}}$ load-support result into a validated multi-family joint estimator.

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